

Complements on the subtori of a group prescheme.

Application to smooth groups

SGA 3, Exposé XV

English draft

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About this draft

This is an unofficial English translation of Exposé XV, “Complements on the subtori of a group prescheme. Application to smooth groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Expo15.pdf` (68 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XV. Complements on the subtori of a group prescheme. Application to smooth groups

by M. Raynaud*

0. Introduction

This exposé complements and partly takes up again exposés XI and XII; knowledge of exposés XIII and XIV is not indispensable. Continuing the effort begun in XII, we shall work with group S -preschemes which are not necessarily affine and not necessarily separated over S .

Sections 1, 2, 3, 4 are devoted to the study of subtori of a group prescheme. One obtains infinitesimal (§ 2) and global (§ 4) lifting theorems, in which points of finite order play an essential role (§ 1).

Sections 5, 6 and 7 are independent of the preceding sections. The consideration of infinitesimal neighborhoods leads to the representability of the functor of smooth subgroups equal to their connected normalizer (§ 5). In §§ 6 and 7, one is interested more particularly in Cartan subgroups.

Finally, in § 8 one gives a necessary and sufficient condition for the functor of subtori of a smooth group, or that of maximal tori, to be representable.

1. Lifting finite subgroups

1. Finite, smooth and central subgroups of multiplicative type

Proposition 1.1. *Let S be an affine scheme, S_0 a closed subscheme of S defined by an ideal of square zero, G a group S -prescheme, H_0 a group subscheme of $G_0 = G \times_S S_0$, smooth over S_0 , of finite type, and of multiplicative type. Then, for there to exist a group subscheme of G , of multiplicative type, lifting H_0 , it is necessary and sufficient that there exist a subscheme H' of G , flat over S , lifting H_0 .*

PROOF. The necessity of the condition is quite clear; let us prove its sufficiency. The group H_0 of multiplicative type is quasi-isotrivial (X 4.5); by Exp. X 2.1, there exist an S -group H of multiplicative type and an S_0 -isomorphism of groups:

$$u_0 : H \times_S S_0 \xrightarrow{\sim} H_0.$$

Since H' is flat over S and $H' \times_S S_0$ is of finite presentation over S_0 (Exp. IX 2.1 b)), H' is of finite presentation over S ; moreover, its fibers are smooth, so H' is smooth over S (EGA IV 17.5.1). Since the scheme S is affine, u_0 therefore lifts to an S -morphism of preschemes:

$$u : H \longrightarrow H'.$$

*This exposé and the next two (Exp. XVI and XVII) do not correspond to oral lectures in the seminar. With a few complements, they develop the substance of (succinct) handwritten notes by A. Grothendieck, written in 1964 on the occasion of the present seminar.

⁰_{xy} version of 1/12/08

It then follows from Exp. III 2.1 and Exp. IX 3.1 that the composite morphism $v_0 : H \times_{S_0} S \xrightarrow{\sim} H_0 \longrightarrow G_0$ also lifts to an S -morphism of groups:

$$v : H \longrightarrow G.$$

Since v_0 is an immersion, the same is true of v . The image of H under v is therefore a group subscheme of G , of multiplicative type, lifting H_0 . $\square$

Proposition 1.2. *Let S be a prescheme, S_0 a subprescheme of S defined by a locally nilpotent sheaf of ideals, G a group S -prescheme, flat and of finite presentation over S , and H_0 a group subscheme of $G_0 = G \times_S S_0$ which is smooth, finite over S_0 , of multiplicative type and central. Then there exists a unique group subscheme H of G , of multiplicative type, lifting H_0 . Moreover, H is central. (See XVII App. III, 1.)*

Proposition 1.2 bis. *Let S, G, S_0, G_0 be as above, H a group S -scheme, of multiplicative type, smooth and finite over S , and $u_0 : H_0 = H \times_S S_0 \longrightarrow G_0$ a central homomorphism. Then u lifts uniquely to a homomorphism $u : H \longrightarrow G$. Moreover, u is central.*

The existence of the lifting u in 1.2 bis follows easily from 1.2 by considering the graph of u_0 . The lifting u is unique and central by Exp. IX 3.4 and Exp. IX 5.1.

PROOF OF 1.2. The uniqueness of H and the fact that H is central follow from Exp. IX 5.6 b) and Exp. IX 3.4 bis. Taking uniqueness into account, to prove the existence of H one may suppose S affine and S_0 defined by an ideal of square zero, and it suffices (1.1) to find a subscheme of G , flat over S , lifting H_0 .

Since H_0 is smooth and finite over S_0 , one may suppose, after restricting S , that there exists an integer $n > 0$, invertible on S , such that $H_0 = {}_nH_0$. Consider the morphism of raising to the n th power in G :

$$u : G \longrightarrow G, \quad x \longmapsto x^n.$$

Again denote by ${}_nG$ the “kernel of u ”, i.e. the inverse image under u of the unit section of G (N.B. u is not in general a group morphism). Let us grant for a moment the following lemma:

Lemma 1.3. *Let k be a field, G a group scheme locally of finite type over k , n an integer prime to the characteristic of k , u the morphism of raising to the n th power in G . Then u is étale at every point x of G which belongs to the center of G .*

Since G is flat and of finite presentation over S , it follows from the preceding lemma and EGA IV 17.8.2 that if x is a point of G projecting to s in S and belonging to the center of G_s , the morphism u is étale at x . If in addition x is a point of ${}_nG$, then ${}_nG$ is étale over S at x . By hypothesis, the group H_0 is central and contained in ${}_nG_0$, so it is in fact contained in the largest open subset V of ${}_nG$ which is étale over S . Since H_0 and $V \times_S S_0 = V_0$ are étale over S_0 , H_0 is an open subset of V_0 (EGA IV 17.8.7 and 17.9.1). But then, the open subprescheme of V having the same underlying space as H_0 is a subscheme of G , flat over S , lifting H_0 .

It remains to prove Lemma 1.3. For this, note that the largest open subset of G on which u is étale is invariant under extension of the base field (EGA IV 17.8.2); this allows one to reduce to the case where x is rational over k . Denote by t_x translation by x , which is a k -automorphism of the scheme G . Since x is in the center of G , one has the commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{u} & G \\ t_x \downarrow & & \downarrow t_x^n \\ G & \xrightarrow{u} & G. \end{array}$$

It therefore suffices to show that u is étale at the origin, but this was seen in VIIA 8.4. $\square$

2. Global lifting of finite groups

Lemma 1.4. *Let A be a local ring, separated and complete for the topology defined by its maximal ideal $\mathfrak{m}$, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/\mathfrak{m}^n)$. Then for every prescheme X (resp. for every S -prescheme X), the canonical map:*

$$(x) \quad \text{Hom}(S, X) \longrightarrow \varprojlim_n \text{Hom}(S_n, X)$$

(resp.:

$$(xx) \quad \Gamma(X/S) \longrightarrow \varprojlim_n \Gamma(X_n/S_n), \quad \text{where } X_n = X \times_S S_n$$

is bijective.

PROOF. (xx) is an easy consequence of (x). Let us prove (x).

Let $u_n : S_n \rightarrow X$ ($n \in \mathbf{N}$) be a coherent system of morphisms. The image y of the closed point of S_n is thus independent of n , and u_n factors through $\text{Spec } \mathcal{O}_y$. By passage to the projective limit, the morphisms u_n define a ring morphism:

$$\tilde{u} : \mathcal{O}_y \longrightarrow \varprojlim_n (A/\mathfrak{m}^n) = A.$$

This shows that (x) is surjective; it is injective as soon as A is separated for the $\mathfrak{m}$ -adic topology. $\square$

Corollary 1.5. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/\mathfrak{m}^n)$, X a scheme finite over S and Y an S -prescheme. Then the canonical map:*

$$\text{Hom}_S(X, Y) \longrightarrow \varprojlim_n \text{Hom}_{S_n}(X_n, Y_n)$$

(where $X_n = X \times_S S_n$ and likewise $Y_n = \dots$) is bijective.

PROOF. Indeed, it follows from EGA II 6.2.5 that X is a finite sum of local S -schemes finite over S . This reduces one to the case where X itself is the spectrum of a complete noetherian local ring. But $\text{Hom}_S(X, Y) = \Gamma(Z/X)$ where Z is the X -prescheme $Y \times_S X$, and one applies the preceding proposition. $\square$

Proposition 1.6. *Let A, S, S_n be as above and let G and M be two group S -preschemes, with M finite over S . Then:*

a) *The canonical map:*

$$\text{Hom}_{S\text{-gr}}(M, G) \longrightarrow \varprojlim_n \text{Hom}_{S_n\text{-gr}}(M_n, G_n)$$

is bijective.

b) *If M is of multiplicative type and G is smooth over S , the canonical map:*

$$\varphi : \text{Hom}_{S\text{-gr}}(M, G) \longrightarrow \text{Hom}_{S_0\text{-gr}}(M_0, G_0)$$

is surjective. Moreover, if $\varphi(u) = \varphi(u') = u_0$, then u and u' are conjugate by an element of $G(S)$ reducing to the unit element of $G(S_0)$.

c) *If M is of multiplicative type and smooth over S , if G is flat of finite type over S , and if $u_0 : M_0 \rightarrow G_0$ is a central homomorphism, u_0 lifts uniquely to a homomorphism*

$$u : M \longrightarrow G.$$

Moreover, u is central if G has connected fibers.

PROOF. a) Follows from 1.5, from the fact that $M \times_S M$ is finite over S , and from the characterization of group morphisms as those making the well-known diagram commute:

$$\begin{array}{ccc} M \times_S M & \xrightarrow{u \times u} & G \times_S G \\ \downarrow & & \downarrow \\ M & \xrightarrow{u} & G. \end{array}$$

b) By Exp. IX 3.6, one can construct a coherent system of homomorphisms $u_n : M_n \rightarrow G_n$ lifting a homomorphism $u_0 : M_0 \rightarrow G_0$. Hence the first assertion of b), taking a) into account.

If now u and u' are two liftings of u_0 , then u_n and u'_n are conjugate by an element g_n of $G(S_n)$ lifting the unit element of $G(S_0)$ (Exp. IX 3.6); *loc. cit.* also implies that one can choose the g_n coherently, hence (1.4) arising from a section g of $G(S)$. The morphisms u and $\text{int}(g)u'$ then agree modulo $\mathfrak{m}^n$ for every n , and therefore agree (1.5).

c) The existence and uniqueness of u follow from a) and 1.2 bis. If G has connected fibers, u is central by Exp. IX 5.6 a). $\square$

2. Infinitesimal lifting of subtori

1. Statement of the theorem

We shall give an infinitesimal lifting theorem for subtori of a group prescheme which does not call upon smoothness hypotheses (unlike Exp. IX 3.6 bis) and which moreover answers a very natural question⁽¹⁾: is it enough to be able to lift “sufficiently many” points of finite order of a subtorus T_0 , to be assured of being able to lift T_0 (infinitesimally, of course)?

Theorem 2.1. *Let S be a noetherian affine scheme, S_0 a closed subscheme of S defined by an ideal J of square zero, G a group S -prescheme of finite type, $G_0 = G \times_S S_0$, T_0 a subtorus of G_0 , q an integer > 0 invertible on S . Suppose that for every integer n equal to a power of q , there exists a subscheme M_n of G , flat over S , such that $M_n \times_S S_0 = {}_nT_0$. Then there exists a subtorus T of G such that $T \times_S S_0 = T_0$.*

Theorem 2.1 will be useful to us through the following two corollaries:

Corollary 2.2. *Let S be a locally noetherian prescheme, S_0 a closed subprescheme of S defined by a locally nilpotent sheaf of ideals, G a group S -prescheme of finite type, T_0 a subtorus of $G_0 = G \times_S S_0$, q an integer > 0 invertible on S ; finally, with the integer n ranging over the powers of q , let (M_n) be a coherent system of group S -subschemes of G , of multiplicative type, lifting the ${}_nT_0$ (N.B. The system of subgroups of multiplicative type (M_n) is said to be coherent if $M_m = {}_m(M_n)$ when the integer m divides n .) Then there exists one and only one subtorus T of G such that $T \times_S S_0 = T_0$ and ${}_nT = M_n$ for every n .*

Corollary 2.3. *Let G be a group S -prescheme, flat and of finite presentation over S , S_0 a closed subprescheme of S defined by a sheaf of ideals of finite type and locally nilpotent, T_0 a central torus of $G_0 = G \times_S S_0$. Then there exists one and only one subtorus T of G lifting T_0 . Moreover, T is central.*

Remark 2.4. We leave to the reader the task of formulating the analogues of statements 2.1, 2.2, 2.3, where instead of lifting a subtorus of G_0 , one is given a torus T over S and proposes to lift a morphism

$$u_0 : T_0 \rightarrow G_0$$

(one reduces to the preceding cases by considering the graph of u_0).

Let us show how one deduces Corollaries 2.2 and 2.3 from 2.1.

Proof of Corollary 2.2. The uniqueness of T follows from Exp. IX 4.8 b) and Exp. IX 4.10. To prove the existence of T , one may therefore reduce to the case where S is affine, hence noetherian, and to the case where S_0 is defined by an ideal of square zero.

Lemma 2.5. *Let G be a group S -prescheme, of finite presentation, and H a group subscheme of G , of multiplicative type, smooth over S . Then $\text{Centr}_G(H)$ is representable by a group subprescheme of G , of finite presentation.*

⁽¹⁾**N.D.E.** The idea of approximating a torus by its finite subgroups appears in Grothendieck’s proof of the connectedness of centralizers of tori, see § 4.6 of BIBLE.

The lemma follows from Exp. VIII 6.5 e), without a smoothness hypothesis on H , when G is separated over S . In the present case, one notes that the assertion to be proved is local for the fpqc topology, which allows one to suppose H diagonalizable, hence of the form $D_S(M)$. One may also suppose S affine, then S noetherian thanks to EGA IV 8. Since H is smooth over S and of finite type, the order q of the torsion subgroup of M is invertible on S (Exp. VIII 2.1 e)). It is then immediate (cf. Exp. IX 4.10) that the subgroups ${}_nH$, with n ranging over the powers of q , are schematically dense in H (Exp. IX 4.1). But ${}_nH$ is a completely split covering of S (that is, is isomorphic to a finite direct sum of copies of S), so $\text{Centr}_G({}_nH) = Z_n$ is representable by the intersection of the centralizers in G of the sections of ${}_nH$ over S . It then suffices to apply the lemma:

Lemma 2.5 bis. *Let S be a noetherian prescheme, G a group S -prescheme of finite type, H a subgroup of multiplicative type of G , H_i an increasing filtered family of subgroups of multiplicative type of G , and suppose that $Z_i = \text{Centr}_G(H_i)$ is representable by a group subprescheme of G . Then the family of the Z_i is stationary. If in addition the H_i are schematically dense in H , one has $Z_i = \text{Centr}_G(H)$ for i sufficiently large.*

To see that the family of the Z_i is stationary, it suffices to show that the family of the underlying sets $\text{ens}(Z_i)$ is stationary. Indeed, the stationary value will be a closed subset of an open subset U of G , and after replacing G by U , one is reduced to studying a decreasing filtered family of closed subpreschemes of a noetherian prescheme. An easy constructibility argument reduces one to the case where S is integral. One must then show that the family of the $\text{ens}(Z_i)$ is stationary over a nonempty open subset of S . Now the generic fiber of G is separated (Exp. VIA 0.3), so, after restricting S , one may suppose G separated over S (EGA IV 8). But then Z_i is closed in G (Exp. VIII 6.5 e)). To establish the last assertion of the lemma, denote by Z the stationary value of the family Z_i . It is clear that $\text{Centr}_G(H)$ is a subfunctor of Z ; let us show that Z centralizes H . Now let E be the subprescheme of $H \times_S Z$ which is the kernel of the pair of morphisms:

$$\begin{aligned} H \times_S Z &\rightrightarrows G \\ (h, c) &\longmapsto c \\ (h, c) &\longmapsto hch^{-1}. \end{aligned}$$

The prescheme E contains $H_i \times_S Z$ for every i . On the other hand, the H_i are flat over S , so (EGA IV 11.10.9) for every point s of S , the $(H_i)_s$ are schematically dense in H_s , and the $H_i \times_S Z$ are schematically dense in $H \times_S Z$. Since G_s is separated, E_s is closed in $(H \times_S Z)_s$ and consequently is equal to it. But then E is closed in $H \times_S Z$, and is therefore equal to $H \times_S Z$. This says that Z centralizes H .

Let us return to the proof of 2.2. By 2.5, $Z_n = \text{Centr}_G(M_n)$ is representable, and by 2.5 bis the decreasing family of subpreschemes Z_n is stationary; let Z be the stationary value. The group Z contains T_0 and the M_n . After replacing G by Z , one may therefore suppose the M_n central.

One is then in the conditions for applying Theorem 2.1, and there exists a subtorus T of G lifting T_0 . The groups ${}_nT$ and M_n are then two liftings of ${}_nT_0$, so they are conjugate (Exp. IX 3.2 bis), and consequently agree, since M_n is central. The torus T answers the question.

PROOF OF COROLLARY 2.3. The uniqueness of T follows from Exp. IX 5.1 bis, and the fact that T is central follows from IX 5.6 b). This remark allows one to reduce by the usual procedure to the case where S is affine (hence S_0 defined by a nilpotent ideal of finite type), then to the case where S is noetherian. Restricting S if necessary, one may suppose that there exists an integer q invertible on S . Corollary 2.3 is then a consequence of 2.2 and 1.2. $\square$

Remark 2.6. One shows easily that Corollary 2.3 remains true if one replaces the torus T_0 by a smooth central subgroup of multiplicative type of G_0 .

2. Proof of 2.1

a) Reduction to the case where $T_0 = \mathbf{G}_{m, S_0}$.

Thanks to 1.1, one may suppose that M_n is a subgroup of multiplicative type. Using Exp. IX 3.2 bis, one may suppose that the family of the M_n is coherent (2.2). The centralizer Z_n of M_n in G is representable (2.5), and the filtered family of the Z_n is stationary (2.5 bis). After replacing G by Z_n for n sufficiently large, one may therefore suppose T_0 and the M_n central. The uniqueness of the lifting of T_0 is then assured (Exp. IX 5.1 bis).

Proceeding as in the proof of 1.1, one may suppose that there exist an S -torus T and an S_0 -isomorphism:

$$u_0 : T \times_S S_0 \xrightarrow{\sim} T_0$$

and it is equivalent to lift u_0 or to lift T_0 . In view of uniqueness, it suffices to prove the existence of a lifting of u_0 after making a faithfully flat affine extension $S' \rightarrow S$ of finite type (fpqc descent), which allows one to suppose $T = \mathbf{G}_{m, S}^r$ (Exp. X 4.5). If the restriction of u_0 to each factor $\mathbf{G}_m$ lifts to an S -morphism, necessarily central, one immediately deduces a lifting of u_0 . In brief, one may suppose $T_0 = \mathbf{G}_{m, S_0}$.

b) Definition of the obstruction to the existence of a lifting of T_0 .

To prove 2.1, it suffices by 1.1 to find a subscheme of G , flat over S , lifting T_0 . We shall see that one can define the obstruction to the existence of such a lifting as an element of a certain $\text{Ext}^1(,)$ of sheaves of modules.

Let U be an open subset of G such that T_0 is closed in U , and again denote by U (resp. U_0) the open subscheme of G (resp. G_0) having U as underlying space. The sheaf $\mathcal{O}_{T_0}$, regarded as a sheaf on U , is therefore a quotient of $\mathcal{O}_{U_0}$. Let h be the canonical epimorphism:

$$h : \mathcal{O}_{U_0} \rightarrow \mathcal{O}_{T_0}.$$

Lemma 2.7. *The canonical map:*

$$h' = \text{id}_J \otimes h : J \otimes_{S_0} \mathcal{O}_{U_0} \rightarrow J \otimes_{S_0} \mathcal{O}_{T_0}$$

factors (obviously uniquely) as $i \circ j_U$, where j_U is the canonical epimorphism:

$$J \otimes_{S_0} \mathcal{O}_{U_0} \rightarrow J\mathcal{O}_U \simeq J\mathcal{O}_{U_0}.$$

PROOF. One must show that $h'(K) = 0$, where K is the kernel of j_U . Now for every integer n equal to a power of q , one has an epimorphism:

$$h_n : \mathcal{O}_{T_0} \rightarrow \mathcal{O}_{nT_0},$$

and since nT_0 lifts to a scheme M_n flat over S , the canonical morphism:

$$j_n : J \otimes_{S_0} \mathcal{O}_{nT_0} \rightarrow J\mathcal{O}_{M_n}$$

is an isomorphism.

The commutative diagram below:

$$\begin{array}{ccccc} K & \longrightarrow & J \otimes_{S_0} \mathcal{O}_{U_0} & \xrightarrow{j_U} & J\mathcal{O}_U \subset \mathcal{O}_U \\ & & \downarrow h' & \swarrow i & \\ & & J \otimes_{S_0} \mathcal{O}_{T_0} & & \\ & & \downarrow h'_n & & \\ & & J \otimes_{S_0} \mathcal{O}_{nT_0} & \xrightarrow[\cong]{j_n} & J\mathcal{O}_{M_n} \subset \mathcal{O}_{M_n} \end{array}$$

proves that $h'(K)$ is contained in $\text{Ker } h'_n$ for every n , hence is contained in $\bigcap_n \text{Ker } h'_n$, and it suffices to show that the latter is zero. Now the sheaf $\mathcal{O}_{T_0}$ is equal to the sheaf $\mathcal{O}_{S_0}[\mathbf{Z}]$, the group algebra of $\mathbf{Z}$ with coefficients in $\mathcal{O}_{S_0}$, whereas $\mathcal{O}_{nT_0}$ is the quotient algebra $\mathcal{O}_{S_0}[\mathbf{Z}/n\mathbf{Z}]$.

Let $a = \sum_{m \in \mathbf{Z}} a_m \otimes m$ be an element of $J \otimes_{S_0} \mathcal{O}_{T_0}$. Thus the a_m are sections of J , almost all zero. Take n sufficiently large that the indices m for which a_m is nonzero have

distinct images in $\mathbf{Z}/n\mathbf{Z}$. Then if $a \in \text{Ker } h'_n$, one necessarily has $a = 0$. This proves that $\bigcap_n \text{Ker } h'_n = 0$, and proves 2.7. $\square$

Let then K_0 be the kernel of h ; consider the following diagram:

$$\begin{array}{ccccccc}
 & & & & 0 & & \\
 & & & & \downarrow & & \\
 & & J\mathcal{O}_U & & K_0 & & \\
 & & \downarrow \wr & & \downarrow & & \\
 0 & \longrightarrow & J\mathcal{O}_{U_0} & \longrightarrow & \mathcal{O}_U & \longrightarrow & \mathcal{O}_{U_0} \longrightarrow 0 \\
 & & \downarrow i & & & & \downarrow h \\
 & & J \otimes_{S_0} \mathcal{O}_{T_0} & & & & \mathcal{O}_{T_0} \\
 & & & & & & \downarrow \\
 & & & & & & 0
 \end{array}$$

The sheaf $\mathcal{O}_U$ defines an element Φ of the group $\text{Ext}_{\mathcal{O}_U}^1(\mathcal{O}_{U_0}, J\mathcal{O}_{U_0})$. Let Ψ be the element of $\text{Ext}_{\mathcal{O}_U}^1(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$ deduced from Φ by the bifactoriality of $\text{Ext}^1(,)$ thanks to the morphisms $K_0 \longrightarrow \mathcal{O}_{U_0}$ and $i : J\mathcal{O}_{U_0} \longrightarrow J \otimes_{S_0} \mathcal{O}_{T_0}$. It follows from Exp. III 4.1 and the infinitesimal flatness criterion (cf. Exp. III 4.3) that there exists a subscheme of U , flat over S , lifting T_0 if and only if Ψ is zero.

But note that T_0 is affine, so it suffices (Exp. III 4.5 and 4.6) to show that there exists locally on U a subscheme flat over S lifting T_0 . In brief, it suffices to show that the image of Ψ in the sheaf:

$$\mathcal{E} = \mathcal{E}xt_{\mathcal{O}_U}^1(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

is zero. We shall again denote by Ψ this element of $\Gamma(U, \mathcal{E})$.

c) Reduction to the case of S local artinian, with algebraically closed residue field.

Since K_0 and $J \otimes_{S_0} \mathcal{O}_{T_0}$ are coherent sheaves, the same holds for the sheaf $\mathcal{E}$; moreover, $\mathcal{E}$ has its support in T_0 , since this is so for $J \otimes_{S_0} \mathcal{O}_{T_0}$. To show that the section f of $\mathcal{E}$ is zero, it suffices to see that for every point u of T_0 , the image of f in the fiber of $\mathcal{E}$ at u is zero. But the formation of $\mathcal{E}xt^i(,)$ of coherent sheaves commutes with flat extensions of the base⁽²⁾, so one is reduced to proving the existence of a lifting of $T_0 \cap \text{Spec } \mathcal{O}_u$ which is flat over S . Let s be the projection of u to S ; one may then replace S by $\text{Spec } \mathcal{O}_s$ and G by $G \times_S \text{Spec } \mathcal{O}_s$.

After again making a faithfully flat extension, one may suppose that $\mathcal{O}_s$ has an algebraically closed residue field (EGA 0_{III} 10.3.1).

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_s$, and $S_n = \text{Spec } \mathcal{O}_s / \mathfrak{m}^n$. Suppose one has shown that for every $n > 0$, the obstruction to lifting $T_0 \times_S S_n = (T_0)_n$ to a subscheme of $G_n = G \times_S S_n$, flat over S_n , is zero, and let T_n be the unique lifting of $(T_0)_n$, flat over S_n , which is a subtorus of G_n . It is clear that if $n > n'$, one has:

$$(T_n) \times_{S_n} S_{n'} = T_{n'}.$$

If then u is a point of T_0 projecting to s , it follows from the lemma below, applied to the coherent system of liftings $(T_n \cap \text{Spec } \mathcal{O}_u)$ of $(T_0)_n \cap \text{Spec } \mathcal{O}_u$, that there does indeed exist a lifting of $T_0 \cap \text{Spec } \mathcal{O}_u$ flat over $\mathcal{O}_s$. One is therefore reduced to proving that Ψ is zero when $S = S_n$ is the spectrum of a local artinian ring with algebraically closed residue field, and to proving:

Lemma 2.8. *Let $A \longrightarrow B$ be a local homomorphism of noetherian local rings, $\mathfrak{m}$ the maximal ideal of A , J an ideal of square zero of A , M a B -module of finite type, $A_0 = A/J$, $B_0 = B/JB$, $M_0 = M/JM$, N_0 a quotient B_0 -module of M_0 which is flat over A_0 . For*

⁽²⁾**N.D.E.** Cf. EGA 0_{III}, 12.3.5.

PROOF OF 2.8. Denote by P_0 the kernel of the epimorphism $M_0 \rightarrow N_0$. For every $n > 0$, one therefore has the following commutative diagram, where the horizontal rows are exact:

$$\begin{array}{ccccccc}
(*)_n & & & & & & 0 \\
& & & & & & \downarrow \\
& & & & & & P_{0,n} \\
& & & & & & \downarrow \\
& & & & & & \downarrow \\
& & 0 \longrightarrow & J_n M_n & \longrightarrow & M_n & \longrightarrow M_{0,n} \longrightarrow 0 \\
& \nearrow & & \downarrow & & \downarrow & \downarrow \\
J_n \otimes_{A_{0,n}} M_{0,n} & & & & & & \\
& \searrow & & \downarrow & & \downarrow & \downarrow \\
& & 0 \longrightarrow & J_n \otimes_{A_{0,n}} N_{0,n} & \longrightarrow & N_n & \longrightarrow N_{0,n} \longrightarrow 0 \\
& & & & & \downarrow & \downarrow \\
& & & & & 0 & 0
\end{array}$$

The Artin–Rees lemma (Bourbaki, *Commutative Algebra*, Chap. 3 § 3 Cor. 1) shows that the filtration defined on JM (resp. $J \otimes_{A_0} M_0$ and $J \otimes_{A_0} N_0$) by the kernels of the morphisms:

$$JM \longrightarrow J_n M_n,$$

(resp. $J \otimes_{A_0} M_0 \longrightarrow J_n \otimes_{A_{0,n}} M_{0,n}$ and $J \otimes_{A_0} N_0 \longrightarrow J_n \otimes_{A_{0,n}} N_{0,n}$)

is $\mathfrak{m}B$ -good, so that by taking the projective limit of the diagrams $(*)_n$ and denoting by $\hat{Q}$ the separated completion of a B -module Q for the $\mathfrak{m}B$ -adic topology, one obtains the following

$$\widehat{(*)}$$

$$\begin{array}{ccccccc}
& & & & & & 0 \\
& & & & & \downarrow & \\
& & & & & \widehat{P}_0 & \\
& & & & & \downarrow & \\
& & & & & \widehat{M}_0 & \rightarrow 0 \\
& & \xrightarrow{\quad} \widehat{JM} & \longrightarrow & \widehat{M} & \longrightarrow & \\
& \nearrow & \downarrow & & \downarrow & & \downarrow \\
J\widehat{\otimes}_{A_0} M_0 & & & & & & \widehat{N}_0 \\
& \searrow & \downarrow & & \downarrow & & \downarrow \\
0 \rightarrow J\widehat{\otimes}_{A_0} N_0 & \rightarrow & \varprojlim_n N_n & \rightarrow & \widehat{N}_0 & \rightarrow & 0 \\
& & \downarrow & & \downarrow & & \\
& & 0 & & 0 & &
\end{array}$$

$$J \otimes_{A_0} M_0 \longrightarrow J \otimes_{A_0} N_0$$
$$\begin{array}{ccc} J \otimes_{A_0} M_0 & \xrightarrow{\text{can.}} & JM \\ & \searrow \quad \swarrow & \\ & J \otimes_{A_0} N_0 & . \end{array}$$
$$0 \longrightarrow JM \longrightarrow M \longrightarrow M_0 \longrightarrow 0$$
$$P_0 \longrightarrow M_0 \quad \text{and} \quad JM \longrightarrow J \otimes_{A_0} N_0.$$

d) Study of $\mathcal{E}$. We therefore suppose that S is the spectrum of a local artinian ring A , with algebraically closed residue field k . Let A_0 be the ring of S_0 , $\mathfrak{m}_0$ the maximal ideal of A_0 . Since S_0 is artinian, T_0 is closed in G (Exp. VIB 1.4.2); one may therefore take the open subset U equal to G , so that:

$$\mathcal{E} = \mathcal{E}xt_{\mathcal{O}_C}^1(K_0, J \otimes_{S_0} \mathcal{O}_{T_0}).$$

Let B_0 be the ring of the affine S_0 -scheme $T_0 = \mathbf{G}_{\mathbf{m}, S_0}$, and $\overline{B}_0$ the ring of the special fiber $T_0 \times_{S_0} \mathrm{Spec}(k) = \mathbf{G}_{\mathbf{m}, k}$ of T_0 . The sheaf $\mathcal{E}$ is a coherent $\mathcal{O}_{T_0}$ -module, hence is defined by a B_0 -module of finite type which we shall denote by E .

Consider the graded module associated with E and the $\mathfrak{m}_0 B_0$ -adic filtration:

$$E_r = \mathfrak{m}_0^r E / \mathfrak{m}_0^{r+1} E.$$

Each E_r is therefore a $\overline{B}_0$ -module of finite type, and $E_r = 0$ for r sufficiently large, since S_0 is artinian.

Let then g be a section of G over S inducing on S_0 a section g_0 of T_0 . Translation (on the left, to fix ideas) by the element g defines an “automorphism of the situation” from the point of view of the obstruction problem under consideration. In particular, to g corresponds an S_0 -automorphism of the sheaf $\mathcal{E}$ leaving the obstruction Ψ fixed. More precisely, g defines a semilinear automorphism of the B_0 -module E (relative to the A_0 -automorphism of B_0 defined by translation by g_0 in the group T_0). By reduction modulo $\mathfrak{m}_0^{r+1}$, g thus defines a semilinear automorphism of E_r (relative to the k -automorphism of $\overline{B}_0$ defined by translation by $g_0 \times_{S_0} \text{Spec}(k)$ in $T_0 \times_{S_0} \text{Spec}(k)$).

Lemma 2.9. *For every integer $r \geq 0$, E_r is a locally free $\overline{B}_0$ -module.*

PROOF. Let x be a point of T_0 , $\kappa(x)$ its residue field, $(E_r)_x$ “the fiber” of E_r at x , equal to $E_r \otimes_{\overline{B}_0} \kappa(x)$, $\ell(x)$ the rank of $(E_r)_x$ over $\kappa(x)$, ℓ the maximum value of $\ell(x)$ as x ranges over the points of T_0 . Denote by L the largest closed subscheme of $\text{Spec } \overline{B}_0 = \mathbf{G}_{\mathfrak{m},k}$ over which E_r is locally free of rank ℓ (TDTE IV Lemma 3.6). Let β be a point of $L(k)$ (there are such points, since L is of finite type over the algebraically closed field k), and let α be a point of $\mathbf{G}_{\mathfrak{m},k}(k)$ of order equal to a power n of q . The point α is therefore rational over k , and since by hypothesis ${}_n T_0$ lifts to a subscheme M_n étale over S , there exists a section a of G over S lifting α and which, over S_0 , is a section of T_0 . The remarks made above then show that the fibers of E_r are k -isomorphic at the points β and $\alpha\beta$ of $\mathbf{G}_{\mathfrak{m},k}(k)$. But the points of order a power of q are dense in $\mathbf{G}_{\mathfrak{m},k}$, and the same holds for their translates by β . Since L is closed in $\mathbf{G}_{\mathfrak{m},k}$ and $\mathbf{G}_{\mathfrak{m},k}$ is reduced, L equals $\mathbf{G}_{\mathfrak{m},k}$ and E_r is locally free over $\overline{B}_0$, of rank ℓ . $\square$

e) End of the proof of Theorem 2.1. Let $K_0(n)$ be the sheaf of ideals of $\mathcal{O}_{G_0}$ defining the closed subscheme ${}_n T_0$. The sheaf $K_0^{(3)}$ is therefore a subsheaf of $K_0(n)$. To simplify, put:

$$R = J \otimes_{S_0} \mathcal{O}_{T_0} \quad \text{and} \quad R(n) = J \otimes_{S_0} \mathcal{O}_{{}_n T_0}.$$

The sheaf $R(n)$ is therefore a quotient of R , and one has the diagram of maps:

$$\begin{array}{ccccccc} & & & & K_0 & & \\ & & & & \downarrow & & \\ & & & & K_0(n) & & \\ & & & & \downarrow & & \\ 0 & \longrightarrow & J\mathcal{O}_G & \longrightarrow & \mathcal{O}_G & \longrightarrow & \mathcal{O}_{G_0} \longrightarrow 0 \\ & & \downarrow & & & & \\ & & R & & & & \\ & & \downarrow & & & & \\ & & R(n) & & & & \end{array}$$

⁽³⁾**N.D.E.** Introduced after Lemma 2.7.

Using the bifactoriality of $\mathcal{E}xt_{\mathcal{O}_G}^1(,)$, one then obtains the following commutative diagram:

$$\begin{array}{ccc} \mathcal{E}xt_{\mathcal{O}_G}^1(\mathcal{O}_{G_0}, J\mathcal{O}_G) & & \\ \searrow & & \\ \mathcal{E}xt_{\mathcal{O}_G}^1(K_0(n), R) & \longrightarrow & \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) = \mathcal{E} \\ \downarrow & & \downarrow \\ \mathcal{E}xt_{\mathcal{O}_G}^1(K_0(n), R(n)) & \longrightarrow & \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n)). \end{array}$$

Again denote by Φ the element of $\mathcal{E}xt_{\mathcal{O}_G}^1(\mathcal{O}_{G_0}, J\mathcal{O}_G)$ defined by the exact sequence:

$$0 \longrightarrow J\mathcal{O}_G \longrightarrow \mathcal{O}_G \longrightarrow \mathcal{O}_{G_0} \longrightarrow 0,$$

so that Ψ is the image of Φ in $\mathcal{E}$. Since ${}_nT_0$ lifts to a subscheme M_n of G , flat over S , the image of Φ in the sheaf $\mathcal{E}xt_{\mathcal{O}_G}^1(K_0(n), R(n))$ is zero (Exp. III 4.1); a fortiori the image of Φ in $\mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n))$, which is also the image of Ψ , is zero.

Lemma 2.10. *The canonical morphism:*

$$\mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) \otimes_{B_0} \mathcal{O}_{{}_nT_0} \longrightarrow \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n))$$

is injective for every integer n .

PROOF. Indeed, the affine scheme $T_0 = \mathbf{G}_{m, S_0}$ has ring:

$$B_0 = A_0[T, T^{-1}].$$

The subscheme ${}_nT_0$ is defined by the vanishing of the following section of B_0 :

$$h(n) = T^n - 1,$$

which is regular (EGA 0_{IV} 15.2.2) and remains regular after every base change $S'_0 \longrightarrow S_0$. One therefore has an exact sequence of sheaves:

$$0 \longrightarrow \mathcal{O}_{T_0} \xrightarrow{h(n)} \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0} \longrightarrow 0.$$

Since ${}_nT_0$ is flat over S , by tensoring with J over $\mathcal{O}_{S_0}$ one obtains the exact sequence:

$$0 \longrightarrow R \xrightarrow{h(n)} R \longrightarrow R(n) \longrightarrow 0,$$

then the exact sequence of $\mathcal{E}xt$:

$$\cdots \longrightarrow \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) \xrightarrow{h(n)} \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R) \longrightarrow \mathcal{E}xt_{\mathcal{O}_G}^1(K_0, R(n)),$$

which completes the proof of the lemma. $\square$

What precedes shows that for every integer n equal to a power of q , the image of Ψ in $E/h(n)E$ is zero. To show that Ψ is zero, it suffices to see that if $\Psi \in \mathfrak{m}_0^r E$, then $\Psi \in \mathfrak{m}_0^{r+1} E$. Let $\bar{\Psi}$ be the image of Ψ in E_r . There exists an element $\Psi(n)$ of E such that:

$$\Psi = \Psi(n)h(n) \quad (n \text{ equal to a power of } q).$$

We noted that the image $\overline{h(n)}$ of $h(n)$ in $\bar{B}_0$ is again $\bar{B}_0$ -regular. Since E_s is locally free over $\bar{B}_0$ for every s (2.9), multiplication by $\overline{h(n)}$ in E_s is injective. One immediately deduces that $\Psi(n)$ is in $\mathfrak{m}_0^r E$. Let $\overline{\Psi(n)}$ be its image in E_r , so that one has the relation:

$$\bar{\Psi} = \overline{h(n)} \overline{\Psi(n)} \quad (n \text{ equal to a power of } q).$$

This shows that the set F of points of $\mathbf{G}_{m, k}$ at which $\bar{\Psi}$ takes the value 0 contains the dense set of points of order a power of q . Moreover, F is closed (since E_r is locally free over $\bar{B}_0$), and $\mathbf{G}_{m, k}$ is reduced, so $\bar{\Psi}$ is zero.

This completes the proof of Theorem 2.1.

3. Characterization of a subtorus by its underlying set

1. Statement of the theorem

Notation. If X is a prescheme, $\text{ens}(X)$ denotes the underlying set of X . If X and S' are two S -preschemes, $X_{S'} = X \times_S S'$, $u : X_{S'} \rightarrow X$ the canonical morphism, E a subset of $\text{ens}(X)$, one denotes by $E_{S'}$ (or $E \times_S S'$) the subset of $\text{ens}(X_{S'})$ equal to $u^{-1}(E)$.

Theorem 3.1. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type, q an integer > 0 invertible on S , E a subset of $\text{ens}(G)$. Consider the following assertions concerning E :*

- (i) *The set E is the underlying set of a subtorus T of G .*
- (ii) *a) For every point s of S , there exists a subtorus T_s of G_s such that $E \cap \text{ens}(G_s) = E_s$ is the underlying set of T_s .*
b) With the integer n ranging over the powers of q , there exists a coherent family (cf. 2.2) M_n of group subschemes of G , of multiplicative type, such that for every point s of S , one has:

$$(M_n)_s = {}_n T_s.$$

- (iii) *a) As in (ii) a).*
b) The set E is locally closed in $\text{ens}(G)$, and the dimension of the fibers of E , relative to S , is locally constant.
- (iv) *a) As in (ii) a).*
b) For every S -scheme S' which is the spectrum of a complete discrete valuation ring with algebraically closed residue field, $E_{S'}$ is the underlying set of a subtorus of $G_{S'}$.

Then one has the following implications:

A) (i) $\iff$ (ii) $\implies$ (iii) $\implies$ (iv).

B) If G is separated over S , one has (iii) $\iff$ (iv), and moreover E is closed.

C) Conditions (i), (ii), (iii) (and also (iv), if G is separated over S) are equivalent in the following two cases:

First case: a) The prescheme S is reduced, or else G is flat over S , and

b) For every point s of S , T_s is a central torus of G_s .

Second case: S is normal.

Moreover, in the two cases above, the torus T with underlying space E is unique.

Remarks 3.2. a) When S is reduced, it is unnecessary in (ii) to suppose that the family M_n is coherent, this condition being implied by the other hypotheses. Indeed, if the integer m divides n , the subgroups ${}_m M_n$ and M_m are étale over S , hence reduced, and have the same underlying space, so they agree.

b) If S is no longer supposed normal, it is no longer true in general that (iii) $\implies$ (i), even when S is reduced, geometrically unibranch, and G is a group scheme smooth over S . Indeed, consider the Borel subgroup of $\text{SL}_{2,S}$ formed by matrices of the form:

$$\begin{pmatrix} t & u \\ 0 & t^{-1} \end{pmatrix},$$

where S is the affine curve over a field k with ring:

$$k[x, y]/(y^2 - x^3).$$

Consider then the set E obtained as follows: over the “cusp of S ” ($x = y = 0$), we take the diagonal torus ($u = 0$). Over the complementary open subset ($x \neq 0$), we take the torus deduced from the diagonal torus by conjugation by the element:

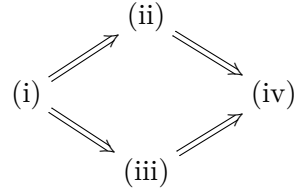
$$\begin{pmatrix} 1 & y/x \\ 0 & 1 \end{pmatrix}.$$

The set E obtained satisfies (iii) a); on the other hand, it is closed in G , and the reduced subscheme with underlying set E has equations:

$$\begin{cases} xu + y(t - t^{-1}) = 0, \\ u^2 - x(t - t^{-1})^2 = 0. \end{cases}$$

It is therefore not a subtorus of G , since the fiber over the cusp is not reduced.

Plan of the proof of 3.1. In a first part, we shall establish the following “easy” implications:



and (iv) $\implies$ [(iii) and E closed] if G is separated over S .

The proof of the more delicate implications will proceed in three stages:

- I) Reduction of the implication (iii) $\implies$ (i) (under the hypotheses of C), first case) to the case where S is normal.
- II) (iii) $\implies$ (ii) if S is normal.
- III) (ii) $\implies$ (i).

2. Proof of the “easy” assertions contained in 3.1

(i) $\implies$ (ii) and (iii) is trivial.

(iii) $\implies$ (iv). After replacing S by S' , we may suppose that S is the spectrum of a discrete valuation ring. Let t be the generic point of S , and s the closed point.

Since E is locally closed, there exists a subprescheme of G which is reduced and whose underlying space is E ; we shall denote it by $\overline{E}$. The generic fiber $\overline{E}_t$ of $\overline{E}$ is therefore equal to the subtorus T_t of rank r of G_t (by (iii) a)). Let E' be the schematic closure of $\overline{E}_t$ in $\overline{E}$. The prescheme E' is irreducible, and its closed fiber E'_s is nonempty (it contains the unit section of G_s), so E'_s has dimension r (EGA IV 14.3.10). But E'_s is then a closed subscheme of $\overline{E}_s$, and the latter has dimension r (by (iii) b)) and is irreducible (since it has the same underlying space as T_s), so E'_s has the same underlying space as $\overline{E}_s$, and consequently $E' = \overline{E}$, which proves that $\overline{E}$ is flat over S .

Let now E'' be the schematic closure of $\overline{E}_t$ in G . Then E'' is a group subprescheme of G , flat over S (Exp. VIII 7.1), containing E' , hence $\overline{E}$. The closed fiber E''_s is an algebraic subgroup of G_s , of dimension r (*loc. cit.*). Since T_s is an irreducible closed subset of G_s , of dimension r , E''_s has the same underlying set as the neutral component $(E''_s)^0$ of E''_s . Let $(E'')^0$ be the “neutral component” of E'' , i.e. the open subgroup of E'' complementary to the union of the irreducible components of E''_s not containing the origin. One therefore has:

$$E = \text{ens}[(E'')^0].$$

Since $\overline{E}$ and E'' are reduced, one even has $\overline{E} = (E'')^0$. Finally, $\overline{E}$ is a group subprescheme of G , flat and of finite type over S , with connected fibers, hence separated (Exp. VIB 5.2), whose generic fiber is a torus T_t , and whose reduced closed fiber is a torus T_s ; but then $\overline{E}$ is a torus (Exp. X 8.8).

(ii) $\implies$ (iv). One may again suppose that S is the spectrum of a discrete valuation ring, and we retain the notation introduced above. The schematic closure in G of T_t is a group subprescheme $\overline{T}$ of G , flat over S , containing M_n for every integer n equal to a power of q . Consequently, the closed fiber of $\overline{T}$ is a closed subscheme of G_s containing $(M_n)_s$ for every n , hence containing T_s . For reasons of dimension, the “neutral component” $\overline{T}^0$ of $\overline{T}$ has E as underlying set, and one concludes as above that $\overline{T}^0$ is a subtorus of G .

(iv) $\implies$ [(iii) and E closed], if G is separated over S . Let us show that E is closed.⁽⁴⁾
Let us first prove the lemma:

Lemma 3.3. *If the conditions stated in 3.1 iv) are satisfied, E is a locally constructible subset of $\text{ens}(G)$.*

PROOF. By the usual method, one is reduced to studying the case where S is noetherian, integral, with generic point η . After restricting S , one may suppose (Exp. VIB 10.10) that there exists a group subscheme of G , say H , flat over S , with connected fibers, whose generic fiber H_η is equal to T_η . To prove 3.3, it then suffices to show that $\text{ens}(H) = E$. Now if s is a point of S , by EGA II 7.1.9 there exists an S -scheme S' , the spectrum of a discrete valuation ring, which one may suppose complete with algebraically closed residue field, whose generic point t' projects to η and whose closed point s' projects to s . By (iv) b), there exists a subtorus $T_{S'}$ of $G_{S'}$ with underlying space $E_{S'}$. The two group subschemes $T_{S'}$ and $H_{S'} = H \times_S S'$ of $G_{S'}$ are flat over S' , have connected fibers, and have the same generic fiber $T_{t'}$, so they agree with the neutral component of the schematic closure of $T_{t'}$ in $G_{S'}$. Consequently $\text{ens}(H_s) = E_s$, hence $\text{ens}(H) = E$, which proves the lemma. $\square$

This being so, knowing that E is locally constructible, to see that E is closed it suffices to show that every specialization in G of a point of E is a point of E . By the usual technique, one is reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. But then, the subtorus of G with underlying space E ((iv) b)) is closed in G , since G is separated (Exp. VIII 7.12).

3. Continuation of the proof of 3.1

I) Reduction of (iii) $\implies$ (i) (C, first case) to the case where S is normal.

a) Reduction to the case of S affine reduced.

We therefore suppose that for every point s of S , E_s is the underlying space of a central subtorus of G_s . The uniqueness of T then follows from Exp. IX 5.1 bis, and moreover T will necessarily be central (*loc. cit.*). In view of uniqueness, to prove the existence of T , we may suppose S noetherian, affine with ring A . If S is not reduced, by hypothesis G is flat over S . By 2.3, it then suffices to solve the problem for S_{red} and $G \times_S S_{\text{red}}$. We may therefore suppose in addition that S is reduced.

b) Reduction to the case where the ring A is of finite type over $\mathbf{Z}$.

Let us first prove two lemmas:

Lemma 3.4. *Let k be a field, G an algebraic k -group, E a subset of G , k' an extension of k , $T_{k'}$ a subtorus of $G_{k'}$ with underlying space $E_{k'}$. Then if $T_{k'}$ is central or k is perfect, E is the underlying set of a subtorus of G_k .*

PROOF. Indeed, by fpqc descent, it suffices to show that the two inverse images of $T_{k'}$ in $G_{k''}$, where $k'' = k' \otimes_k k'$, agree. Now they have the same underlying space, namely the inverse image of E . If $T_{k'}$ is central, the lemma is then a consequence of Exp. IX 5.1 bis. If k is perfect, k'' is reduced, and the two inverse images of $T_{k'}$, being smooth over k'' , are reduced, hence agree. $\square$

Remark 3.5. It follows from the preceding lemma that in the statement of 3.1 (iv), property (iv) a) is a consequence of (iv) b) in the following two cases:

- (1) One supposes that the residue fields of the points of S are perfect.
- (2) For every S' as in (iv) b), one supposes that the torus with underlying space equal to $E_{S'}$ is central in $G_{S'}$.

Lemma 3.6. *Let S be a prescheme which is a projective limit of affine schemes S_i (cf. EGA IV 8), H a group S -scheme of multiplicative type and of finite type. Then there exist an index*

⁽⁴⁾**N.D.E.** The fact that (iv) $\implies$ (iii) will appear in the course of the proof.

i , a group S_i -scheme H_i of multiplicative type and of finite type, and an S -isomorphism:

$$H_i \times_{S_i} S \xrightarrow{\sim} H.$$

If in addition H is isotrivial, one may suppose H_i isotrivial.

PROOF. Since H is of finite type over S , H is even of finite presentation over S (Exp. IX 2.1 b)), so there exist an index ℓ and a group S_ℓ -scheme H_ℓ such that $H_\ell \times_{S_\ell} S$ is isomorphic to H (Exp. VIB 10). Putting $H_i = \prod H_{\ell S_i} S_\ell$, one therefore has $H \simeq \prod H_i S_i$ for every $i \geq \ell$.

Since H is of finite type over S , H is quasi-isotrivial (Exp. X 4.5), hence trivialized by an étale surjective morphism $S' \rightarrow S$. Using the quasi-compactness of S , one sees easily that there exists a covering of S by finitely many affine open subsets S_α and, for every α , an étale, surjective morphism of finite presentation $S'_\alpha \rightarrow S_\alpha$ trivializing $H|_{S_\alpha}$. This covering (S_α) of S then comes from a covering ($S_{i,\alpha}$) of S_i for i sufficiently large (EGA IV 8). After replacing S_i by $S_{i,\alpha}$ and S by S_α , one may therefore suppose that H is trivialized by an étale surjective morphism $S' \rightarrow S$ of finite presentation. For i sufficiently large, there then exist a prescheme S'_i , an étale surjective morphism $S'_i \rightarrow S_i$ of finite presentation, and an S -isomorphism $S'_i \times_{S_i} S \rightarrow S'$ (EGA IV 17.16). Put then, for j sufficiently large:

$$S'_j = S'_i \times_{S_i} S_j, \quad H'_j = H_j \times_{S_j} S'_j, \quad H' = H \times_S S'.$$

In view of the choice of S' , there exist an abelian group M of finite type and an S -isomorphism of group schemes $D_{S'}(M) \xrightarrow{\sim} H'$. Since the S'_i are quasi-compact and $S' = \varprojlim S'_i$, it follows from Exp. VIB 10 that there exist an index j and an S'_j -isomorphism of group schemes:

$$D_{S'_j}(M) \xrightarrow{\sim} H'_j.$$

But this says that H_j is a quasi-isotrivial group of multiplicative type.

When H is isotrivial, one proceeds in an analogous manner, using a trivialization of H by a finite étale morphism $S' \rightarrow S$. $\square$

This being so, we can carry out the announced reduction b). The ring A of S is the filtered inductive limit of its subrings A_i of finite type over $\mathbf{Z}$. Let $S_i = \text{Spec } A_i$, $u_j : S \rightarrow S_j$ and $u_{jk} : S_k \rightarrow S_j$ ($k \geq j$) be the transition morphisms. Since S is noetherian and G is of finite presentation over S , there exist an index i and a group S_i -prescheme G_i , of finite type over S_i , such that G is S -isomorphic to $G_i \times_{S_i} S$. Likewise, since E is locally closed in G , one may suppose that there exists a locally closed subset E_i of G_i such that $E = E_i \times_{S_i} S$ (EGA IV 8.3.11). For every $j > i$, denote by $G_j = G_i \times_{S_i} S_j$ and $E_j = E_i \times_{S_i} S_j$, and let Q_j be the set of points s of S_j such that $(E_j)_s$ is the underlying set of a central subtorus of $(G_j)_s$.

It follows from 3.4 that $Q_k = u_{jk}^{-1}(Q_j)$ for $k \geq j$, and by hypothesis $\text{ens}(S) = u_j^{-1}(Q_j)$ for $j \geq i$. Moreover, I say that Q_j is ind-constructible (EGA IV 1.9.4). Indeed, since S_j is noetherian, it suffices (EGA IV 1.9.10) to see that if S is an integral noetherian scheme with generic point η , and if E_η is the underlying set of a central subtorus T_η of G_η , there exists a neighborhood U of η such that for every point s of U , E_s has the same property. Now after restricting S , one may suppose that the center Z of G is representable and that there exists a group subscheme T of G whose generic fiber is T_η (VIB § 10). But then $\text{ens}(T)$ and E are two locally closed subsets of $\text{ens}(G)$ which agree on the generic fiber, so one can find a neighborhood U of η such that, over U , $\text{ens}(T) = E$ (EGA IV 8.3.11). For the same reasons, one may suppose that over U , T is central, since T is a torus (3.6).

Knowing now that Q_j is ind-constructible, it follows from EGA IV 8.3.4 that $Q_j = \text{ens}(S_j)$ for j sufficiently large. We may then replace S, G, E by S_j, G_j, E_j , which reduces one to the case where S is a reduced affine scheme of finite type over $\mathbf{Z}$.

c) Reduction to the case where S is the spectrum of a reduced complete noetherian local ring.

Because of the uniqueness of the torus with underlying space E , the usual technique (EGA IV 8) and Lemma 3.6 allow one to replace S by the spectrum of the local ring A of a point of S . Let $\hat{S}$ be the spectrum of the completion $\hat{A}$ of A for the topology defined by

the maximal ideal. Since A is the localization of an algebra of finite type over $\mathbf{Z}$, $\widehat{S}$ is again reduced (EGA IV 7.6.5). I say that it suffices to solve the problem after extension of the base $\widehat{S} \rightarrow S$. Indeed, if $\widehat{T}$ denotes the subtorus of $G_{\widehat{S}}$ with underlying space $E_{\widehat{S}}$, its two inverse images in $G_{\widehat{S} \times_S \widehat{S}}$ are two central subtori having the same underlying space, hence agreeing (Exp. IX 5.1 bis); and by fpqc descent, $\widehat{T}$ comes from a torus T of G answering the question (cf. 3.4).

d) A descent lemma.

Recall the following properties of finite morphisms, which were pointed out in TDTE I: let S and S' be two preschemes and $u : S' \rightarrow S$ a finite morphism. Then:

- (1) The morphism u is an epimorphism if and only if the canonical sheaf morphism:

$$\mathcal{O}_S \rightarrow u_*(\mathcal{O}_{S'})$$

is injective.

- (2) The morphism u is an effective epimorphism (Exp. IV 1.3) if and only if the canonical diagram:

$$0 \rightarrow \mathcal{O}_S \rightarrow u_*(\mathcal{O}_{S'}) \rightrightarrows u_*(\mathcal{O}_{S'}) \otimes_{\mathcal{O}_S} u_*(\mathcal{O}_{S'})$$

is exact.

- (3) If in addition S is noetherian and u is an epimorphism, u is the composite of a finite sequence of finite effective epimorphisms.

We are then in a position to prove the following lemma, whose proof uses a technique of nonflat descent:

Lemma 3.7. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type, S' a prescheme, and $u : S' \rightarrow S$ a finite morphism. For every S -prescheme T , denote by $M(T)$ the set of subgroups of multiplicative type of G_T (resp. the set of central subgroups of multiplicative type of G_T), so that M is naturally a contravariant functor defined on $(\text{Sch})/S$. Then if u is an effective epimorphism (resp. an epimorphism), the diagram of sets:*

$$(*) \quad M(S) \rightarrow M(S') \rightrightarrows M(S' \times_S S')$$

is exact.

Proof. i) *Reduction to the case where u is an effective epimorphism.* We are therefore interested in the functor of central subgroups of multiplicative type of G . The injectivity of $M(S) \rightarrow M(S')$ is a question local in nature on S , and once this injectivity is granted, the exactness of $(*)$ becomes a problem local in nature on S . One may therefore suppose S affine noetherian.

Let us study the case where $u : S \rightarrow S'$ is the composite of two finite epimorphisms:

$$S' \xrightarrow{v} S'' \xrightarrow{w} S.$$

I say that if the two diagrams:

$$(*)' \quad M(S'') \rightarrow M(S') \rightrightarrows M(S' \times_{S''} S'),$$

$$(*)'' \quad M(S) \rightarrow M(S'') \rightrightarrows M(S'' \times_S S'')$$

are exact, the same holds for $(*)$.

Indeed, the injectivity of $M(S) \rightarrow M(S')$ is clear. If now T' is an element of $\text{Ker}[M(S') \rightrightarrows M(S' \times_S S')]$, a fortiori T' belongs to $\text{Ker}[M(S') \rightrightarrows M(S' \times_{S''} S')]$, so by the exactness of $(*)'$, it comes from a unique element T'' of $M(S'')$. It suffices to show that T'' belongs to $\text{Ker}[M(S'') \rightrightarrows M(S'' \times_S S'')]$, for in view of the exactness of $(*)''$, T'' will descend to S . Let therefore T''_1 and T''_2 be the two inverse images of T'' in $M(S'' \times_S S'')$. Since they are two central subgroups of multiplicative type of $G_{S'' \times_S S''}$, to show that $T''_1 = T''_2$, it suffices to see

that they have the same fibers (Exp. IX 5.1 bis). Consider the commutative diagram:

$$\begin{array}{ccc} S' & \xleftarrow{\quad} & S' \times_S S' \\ v \downarrow & & \downarrow v \times v \\ S'' & \xleftarrow{\quad} & S'' \times_S S'' \end{array}$$

The morphism v is a finite epimorphism, hence is dominant (property (a) above) and closed, so is surjective, and the same holds for $v \times v$. Let then x'' be a point of $S'' \times_S S''$, and x' a point of $S' \times_S S'$ over x'' . It follows from the commutativity of the diagram above that the two inverse images of $(T_1'')_{x''}$ and $(T_2'')_{x''}$ in $G_{x'}$ agree, hence $(T_1'')_{x''} = (T_2'')_{x''}$ (EGA IV 2.2.15).

What precedes and an immediate induction on the number of factors of a factorization of $u : S' \rightarrow S$ into effective epimorphisms (property c) recalled above) show that to prove the exactness of $(*)$ in the case of central subgroups of multiplicative type, one may restrict to the case where u is an effective epimorphism. Finally, using Exp. IX 5.1 bis once more, one sees that it suffices to prove 3.7 when M is the functor of subgroups of multiplicative type of G and u an effective epimorphism.

ii) *Injectivity of $M(S) \rightarrow M(S')$.* Since $u : S' \rightarrow S$ is an epimorphism, the canonical morphism:

$$\mathcal{O}_S \rightarrow u_*(\mathcal{O}_{S'})$$

is injective; moreover, an S -group of multiplicative type is flat over S ; the injectivity of $M(S) \rightarrow M(S')$ is therefore a consequence of the following more general lemma:

Lemma 3.8. *Let $f : X \rightarrow S$ and $g : S' \rightarrow S$ be two morphisms of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, $X' = X \times_S S'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, $Q(\mathcal{F})$ the set of quotient $\mathcal{O}_X$ -modules $\mathcal{G}$ of $\mathcal{F}$ which are quasi-coherent and flat over S , $Q(\mathcal{F}')$ the analogue relative to $\mathcal{F}', X'$ and S' . Suppose g quasi-compact and $\mathcal{O}_S \rightarrow g_*(\mathcal{O}_{S'})$ injective; then the canonical map:*

$$\begin{aligned} Q(\mathcal{F}) &\rightarrow Q(\mathcal{F}') \\ \mathcal{G} &\mapsto \mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \end{aligned}$$

is injective.

Indeed, one may suppose S affine, then X affine. Since the morphism g is quasi-compact, S' is then the union of finitely many affine open subsets S'_i . After replacing S' by $\coprod S'_i$ (an operation preserving the injectivity of $\mathcal{O}_S \rightarrow g_*(\mathcal{O}_{S'})$), one may suppose S' affine. One is then reduced to proving the following lemma, whose proof is immediate:

Lemma 3.9. *Let $A \rightarrow A'$ be an injective ring homomorphism, M an A -module, $N = M/P$ a quotient A -module flat over A , $M' = M \otimes_A A'$, $N' = N \otimes_A A' = M'/P'$. Then P is the inverse image of P' under the canonical homomorphism $M \rightarrow M'$ (so N and P are known when one knows N' and P').*

iii) *Exactness of 3.7 (*) at $M(S')$.* Let T' be an element of $\text{Ker}[M(S') \rightrightarrows M(S' \times_S S')]$. Suppose one has proved iii) when S is the spectrum of a noetherian local ring. For every point s of S , there therefore exists a subgroup of multiplicative type T_s of $G \times_S \text{Spec}(\mathcal{O}_{S,s})$ arising by descent from $T' \times_{S'} u^{-1} \text{Spec}(\mathcal{O}_{S,s})$. By Exp. VIB § 10 and 3.6, there exist a neighborhood U of s in S and a group subscheme T of G over U , of multiplicative type, extending T_s . Let U' be the inverse image of U in S' . One then knows two subschemes of $G'|_{U'}$: $T'|_{U'}$ and $T \times_U U'$, agreeing on $u^{-1}(\text{Spec}(\mathcal{O}_{S,s}))$. If one regards $u^{-1}(\text{Spec}(\mathcal{O}_{S,s}))$ as the projective limit of the schemes $u^{-1}(V)$, where V ranges over the open neighborhoods of s in U , it follows from EGA IV 8 that there exists an open subset V of U containing s such that $T \times_U U'$ and T' agree over $u^{-1}(V)$. Thus, under the hypotheses made, T' descends locally on S , but because of the uniqueness proved in ii), T' then descends globally on S . In brief, it suffices to prove iii) when S is the spectrum of a noetherian local ring.

Denote then by $\widehat{S}$ the spectrum of the completion of the ring of S , and let $S'' = S' \times_S S'$, $\widehat{S}' = \widehat{S} \times_S S'$, $\widehat{S}'' = \widehat{S} \times_S S'' = \widehat{S}' \times_S \widehat{S}'$. I say that it suffices to show that the diagram:

$$(*) \quad M(\widehat{S}) \longrightarrow M(\widehat{S}') \rightrightarrows M(\widehat{S}'')$$

is exact at $M(\widehat{S}')$. This follows from the commutative diagram below, where the second row is exact at $M(\widehat{S}')$ by hypothesis, where the first two columns are exact (fpqc descent), and where the map f is injective, as follows from ii) applied to the finite epimorphism:

$$\widehat{S}' \times_{S'} \widehat{S}' \longrightarrow \widehat{S} \times_S \widehat{S}$$

deduced from $S' \longrightarrow S$ by the flat base change $\widehat{S} \times_S \widehat{S} \longrightarrow S$:

$$\begin{array}{ccccc} M(S) & \longrightarrow & M(S') & \rightrightarrows & M(S'') \\ \downarrow & & \downarrow & & \downarrow \\ M(\widehat{S}) & \longrightarrow & M(\widehat{S}') & \rightrightarrows & M(\widehat{S}'') \\ \Downarrow & & \Downarrow & & \\ M(\widehat{S} \times_S \widehat{S}) & \xrightarrow{f} & M(\widehat{S}' \times_{S'} \widehat{S}') & & \end{array}$$

(the diagram chasing is left to the reader). It follows from characterization b) of finite effective epimorphisms that the morphism $\widehat{S}' \longrightarrow \widehat{S}$, deduced from $S' \longrightarrow S$ by the flat base change $\widehat{S} \longrightarrow S$, is again a finite effective epimorphism. One is therefore reduced to the case where in addition S is the spectrum of a complete noetherian local ring. Denote by S_0 the reduced subscheme of S whose underlying space is the closed point of S . Let T' be an element of $\text{Ker}[M(S') \rightrightarrows M(S'')]$, T'' its image in $M(S'')$. Since $S'_0 = S' \times_S S_0$ is faithfully flat over S_0 , there exists an S -subgroup of multiplicative type T_0 of G_{S_0} whose inverse image in $G_{S'_0}$ is $T'_{S'_0}$ (fpqc descent). But S is local noetherian complete, so there exist an S -group T of multiplicative type and an S_0 -morphism $u_0 : T \times_S S_0 \longrightarrow T_0$ (Exp. X 3.3). The inverse image u'_0 of u_0 over S'_0 extends uniquely to an S' -isomorphism $u' : T_{S'} \longrightarrow T'$, again by Exp. X 3.3 (note that S' , being finite over S local complete, is the sum of finitely many complete local schemes). The two inverse images of u' over S'' are two morphisms from $T_{S''}$ to T'' which agree on $S_0 \times_S S''$, hence agree (*loc. cit.*). Since T is flat over S and $S' \longrightarrow S$ is a finite effective epimorphism, it follows from TDTE I page 8 that the diagram:

$$\text{Hom}_S(T, G) \longrightarrow \text{Hom}_{S'}(T_{S'}, G_{S'}) \rightrightarrows \text{Hom}_{S''}(T_{S''}, G_{S''})$$

is exact. Thus u' comes from a morphism of preschemes $u : T \longrightarrow G$. Likewise, $T \times_S T$ is flat over S , and consequently the map:

$$\text{Hom}_S(T \times_S T, G) \longrightarrow \text{Hom}_{S'}(T_{S'} \times_{S'} T_{S'}, G_{S'})$$

is injective. One immediately deduces that u is a group morphism, since this is so for u' . Moreover, u is a monomorphism. Indeed, note first that $\text{Ker } u$ is flat over S , for to establish this fact one may suppose S local artinian (EGA 0_{III} 10.2.2), hence G separated (Exp. VIA 0.3), but then $\text{Ker } u$ is of multiplicative type (Exp. IX 6.8), hence is flat over S . Since $\text{Ker}(u) \times_S S' = \text{Ker}(u') = 0$, one has $\text{Ker}(u) = 0$ (3.8). But u , being a monomorphism, is an immersion (Exp. VIII, Remarks 7.13), and the image group $u(T)$ is indeed an element of $M(S)$ whose image in $M(S')$ is T' , which completes the proof of 3.7.

e) End of the proof of I).

By reduction c), one is reduced to the case where S is the spectrum of a reduced complete noetherian local ring A . Let S' be the spectrum of the normalization A' of A , which is finite over A by Nagata (EGA 0_{IV} 23.1.5); $S' \longrightarrow S$ is an epimorphism, since A maps injectively into A' . Suppose then that there exists a torus T' of $G_{S'}$ with underlying space $E_{S'}$. The two inverse images of T' in $G_{S' \times_S S'}$ are two central subtori having the same underlying space, hence agreeing (Exp. IX 5 bis); so by 3.7, T' comes from a central subtorus T of G_S which

obviously has E as underlying space. It therefore suffices to prove the existence of T' , which reduces one to the case where S is normal and completes the proof of I).

II) Proof of (iii) $\implies$ (ii) when S is normal.

We may restrict to the case where S is integral; let t be its generic point. For every integer n equal to a power of q , let ${}_nG$ be the subprescheme of G which is the “kernel” of raising to the n th power in G . Since E is locally closed in G (by (iii) b)), the intersection of $\text{ens}({}_nG)$ with E is locally closed in $\text{ens}(G)$; denote then by $E(n)$ the reduced subprescheme of G with underlying space $\text{ens}({}_nG) \cap E$.

Let us show that the structural morphism $E(n) \longrightarrow S$ is separated and universally open. For these two properties, one has a valuative criterion (EGA II 7.2.3 and EGA IV 14.5.8). Let therefore S' be an S -scheme which is the spectrum of a complete discrete valuation ring with algebraically closed residue field. We have proved that 3.1 (iii) $\implies$ 3.1 (iv), so there exists a subtorus T' of $G_{S'}$ with underlying space $E_{S'}$. Now ${}_nT'$ is finite and étale over S' , hence is separated and universally open over S' , and the same holds for $E(n) \times_S S'$, which has the same underlying space as ${}_nT'$.

Moreover, the fibers of $E(n)$ have the same number of geometric points, namely r^n if r is the rank of the torus T_t . Finally, the generic fiber $E(n)_t$, being reduced, is equal to ${}_nT_t$, hence is étale over $\text{Spec}(t)$. Since S is normal, it then follows from SGA I 10.11 that $E(n)$ is an étale covering of S .

If s is a point of S , $E(n)_s$ is étale over $\text{Spec } \kappa(s)$, hence is reduced and consequently agrees with the group of multiplicative type ${}_nT_s$. Let us show that $E(n)$ is a group subprescheme of G . Indeed, let m be the morphism

$$E(n) \times_S E(n) \longrightarrow G$$

induced by multiplication in G . The map $\text{ens}(m)$ underlying m factors through $E(n)$, so m factors through the prescheme $E(n)$, since the first member $E(n) \times_S E(n)$ is étale over S , hence reduced. It then follows from Exp. X 4.8 a) that $E(n)$ is a subgroup of multiplicative type. As already noted (3.2 a)), the family of subgroups $E(n)$ is necessarily coherent. We have therefore proved that (iii) $\implies$ (ii) when S is normal.

III) Proof of (ii) $\implies$ (i).

In fact, we shall show that there exists a unique subtorus T of G with underlying space equal to E and such that ${}_nT = M_n$ for every n equal to a power of q . The uniqueness of T follows simply from Exp. IX 4.8 b). To establish the existence of T , taking uniqueness into account, we may suppose successively that:

- a) S is noetherian.
- b) The M_n are central subgroups. Indeed, it suffices to replace G by $Z_n = \text{Centr}_G(M_n)$ for n sufficiently large (2.5 and 2.5 bis).
- c) S is the spectrum of a local ring. Suppose indeed that the problem is solved after every base change $\text{Spec}(\mathcal{O}_{S,s}) \longrightarrow S$, where s ranges over the points of S . Let T_s be the subtorus of $G \times_S \text{Spec}(\mathcal{O}_{S,s})$ thus obtained. For every s , there then exist an open neighborhood U of s and a subtorus T of $G|_U$ extending T_s (Exp. VIB § 10 and 3.6). We have proved that 3.1 (ii) $\implies$ 3.1 (iv); since S is noetherian, E is therefore constructible (3.3). Consequently, after restricting U , one may suppose that $\text{ens}(T) = E \times_S U$ (EGA IV 9.5.2). But then, for every integer n equal to a power of q , ${}_nT$ and $M_n \times_S U$ are two subgroups of multiplicative type of $G|_U$ having the same fibers, hence agreeing, since M_n is central (Exp. IX 5.3 bis). In brief, under the hypotheses made, there exists a solution locally on S , hence, because of uniqueness, there exists a global solution.
- d) S is the spectrum of a complete noetherian local ring and, if s is the closed point, T_s is trivial. This follows from EGA 0_{III} 10.3.1 and fpqc descent.
- e) S is reduced. One applies 2.2.
- f) S is normal. One applies Nagata’s finiteness theorem (EGA 0_{IV} 23.1.5) and Lemma 3.7 in the case of central subtori.

These reductions having been made, since T_s is trivial, $(M_n)_s$ is trivial, hence M_n is trivial (Exp. X 3.3). If t is a point of S , the subgroups ${}_nT_t$ of T_t are therefore trivial for every n equal to a power of q , and it follows easily from Exp. X 1.4 that T_t itself is trivial. It then suffices to prove the lemma:

Lemma 3.10. *Under the hypotheses of 3.1 (ii), suppose in addition that S is noetherian and normal, and that for every point s of S , T_s is a trivial torus. Then there exists a unique subtorus T of G with underlying space equal to E , such that ${}_nT = M_n$ for every n equal to a power of q . Moreover, T is trivial.*

PROOF. The uniqueness of T follows from the fact that T , being smooth over S , is reduced. To prove existence, one may suppose S irreducible with generic point η . Let r be the rank of T_η , $T' = \mathbf{G}_{m,S}^r$, u_η an isomorphism of T'_η onto T_η , ${}_nu_\eta$ the restriction of u_η to ${}_nT'_\eta$. Since ${}_nT'$ and M_n are trivial, there exists a unique extension ${}_nu$ of ${}_nu_\eta$ to an S -isomorphism of ${}_nT'$ onto M_n . I say that for every point s of S , there exists a group isomorphism, necessarily unique:

$$u_s : T'_s \xrightarrow{\sim} T_s$$

extending ${}_nu_s$ for every n equal to a power of q . Indeed, let S_1 be an S -scheme, the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point t_1 projects to η and whose closed point s_1 projects to s (EGA II 7.1.9). It follows from the proof of 3.1 (ii) $\implies$ 3.1 (iv) that there exists a subtorus T^1 of G_{S_1} such that $(M_n)_{S_1} = {}_nT^1$ for every n equal to a power of q . Since S_1 is normal, T^1 is isotrivial (Exp. X 5.16), and it follows from the classification of isotrivial tori (Exp. X 1.2) and SGA1 V 8.2 that T^1 , having trivial generic fiber, is trivial. One can therefore extend the isomorphism $u_\eta \times_\eta t_1$ to an S_1 -isomorphism of group schemes:

$$u^1 : T' \times_S S_1 \xrightarrow{\sim} T^1.$$

The restriction of u^1 to ${}_nT' \times_S S_1$, on the one hand, and ${}_nu \times_S S^1$, on the other hand, agree on the generic fiber by construction, hence agree. The restriction $u^1_{s_1}$ of u^1 to the closed fiber therefore gives the required extension after extension of the residue field $\kappa(s) \longrightarrow \kappa(s_1)$. It then follows from Exp. IX 4.8 a) and fpqc descent that $u^1_{s_1}$ descends to $\kappa(s)$ as a group isomorphism $u_s : T'_s \xrightarrow{\sim} T_s$ answering the question.

Moreover, T' is smooth over S , which is normal, hence is normal. It then follows from an easy criterion for extending rational maps (EGA IV₄ 20.4.6) that, for there to exist an S -morphism $u : T' \longrightarrow G$ whose restriction to T'_s , for every point s of S , is the composite morphism:

$$T'_s \xrightarrow{u_s} T_s \longrightarrow G_s,$$

it is necessary and sufficient that this be so after every base change $S_1 \longrightarrow S$, where S_1 is the spectrum of a complete discrete valuation ring with algebraically closed residue field.

Now in the present case, an argument analogous to the one just made shows that the extension condition is indeed satisfied. Denote by u the S -morphism $T' \longrightarrow G$ thus obtained.

Let us show that u is indeed a group morphism. Let $m_{T'}$ (resp. m_G) be the morphism defining multiplication in T' (resp. G). One must verify that $m_G \circ (u \times_S u) = u \circ m_{T'}$. Now the coincidence subscheme of these two morphisms is a subscheme of $T' \times_S T'$ containing the fibers (since u_S is a group morphism), hence having the same underlying space as $T' \times_S T'$, hence equal to the latter, since $T' \times_S T'$ is smooth over S , thus reduced.

Finally, note that u is a monomorphism (since this is so on the fibers), hence is an immersion (Exp. VIII 7.9). The image of T' under u is then a subtorus of G having E as underlying set. $\square$

This completes the proof of Theorem 3.1.

4. Characterization of a subtorus T by the subgroups ${}_nT$

1. Statement of the main theorem

Theorem 4.1. *Let S be a connected locally noetherian prescheme, G a group S -prescheme of finite type over S , q an integer > 1 invertible on S , r a positive integer. For every integer n equal to a power of q , let $M(n)$ be a group subscheme of G , of multiplicative type and of type $(\mathbf{Z}/n\mathbf{Z})^r$. One supposes that:*

a) *The family of subgroups $M(n)$ is coherent, that is, if the integer m divides n , one has:*

$${}_mM(n) = M(m).$$

b) *There exist a point s of S and a subtorus T_s of G_s such that:*

$$M(n)_s = {}_nT_s \quad \text{for every } n.$$

c) *For every point t of S , there exists a closed affine subscheme $F_{\bar{t}}$ of $G_{\bar{t}}$ containing $M(n)_{\bar{t}}$ for every n .*

Then there exists one and only one subtorus T of G such that ${}_nT = M(n)$ for every n equal to a power of q .

One has an analogous theorem concerning the lifting of morphisms:

Theorem 4.1 bis. *Let S, G, q be as above, T an S -torus, and for every integer n equal to a power of q , let $u(n)$ be a group S -morphism ${}_nT \rightarrow G$. One supposes that:*

a) *The family of morphisms $u(n)$ is coherent, i.e. if m divides n , one has:*

$$u(m) = u(n)|_{{}_mT}.$$

b) *There exist a point s of S and a group morphism:*

$$u_s : T_s \rightarrow G_s$$

such that $u_s|_{{}_nT_s} = u(n)_s$ for every n equal to a power of q .

c) *For every point t of S , there exists a closed affine subscheme $F_{\bar{t}}$ of $G_{\bar{t}}$ containing $u(n)_{\bar{t}}({}_nT_{\bar{t}})$ for every n .*

Then there exists a unique group morphism $u : T \rightarrow G$ such that for every n equal to a power of q , the restriction of u to ${}_nT$ is equal to $u(n)$.

Remark 4.2. Using the lower semicontinuity of the abelian rank of a group prescheme flat of finite type over the spectrum of a discrete valuation ring (cf. Exp. X 8.7), one may, in the statements of 4.1 and 4.1 bis, weaken condition c) by merely asking that the required closed affine subscheme $F_{\bar{t}}$ exist for every maximal point t of S .

Let us show how 4.1 bis follows from 4.1. Let $G' = G \times_S T$. For every integer n equal to a power of q , consider the group morphism:

$$v(n) : {}_nT \rightarrow G'$$

whose projections onto G and T are respectively $u(n)$ and the canonical immersion ${}_nT \rightarrow T$. The morphism $v(n)$ is therefore an immersion; let $M(n)$ be the image subgroup. It is clear that the family of subgroups $M(n)$ is coherent in the sense of 4.1, that the group $M(n)_s$ is equal to ${}_nT'_s$, where T'_s is the subtorus of G'_s which is the graph of u_s , and that for every point t of S , the closed affine subscheme $F_{\bar{t}} \times_{\bar{t}} T_{\bar{t}}$ of $G'_{\bar{t}}$ contains the subgroups $M(n)_{\bar{t}}$ for every n . By 4.1, there therefore exists a subtorus T' of G' such that ${}_nT' = M(n)$ for every n equal to a power of q . Let f be the restriction to T' of the projection of G' onto T , and $f(n)$ the restriction of f to $M(n)$. One has:

$$f(n) \circ v(n) = \text{id}_{{}_nT}.$$

The fiber at s of T' is the torus already denoted by T'_s , equal to the graph of u_s (this follows from Exp. IX 4.8 b)), so f_s is an isomorphism. But $\text{Ker } f$ and $\text{Coker } f$ are groups of

multiplicative type (Exp. IX 2.7), of constant type since S is connected, hence reduced to the unit group, and f is an isomorphism. Let v be the inverse isomorphism of f . One has:

$$v|_{nT} = v(n).$$

Consequently, the composite of v and the projection of G' onto G is a morphism $u : T \rightarrow G$ answering the question. What precedes proves the existence of the morphism u ; as for uniqueness, it follows in any case from Exp. IX 4.8 a).

2. Application

We propose to generalize Theorem 7.1 of Exp. IX.

Let A be a complete noetherian local ring, I its maximal ideal, $S = \operatorname{Spec} A$, $S_m = \operatorname{Spec}(A/I^m)$. For every prescheme X , put $X_m = X \times_S S_m$.

Let then G be a group S -prescheme of finite type, T an S -torus, q an integer invertible on S , and $u_m : T_m \rightarrow G_m$ ($m \in \mathbf{N}$) a coherent family of group morphisms. With the integer n ranging over the powers of q , denote by $u_m(n)$ the restriction of u_m to ${}_nT_m$, and by $u(n)$ the unique group morphism:

$$u(n) : {}_nT \rightarrow G$$

extending the morphisms $u_m(n)$ for every m (1.6 a)). We shall say that the family (u_m) , $m \in \mathbf{N}$, is admissible if for every point t of S , there exists a closed affine subprescheme $F_{\bar{t}}$ of $G_{\bar{t}}$ containing $u(n)_{\bar{t}}({}_nT_{\bar{t}})$ for every n equal to a power of q (this property is independent of q , as we shall see).

Proposition 4.3. *With the notation above, the canonical map:*

$$\operatorname{Hom}_{S\text{-gr}}(T, G) \rightarrow \varprojlim_m \operatorname{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

induces an isomorphism from the first member onto the subset of the second member formed by the “admissible” coherent families.

Indeed, it suffices to apply 4.1 bis, taking as s the closed point of S .

Corollary 4.4. *With the notation above, suppose in addition that G has affine fibers; then the canonical map:*

$$\operatorname{Hom}_{S\text{-gr}}(T, G) \rightarrow \varprojlim_m \operatorname{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

is an isomorphism.

Indeed, if G has affine fibers, every coherent family is “admissible”.

Remark 4.5. When G is separated, in 4.3 one may replace the ring A by a noetherian I -adic ring; indeed, one may use EGA III 5.4.1 instead of 1.6 a).

3. Proof of 4.1

Lemma 4.6. *Let k be a field, G an algebraic k -group, q an integer prime to the residual characteristic of k , r an integer > 0 , $M(n)$ (with n ranging over the powers of q) a coherent family of subgroups of G , of multiplicative type and of type $(\mathbf{Z}/n\mathbf{Z})^r$. Then there exists a smallest closed subscheme H of G containing the $M(n)$ for every n . Moreover, H is an algebraic subgroup of G , smooth, connected and commutative, “whose formation is compatible with extension of the base field”.*

Let M be the subset of $\operatorname{ens}(G)$ which is the union of the underlying sets of the subgroups $M(n)$ for every n , and H the reduced closed subscheme of G whose underlying space is the closure of M . Since $M(n)$ is étale over k , hence reduced, $M(n)$ is contained in H , and consequently H is the smallest closed subscheme of G containing $M(n)$ for every n . Let now $\bar{k}$ be an algebraically closed extension of k . By construction, the subschemes $M(n)$ are schematically dense in H (Exp. IX 4.1), so the $M(n)_{\bar{k}}$ are schematically dense in $H_{\bar{k}}$ (Exp. IX 4.5); moreover, $M(n)_{\bar{k}}$ is reduced, and it follows that $H_{\bar{k}}$ is necessarily equal to the closed

reduced subscheme of $G_{\bar{k}}$ whose underlying space is the closure of $M_{\bar{k}}$. This proves that H is geometrically reduced (EGA IV 4.6.1). Since the family $M(n)$ is coherent, M is stable under the group law; moreover, $M \times_k M$ is dense in $\text{ens}(H \times_k H)$ and $H \times_k H$ is reduced (EGA IV 4.6.5); one immediately deduces that H is an algebraic subgroup of G . Moreover, H is smooth over S , since it is geometrically reduced (Exp. VIA 1.3.1), and H is commutative since the $M(n)$ are commutative. It remains to see that H is connected. Let H^0 be the neutral component of H , m the number of geometric points of H/H^0 , q^s the exponent of q in the prime factorization of m . For every integer n equal to a power of q , $q^s M(n)$ is then contained in H^0 . But the family $M(n)$ is coherent and $M(n)$ is of type $(\mathbf{Z}/n\mathbf{Z})^r$, so $q^s M(nq^s) = M(n)$. Thus H^0 contains $M(n)$ for every n , and consequently is equal to H .

This being so, we shall make condition c) of 4.1 more precise. Indeed, by what precedes, we may consider the smallest closed subscheme H_t of G_t containing $M(n)_t$ for every n . Since the formation of H_t commutes with extension of the base field (4.6), $H_{\bar{t}} = H_t \times_t \bar{t}$ is contained in the affine closed subscheme $F_{\bar{t}}$, hence is affine. In brief, H_t is affine. On the other hand, we know (4.6) that H_t is a smooth, connected, commutative algebraic group. It then follows from the structure of smooth commutative connected affine algebraic groups over an algebraically closed field (BIBLE 4 Th. 4) that $H_{\bar{t}}$ is the direct product of a torus $T_{\bar{t}}$ and a unipotent group $U_{\bar{t}}$. But then $M(n)_{\bar{t}}$ is necessarily contained in $T_{\bar{t}}$, so $H_{\bar{t}} = T_{\bar{t}}$, and consequently H_t is a torus.

We may now begin the proof of 4.1.

- a) Uniqueness of the solution: it suffices to apply Exp. IX 4.8 b).
- b) Case where S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field k . Denote by K the fraction field of A , s the closed point of S , t the generic point, J the maximal ideal of A , $S_m = \text{Spec}(A/J^m)$, $X_m = X \times_S S_m$.

Let us distinguish two cases:

First case: The point s of 4.1 b) is the generic point t of S . Let then T' be the schematic closure in G of the torus T_t . The closed fiber T'_s is therefore an algebraic subgroup of G_s , of dimension r , containing $M(n)_s$ for every n , so T'_s contains H_s . But H_s is a torus containing $M(n)_s$, so H_s has rank at least r . Consequently H_s has the same underlying space as the neutral component of T'_s . The “neutral component” $(T')^0$ of T' is then a group subscheme of G , flat, separated (VIB 5.2), whose generic fiber T_t is a torus and whose reduced closed fiber H_s is a torus. But then $(T')^0$ is a torus (Exp. X 8.8), which we denote by T . The groups ${}_n T$ and M_n are smooth over S , hence reduced, and since they have the same underlying space, they agree. Thus the torus T is the solution of the problem.

Second case: The point s of 4.1 b) is the closed point s of S . After replacing G by the schematic closure in G of the smallest algebraic subgroup H_t containing the family $M(n)_t$ (4.6), we may suppose G_t affine.

For every integer $m \geq 0$, it follows from 2.2 that there exists a unique subtorus T_m of G_m lifting T_s and such that for every integer n equal to a power of q , one has ${}_n T_m = M(n)_m$. Moreover, let $T' = \mathbf{G}_{m,S}^r$. Since k is algebraically closed, T_s is trivial, and there exists a k -isomorphism $u_s : T'_s \xrightarrow{\sim} T_s$. The morphism u_s lifts uniquely to an S_m -isomorphism $u_m : T'_m \rightarrow T_m$ (Exp. IX 3.3). The family of morphisms u_m , $m \in \mathbf{N}$, defines in the limit a morphism $\hat{u}$ of the formal completions $\widehat{T'}$ and $\widehat{G}$ of T' and G along their closed fiber:

$$\begin{array}{ccc} \widehat{T'} & \xrightarrow{\hat{u}} & \widehat{G} \\ i \downarrow & & \downarrow j \\ T' & & G, \end{array}$$

where i and j denote the canonical morphisms.

We shall show that the morphism $\hat{u}$ is algebraizable. For this, we shall reduce to the case where the group G is affine.

Lemma 4.7. *Let S be the spectrum of a discrete valuation ring A , X and Y two S -preschemes, quasi-compact, quasi-separated and flat over S . Then the canonical map:*

$$\Gamma(X, \mathcal{O}_X) \otimes_A \Gamma(Y, \mathcal{O}_Y) \longrightarrow \Gamma(X \times_S Y, \mathcal{O}_{X \times_S Y})$$

is an isomorphism.

PROOF. Let $f : X \rightarrow S$ (resp. $g : Y \rightarrow S$) be the structural morphism. Since X (resp. Y) is quasi-compact and quasi-separated, it follows from EGA I 9.2.2 and EGA IV 1.7.4 that $f_*(\mathcal{O}_X)$ (resp. $g_*(\mathcal{O}_Y)$) is a quasi-coherent $\mathcal{O}_S$ -algebra, hence corresponds to an affine S -scheme $\tilde{X}$ (resp. $\tilde{Y}$). By hypothesis Y is flat over S ; it then follows from EGA III 1.4.15, taking EGA IV 1.7.21 into account, that the canonical map (deduced from the canonical morphism $X \rightarrow \tilde{X}$):

$$\Gamma(\tilde{X} \times_S Y, \mathcal{O}_{\tilde{X} \times_S Y}) \longrightarrow \Gamma(X \times_S Y, \mathcal{O}_{X \times_S Y})$$

is an isomorphism. But since X is flat over S , $\tilde{X}$ is flat over S (for flat over A is equivalent to torsion-free). Applying EGA III 1.4.15 once more, interchanging the roles of X and Y , one obtains an isomorphism:

$$\begin{aligned} \Gamma(\tilde{X}, \mathcal{O}_{\tilde{X}}) \otimes_A \Gamma(\tilde{Y}, \mathcal{O}_{\tilde{Y}}) &\simeq \Gamma(\tilde{X} \times_S \tilde{Y}, \mathcal{O}_{\tilde{X} \times_S \tilde{Y}}) \\ &\xrightarrow{\sim} \Gamma(\tilde{X} \times_S Y, \mathcal{O}_{\tilde{X} \times_S Y}), \end{aligned}$$

whence the lemma. $\square$

In the present case, the S -group G is flat over S and of finite type, hence quasi-compact and quasi-separated. One may therefore apply the lemma to $G \times_S G$:

$$\Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G) \xrightarrow{\sim} \Gamma(G \times_S G, \mathcal{O}_{G \times_S G}).$$

To the morphism $m_g : G \times_S G \rightarrow G$ defining multiplication in G , there therefore corresponds a morphism:

$$\begin{aligned} \Gamma(G, \mathcal{O}_G) &\longrightarrow \Gamma(G \times_S G, \mathcal{O}_{G \times_S G}) \\ &\xrightarrow{\sim} \Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G), \end{aligned}$$

hence an S -morphism:

$$m_{\tilde{G}} : \tilde{G} \times_S \tilde{G} \longrightarrow \tilde{G},$$

where $\tilde{G}$ denotes the affine S -scheme with A -algebra $\Gamma(G, \mathcal{O}_G)$. From there, it is formal to verify that $m_{\tilde{G}}$ gives $\tilde{G}$ a structure of group S -scheme such that the canonical morphism $v : G \rightarrow \tilde{G}$ is a morphism of S -groups.

Remark 4.8. $\tilde{G}$ plays the role of a largest “affine quotient” of G ; moreover, one can show that $\tilde{G}$ is indeed a quotient of G for fpqc, hence is of finite type over S (XVII App. III, 2).

In the case under consideration, the generic fiber G_t of G is affine, so it follows from EGA I 9.3.3 that v_t is an isomorphism. Since $\tilde{G}$ is affine, the coherent family of morphisms:

$$w_m = v_m u_m : T'_m \longrightarrow \tilde{G}_m \quad (m \in \mathbf{N})$$

comes from a unique morphism of S -groups (Exp. IX 7.1):

$$w : T' \longrightarrow \tilde{G}.$$

Let then T_t be the subtorus of G_t equal to $v_t^{-1} w_t(T'_t)$. The torus T_t therefore has rank at most r (as the image of a torus of rank r). Let us show that T_t contains $M(n)_t$ for every n . Indeed, let $u(n)_m$ be the S_m -isomorphism $({}_n T')_m \xrightarrow{\sim} M(n)_m$ obtained by restricting u_m to $({}_n T')_m$. The coherent family of morphisms $u(n)_m$ comes from a unique S -isomorphism $u(n) : {}_n T' \xrightarrow{\sim} M(n)$ (for $M(n)$ is finite over S). For every integer $m \geq 0$, one then has the equalities:

$$w_m|_{({}_n T')_m} = (v_m u_m)|_{({}_n T')_m} = (v \circ u(n))_m.$$

Consequently $w|_{nT'} = v \circ u(n)$ (1.6 a)). In particular, one has:

$$w_t|_{(nT')_t} = v_t \circ u(n)_t,$$

hence $v_t^{-1}w_t(nT')_t = u(n)_t(nT')_t = M(n)_t$.

This indeed proves that T_t contains $M(n)_t$, and implies that T_t has rank r . One concludes as in the first case already studied, by considering the schematic closure in G of T_t , say T'' . Since T_t contains $M(n)_t$, T''_s contains $M(n)_s$, hence contains T_s (density theorem). On the other hand, since T'' is flat over S and T_t has dimension r , T''_s has dimension r (Exp. VIB 4). In brief, T_s has the same underlying space as the neutral component of T''_s , and one concludes as in the first case that the neutral component of T'' is a subtorus of G answering the question.

c) End of the proof of 4.1.

To prove the existence of the subtorus T , one may suppose S reduced (2.2). Taking 3.1 (ii) $\implies$ (i) into account, it then suffices to prove that the set U of points x of S such that there exists a subtorus T_x of G_x with $nT_x = M(n)_x$ for every n equal to a power of q is equal to $\text{ens}(S)$. The torus T_x in question is necessarily unique, and by fpqc descent it suffices to prove its existence after extension of the residue field of x (Exp. IX 4.8 b)).

This being so, since S is locally noetherian and connected, and U is nonempty (it contains the point s of 4.1 b)), one is reduced, by an immediate argument, to proving that if x and x' are two points of S , with x a specialization of x' , and if one of the two points belongs to U , then both points are in U . By the usual technique (EGA II 7.1.9), one reduces to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, a case treated in b).

This completes the proof of Theorem 4.1.

5. Representability of the functor: smooth subgroups identical to their connected normalizer

Proposition 5.1. *Let S be a prescheme, G a group S -prescheme of finite presentation, H a group subprescheme of G , smooth with connected fibers. Then:*

- a) *The normalizer N of H in G is representable by a closed subprescheme of G , of finite presentation over S .*
- b) *The following conditions are equivalent:*
 - i) *The canonical immersion $H \longrightarrow N$ is an open immersion.*
 - ii) *The group N is smooth along the unit section, and its neutral component, which is then representable (Exp. VIB 3.10), is equal to H .*
 - iii) *For every point s of S , one has $H_s = (N_s)^0$.*

PROOF. The group prescheme H is locally of finite presentation over S (H is smooth over S) and has connected fibers, hence is of finite presentation over S (Exp. VIB 5.3.3). Assertion a) is then a consequence of Exp. XI 6.11. The equivalence of the conditions appearing in b) is recorded here for reference, and was proved in VIB 6.5.1. $\square$

This being so, we may state the main theorem of this section:

Theorem 5.2. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , $\mathcal{L}_G$ (or simply $\mathcal{L}$ if there is no ambiguity) the S -functor such that for every S -prescheme S' one has:*

$$\begin{aligned} \mathcal{L}_G(S') &= \text{set of group subpreschemes of } G_{S'}, \text{ smooth over } S', \\ &\text{with connected fibers, identical to their connected normalizer.} \end{aligned}$$

Then the functor $\mathcal{L}$ is representable by an S -prescheme, the union of an increasing family $(U_i)_{i \in \mathbb{N}}$ of open subpreschemes, quasi-projective and of finite presentation over S , hence a fortiori separated over S .

First reductions. For every integer $r \geq 0$, let $\mathcal{L}_r$ be the subfunctor of $\mathcal{L}$ such that for every S -prescheme S' one has:

$$\begin{aligned} \mathcal{L}_r(S') = & \text{set of group subpreschemes of } G_{S'}, \text{ smooth,} \\ & \text{with connected fibers, identical to their connected normalizer} \\ & \text{and of relative dimension } r. \end{aligned}$$

Since the dimension of the fibers of a smooth group is a locally constant function, the canonical monomorphism $\mathcal{L}_r \rightarrow \mathcal{L}$ is representable by an immersion both open and closed. It therefore suffices to show that for every r , $\mathcal{L}_r$ is representable by an S -prescheme having the properties above, for $\mathcal{L}$ will then be representable by the sum S -prescheme $\coprod_{r \in \mathbb{N}} \mathcal{L}_r$.

For every integer $n \geq 0$, every S -prescheme S' and every group subprescheme H of $G_{S'}$, we shall denote by $H^{(n)}$ the n th normal invariant of the unit section $S' \rightarrow H$ of H (EGA IV 16.1.2), so that $H^{(n)}$ is a sheaf of $\mathcal{O}_{S'}$ -modules, of finite type, corresponding to the n th infinitesimal neighborhood of the unit section of H . If H is smooth over S' of relative dimension r , $H^{(n)}$ is a locally free $\mathcal{O}_S$ -module whose rank $\varphi(n, r)$ depends only on n and r . Moreover, since H is a subprescheme of $G_{S'}$, one has a canonical epimorphism, compatible with extension of the base:

$$G^{(n)} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \simeq G_{S'}^{(n)} \rightarrow H^{(n)}.$$

Introduce then the projective S -scheme:

$$P_{\varphi(n, r)} = \text{Grass}_{\varphi(n, r)}((G)^{(n)})$$

(EGA I 2nd ed. 9.7; cf. also Séminaire Cartan, 1960/61, Exp. No. 14 by A. Grothendieck). It then follows from the preceding remarks that the map:

$$H \mapsto H^{(n)}$$

defines a canonical morphism:

$$u_{n, r} : \mathcal{L}_r \rightarrow P_{\varphi(n, r)}.$$

The group G acts naturally on $G^{(n)}$, hence on $P_{\varphi(n, r)}$, by means of the representation:

$$\text{int} : G \rightarrow \underline{\text{Aut}}_{S\text{-gr}}(G), \quad g \mapsto \text{int}(g).$$

Moreover, if S' is a quasi-compact S -prescheme and H an element of $\mathcal{L}_r(S')$, one knows (Exp. XI 6.11) that for n sufficiently large, one has:

$$N = \underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}}(H^{(n)}).$$

For each integer $n \geq 0$, introduce the subfunctor $\mathcal{L}_r^n$ of $\mathcal{L}_r$ such that for every S -prescheme S' one has:

$$\begin{aligned} \mathcal{L}_r^n(S') = & \text{set of subgroups } H \text{ of } G_{S'} \text{ belonging to } \mathcal{L}_r(S') \\ & \text{and such that } \underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}}(H^{(n)}). \end{aligned}$$

Representability of $\mathcal{L}_r^n$. Since the integer r is fixed until the end of the proof of 5.2, we shall omit it from the notation. Thus we shall write $\mathcal{L}, \mathcal{L}^n, P_n, u_n$ instead of $\mathcal{L}_r, \mathcal{L}_r^n, P_{\varphi(n, r)}, u_{n, r}$.

Let v_n be the restriction of u_n to the subfunctor $\mathcal{L}^n$ of $\mathcal{L}$. It follows from the definition of $\mathcal{L}^n$ and 5.1 b) ii) that v_n is a monomorphism. In fact one has the following lemma:

Lemma 5.3. *The morphism v_n is an immersion of finite presentation. A fortiori, $\mathcal{L}^n$ is representable by an S -prescheme, quasi-projective and of finite presentation over S .*

After replacing S by P_n , one is reduced by the usual technique to proving the following assertion: let $Q \in P_n(S)$, and consider the subfunctor $\mathcal{F}$ of the functor h_S represented by

the final object S of $(\text{Sch})/S$, such that for every S -prescheme S' one has:

$$\begin{aligned} \mathcal{F}(S') &= h_S(S') \text{ (a set with one element) if there exists } H \in \mathcal{L}^n(S') \\ &\quad \text{such that } H^{(n)} = Q_{S'}, \\ \mathcal{F}(S') &= \emptyset \text{ otherwise.} \end{aligned}$$

Then the canonical monomorphism $\mathcal{F} \rightarrow S$ is an immersion of finite presentation.

Let us begin by transforming the definition of the functor $\mathcal{F}$. For this, note that the normalizer of Q in G is representable by a group subprescheme N of finite presentation over S (namely the inverse image of the point Q of $P_n(S)$ under the morphism:

$$G \rightarrow P_n, \quad g \mapsto g \cdot Q.$$

) I say that the functor $\mathcal{F}$ agrees with the following subfunctor of h_S :

$$\left\{ \begin{array}{l} \mathcal{F}(S') = h_S(S') \text{ if one has:} \\ \quad \text{(a) } N_{S'} \text{ is smooth along the unit section and of relative dimension } r. \\ \quad \text{(b) } (N_{S'})^{(n)} \text{ (then canonically an element of } P_n(S')) \text{ equals } Q_{S'}. \\ \mathcal{F}(S') = \emptyset \text{ otherwise.} \end{array} \right.$$

Indeed, let us temporarily denote by $\mathcal{F}_1$ the functor $\mathcal{F}$ in its first form, and by $\mathcal{F}_2$ the functor $\mathcal{F}$ in its second form. Then:

$$\text{i) } \mathcal{F}_1(S') = h_S(S') \implies \mathcal{F}_2(S') = h_S(S').$$

Indeed, let $H \in \mathcal{F}^n(S')$ be the subgroup of $G_{S'}$ such that $H^{(n)} = Q_{S'}$. One therefore has:

$$\underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}}(H^{(n)}) = \underline{\text{Norm}}_{G_{S'}}(Q_{S'}) = N_{S'}.$$

Thus by 5.1 b) ii), $N_{S'}$ is smooth along the unit section and its neutral component is H . Consequently $N_{S'}$ has relative dimension r , and since H is open in $N_{S'}$ (by 5.1 b) i)), one has $(N_{S'})^{(n)} = H^{(n)} = Q_{S'}$. In brief, $\mathcal{F}_2(S') = h_S(S')$.

$$\text{ii) } \mathcal{F}_2(S') = h_S(S') \implies \mathcal{F}_1(S') = h_S(S').$$

By hypothesis, $N_{S'}$ is smooth along the unit section, of relative dimension r ; its neutral component is therefore representable (Exp. VIB 3.10) by a group subprescheme H , smooth over S' , with connected fibers of dimension r . Since H is invariant in $N_{S'}$ and open in $N_{S'}$, one has the following inclusions:

$$\begin{aligned} N_{S'} &\subset \underline{\text{Norm}}_{G_{S'}}(H) \subset \underline{\text{Norm}}_{G_{S'}}(H^{(n)}) \\ &= \underline{\text{Norm}}_{G_{S'}}(N_{S'}^{(n)}) = \underline{\text{Norm}}_{G_{S'}}(Q_{S'}) = N_{S'}. \end{aligned}$$

The inclusions above are therefore equalities. The first inclusion then shows that H is equal to its connected normalizer, and the second shows that H is an element of $\mathcal{L}^n(S')$. This says that $\mathcal{F}_1(S') = h_S(S')$.

Implications i) and ii) give $\mathcal{F}_1 = \mathcal{F}_2$. We retain the second definition of the functor $\mathcal{F}$, and shall first “represent condition a)” by an immersion of finite presentation. For this, it suffices to apply:

Lemma 5.4. *Let S be a prescheme, X an S -prescheme locally of finite presentation over S , $\sigma : S \rightarrow X$ a section of X , r an integer ≥ 0 , and $L : ((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ the subfunctor of S defined as follows:*

$$\left\{ \begin{array}{l} L(S') = h_S(S') \text{ if } X_{S'} \text{ is smooth along the section } \sigma_{S'} \\ \quad \text{and of relative dimension } r \text{ at the points of } \sigma_{S'}(S'). \\ L(S') = \emptyset \text{ otherwise.} \end{array} \right.$$

Then:

a) *The monomorphism $L \rightarrow S$ is an immersion of finite presentation.*

b) Let J be the conormal sheaf relative to the immersion $S \rightarrow X$ (EGA IV 16.1.2); suppose that for every point s of S , $J \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ has rank at most r ; then the immersion $L \rightarrow S$ is a closed immersion.

PROOF. The functor L is local in nature on S , which reduces one to the case where S is affine. After replacing X by a neighborhood of $\sigma(S)$, one may suppose X of finite presentation over S , then (EGA IV 8.9) S noetherian (note, in case b), that formation of the conormal sheaf commutes with extension of the base (EGA IV 16.6.4), and that the rank of the fibers of J is a constructible function on S). This being so, for every S -prescheme S' , denote by $J' = J_{S'}$ the conormal sheaf relative to the section $\sigma_{S'}$; let $S^\bullet(J')$, canonically isomorphic to $S^\bullet(J)_{S'}$, be the symmetric algebra of J' over $\mathcal{O}_{S'}$, $\text{Gr}_\bullet(\sigma_{S'})$ the sheaf of graded $\mathcal{O}_{S'}$ -algebras associated with $\sigma_{S'}$ (EGA IV 16), and for every integer $n \geq 0$ let $\sigma_{n,S'} : S^n(J') \rightarrow \text{Gr}_n(\sigma_{S'})$ be the canonical epimorphism.

It follows from EGA IV 17.12.3 and EGA 0_{IV} 19.5.4 that, for $X_{S'}$ to be smooth along the section $\sigma_{S'}$ and of relative dimension r at the points of $\sigma_{S'}(S')$, it is necessary and sufficient that:

- i) J' be a locally free $\mathcal{O}_{S'}$ -sheaf of rank r .
- ii) For every integer $n \geq 1$, $\sigma_{n,S'}$ be an isomorphism.

Now it follows from TDTE IV, Lemma 3.6, that “the functor making J locally free of rank r ” is representable by a subprescheme S_1 , closed in S in case b). Replacing S by S_1 , one is reduced to the case where J is locally free of rank r . Let us then proceed by induction on the integer n . Suppose one has represented by a closed subprescheme S_{n-1} of S the “functor making the morphisms $\sigma_{q,S'}$ injective for every integer $q \leq n-1$ ”, and let us show that the “subfunctor⁽⁵⁾ making $\sigma_{n,S'}$ injective” is representable by a closed subprescheme S_n of S_{n-1} . Replacing S by S_{n-1} , we may suppose that $\sigma_{q,S}$ is bijective for $q \leq n-1$. But then, the $(n-1)$ st normal invariant $X^{(n-1)}$ relative to the section σ_s admits a composition series $X^{(0)}, \dots, X^{(n-1)}$ whose successive quotients $\text{Gr}_0(\sigma_s), \dots, \text{Gr}_{n-1}(\sigma_s)$ are locally free over $\mathcal{O}_{S'}$, hence flat, so $X^{(n-1)}$ is flat over $\mathcal{O}_S$. Since formation of $X^{(i)}$ commutes with extension of the base (EGA IV 16), one has for every S -prescheme S' :

$$\begin{aligned} \text{Gr}_n(\sigma_{S'}) &= \text{Ker}[X_{S'}^{(n)} \rightarrow X_{S'}^{(n-1)}] = (\text{Gr}_n(\sigma))_{S'}, \\ \text{and} \quad \sigma_{n,S'} &= (\sigma_{n,S})_{S'}. \end{aligned}$$

Thus the functor of interest is “the one making the morphism $\sigma_{n,S}$ injective”. This functor is local in nature on S , which allows one to suppose S affine and $S^n(J)$ free over $\mathcal{O}_S$. But it is then clear that the functor in question is representable by the closed subscheme of S defined by the ideal generated by the coordinates of $\text{Ker } \sigma_{n,S}$ relative to a basis of $S^n(J)$.

Since S is noetherian, the decreasing sequence $\{S_n\}$ of closed subpreschemes of S_1 is stationary, and the stationary value represents the functor L , which completes the proof of 5.4. $\square$

Let us return to the question of the representability of $\mathcal{L}^n$. Replacing S by a suitable subprescheme S_1 , one may therefore suppose N smooth along the unit section and of relative dimension r . The functor $\mathcal{L}^n$ is then the “coincidence functor” of two sections of P_n over S , h and g , corresponding to the sheaves Q and $N^{(n)}$ (condition b) occurring in the definition of the functor $\mathcal{F}_2$ above). It is therefore representable by the closed subprescheme of S , the inverse image of the diagonal of $P_n \times_S P_n$ under the morphism $h \times_S g$ of finite presentation. This completes the proof of 5.3.

Study of the transition morphisms $\mathcal{L}^n \rightarrow \mathcal{L}^m$. If a subgroup H of $G_{S'}$ belongs to $\mathcal{L}^n(S')$, it belongs a fortiori to $\mathcal{L}^m(S')$ for every $m \geq n$, whence natural monomorphisms:

$$u_n^m : \mathcal{L}^n \rightarrow \mathcal{L}^m \quad \text{for } m \geq n.$$

⁽⁵⁾**N.D.E.** “Functor” has been replaced by “subfunctor”. Should one write “making $\sigma_{q,S'}$ injective for $q \leq n$ ”?

Lemma 5.5. *The morphism u_n^m is an open immersion.*

PROOF. After changing S , one is reduced to the following problem: let $H \in \mathcal{L}^m(S)$,

$$N = \underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G(H^{(m)}), \quad N' = \underline{\text{Norm}}_G(H^{(n)}),$$

and let $\mathcal{D} : ((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ be the coincidence functor of N and N' , defined by $\mathcal{D}(S') = h_S(S')$ if $N_{S'} = N'_{S'}$, and $\mathcal{D}(S') = \emptyset$ otherwise. One must show that $\mathcal{D} \rightarrow S$ is an open immersion. Now I say that $\mathcal{D}$ is also the subfunctor of S which “makes the immersion $H \rightarrow N'$ open”. Indeed, if $N_{S'} = N'_{S'}$, then $H_{S'} \rightarrow N'_{S'}$ is indeed an open immersion, since this is so for $H_{S'} \rightarrow N_{S'}$ (Prop. 5.1). Conversely, if $H_{S'} \rightarrow N'_{S'}$ is an open immersion, since H has connected fibers, $H_{S'}$ is the neutral component of $N'_{S'}$ (Exp. VIB 3.10), and consequently is invariant in $N'_{S'}$, hence $N'_{S'} \subset N_{S'}$. Since in any case N' contains N , one indeed has $N_{S'} = N'_{S'}$. The group preschemes H and N' are of finite presentation over S , and H is flat over S ; the fact that $\mathcal{D} \rightarrow S$ is an open immersion then follows from Exp. VIB 2.6. $\square$

End of the proof of 5.2. Since the functors $\mathcal{L}^n$ are representable, and the transition morphisms u_n^m are compatible with each other and representable by open immersions, there exists an S -prescheme X , the union of an increasing sequence of open subsets $(X_i)_{i \in \mathbb{N}}$, such that X_i represents the functor $\mathcal{L}^i$, and such that if one identifies $\mathcal{L}^i$ with X_i , the inclusion $X_i \rightarrow X_j$ ($j \geq i$) is identified with u_i^j . To conclude that X represents the functor $\mathcal{L}$, it then suffices to note that, in the category of sheaves on $(\text{Sch})/S$ endowed with the Zariski topology (Exp. IV 6.1), one has:

$$X = \varinjlim X_i \quad \text{and} \quad \mathcal{L} = \varinjlim \mathcal{L}^i.$$

Remark 5.6. With the preceding notation, suppose in addition that S has all its residual characteristics zero. Then for every integer $r \geq 0$, the functor $\mathcal{L}_r$ is equal to $\mathcal{L}_r^1$, hence is representable by an S -prescheme of finite presentation and quasi-projective over S . Indeed, it suffices to show that if $H \in \mathcal{L}_r(S')$, the canonical immersion:

$$H \rightarrow N = \underline{\text{Norm}}_{G_{S'}}(H^{(1)})$$

is an open immersion (for this implies $N = \underline{\text{Norm}}_{G_{S'}}(H)$, hence $H \in \mathcal{L}_r^1(S')$). Since H is flat over S , and H and N are of finite presentation over S , it suffices (Exp. VIB 2.6) to show that for every point s of S , $H_s \rightarrow N_s$ is an open immersion. Now it follows easily⁽⁶⁾ from Cartier’s theorem (Exp. VIB 1.6) that if G is an algebraic group over a field of characteristic 0, and H is a connected algebraic subgroup, one has:

$$\underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G(\text{Lie } H) = \underline{\text{Norm}}_G(H^{(1)}).$$

On the other hand, if S has nonzero residual characteristics, the subfunctors $\mathcal{L}_r^n$ of $\mathcal{L}_r$ may form a strictly increasing sequence (even when S is quasi-compact), and in this case $\mathcal{L}_r$ is not representable by an S -prescheme quasi-compact over S . Take, for example, the algebraic group G , defined over a field k of characteristic $p > 0$, equal to the semidirect product of the torus $T = \mathbf{G}_m \times \mathbf{G}_m$ by the unipotent group $U = \mathbf{G}_a \times \mathbf{G}_a$, with the action of T on U defined by $(t, t', u, u') \rightarrow (tu, t'u')$. For every integer $n > 0$, consider then the smooth connected subgroup U_n of U with equation $u' = u^{p^n}$, and the subtorus T_n of T with equation $t' = t^{p^n}$. It is immediate to verify that T_n acts on U_n , and that the subgroup G_n of G equal to $T_n \cdot U_n$ is smooth, connected and identical to its normalizer in G . Now all the groups G_n , for $n \geq m$, are distinct and have the same infinitesimal neighborhood of order p^m .

Remark 5.7. There exists on $\mathcal{L}$ a canonical invertible sheaf L whose restriction to every open subset U of $\mathcal{L}$, quasi-compact over S , is S -ample. Indeed, consider the group subprescheme H of $G_{\mathcal{L}}$, smooth over $\mathcal{L}$, with connected fibers, equal to its connected normalizer, and

⁽⁶⁾**N.D.E.** One can apply, for example, Proposition II.6.1 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North Holland (1970).

universal for these properties. I say that one may take for L the sheaf $(\det(\mathrm{Lie} H))^{-1}$ (recall that if F is a sheaf of $\mathcal{O}_S$ -modules on a prescheme S , locally free of finite rank, $\det(F)$ denotes the invertible $\mathcal{O}_S$ -module whose restriction to the open and closed subprescheme S_r of S ($r \geq 0$) where F has rank r is equal to $\bigwedge^r(F)$). We retain the notation of the proof of 5.2. To prove the assertion made about L , we may restrict to the functor $\mathcal{L}_r^n$ and prove that $L|_{\mathcal{L}_r^n}$ is S -ample. Consider the canonical immersion $v_n : \mathcal{L}_r^n \rightarrow P_n$, and let Q be the locally free sheaf on P_n , universal for the Grassmannian P_n . By construction, one has $v_n^*(Q) = H^{(n)}$ (where H now denotes the group subprescheme of $G_{\mathcal{L}_r^n}$ universal for the functor $\mathcal{L}_r^n$). Now $\det(Q)$ is the canonical ample sheaf on P_n (EGA I 2nd ed., 9.7), so $\det H^{(n)}$ is ample relative to $\mathcal{L}_r^n$ (EGA II 4.6.13 i) bis). Again denote by J the conormal sheaf, equal to $H^{(1)}$, and by $S^q(J)$ the homogeneous part of degree q of the symmetric algebra of J over $\mathcal{O}_{\mathcal{L}_r^n}$. Since H is smooth over $\mathcal{L}_r^n$, it is immediate that one has a canonical isomorphism:

$$\det H^{(n)} \simeq \prod_{1 \leq q \leq n} \det S^q(J).$$

On the other hand, one proves that for every locally free sheaf J of rank r and every integer $q > 0$, there exists a canonical isomorphism:

$$\det S^q(J) \simeq (\det J)^{\otimes s},$$

where $s > 0$ is an integer depending only on r and q . Finally, one obtains $\det H^{(n)} \simeq (\det J)^{\otimes s}$ for a suitable integer $s > 0$, so (EGA II 4.5.6) $\det J = (\det \mathrm{Lie} H)^{-1}$ is indeed S -ample.

Remark 5.8. Let S be a prescheme, G and H two group S -preschemes of finite presentation over S , $i : H \rightarrow G$ a group S -homomorphism which is a monomorphism. If H is smooth over S , with connected fibers, one knows (Exp. XI 6.11) that $N = \underline{\mathrm{Norm}}_G(H)$ is representable by a closed group subprescheme of G , of finite presentation over S . Suppose in addition that N is smooth along the unit section and has the same relative dimension over S as H . The neutral component N^0 of N is then representable by an open group subprescheme of N , smooth over S (Exp. VIB 3.10). The monomorphism i obviously factors through N^0 . In fact one has $H = N^0$. Indeed, for every point s of S , one has $H_s = (N^0)_s$, these two algebraic groups being connected, smooth and of the same dimension. Since H is flat over S , one deduces that $H \rightarrow N^0$ is an isomorphism (EGA IV 17.9.5). Finally, H is a group subprescheme of G . One has therefore shown that the functor $\mathcal{L}$ introduced in this section is identical to the functor of subgroups H of G , smooth over S , with connected fibers, and equal to their connected normalizer.

6. Functor of Cartan subgroups and functor of parabolic subgroups

When G is a smooth connected algebraic group defined over an algebraically closed field k , one has defined the subtori of G , maximal subtori, Cartan subgroups (Exp. XII 1), Borel subgroups (Exp. XIV 4.1), and parabolic subgroups (Exp. XIV 4.8 bis). We extend these notions to the case of a group prescheme over an arbitrary base as follows:

Definition 6.1. Let S be a prescheme, G a group S -prescheme of finite presentation over S , H a group S -prescheme, $i : H \rightarrow G$ an S -monomorphism making H a subgroup of G . We shall say that H is a subtorus of G (resp. a maximal subtorus of G , a Cartan subgroup, a Borel subgroup, a parabolic subgroup) if:

- i) H is smooth over S .
- ii) For every geometric point $\bar{s}$ over S , $H_{\bar{s}}$ is a subtorus of $(G_{\bar{s}})_{\mathrm{red}}^0$ (resp. a maximal subtorus, a Cartan subgroup, a Borel subgroup, a parabolic subgroup).

Remarks 6.1 bis. a) If the S -group H is a subtorus of G (resp. ...), its fibers are connected, and consequently H is of finite presentation over S (Exp. VIB 5.3.3).

b) If H is a subtorus of G , then H is a torus in the sense of Exp. IX, as follows immediately from Exp. X 8.1. Moreover, the monomorphism $i : H \rightarrow G$ is an immersion (cf. 8.3 below).

c) If G is smooth over S , with connected fibers, and H is a Cartan subgroup of G (resp. a Borel subgroup, a parabolic subgroup), the monomorphism $i : H \rightarrow G$ is an immersion, so that our definitions agree with those introduced in Exp. XII and Exp. XIV. Indeed, H is then identical to its connected normalizer (by XII 6.6 c), XIV 4.8 and 4.8 bis), and it suffices to apply 5.8.

Definition 6.1 ter. Let S be a prescheme, G a group S -prescheme locally of finite type, s a point of S . The dimension of the Cartan subgroups of $(G_{\bar{s}})_{\text{red}}^0$ is called the nilpotent rank of G at the point s , and denoted by $\rho_n(s)$. One defines analogously the reductive rank $\rho_r(s)$, the unipotent rank $\rho_u(s)$, the abelian rank $\rho_{\text{ab}}(s)$ (cf. Exp. X 8.7).

If now G is a smooth connected algebraic group defined over an algebraically closed field k , recall that the radical of G , denoted by $\text{rad}(G)$, is the largest algebraic subgroup of G which is invariant, smooth, connected and solvable; $G/\text{rad}(G)$ is then semisimple (use Exp. XII 6.1 to reduce to the case of G affine). If G is in addition affine, one defines the unipotent radical $\text{rad}^u(G)$ of G as the largest invariant, smooth, connected and unipotent algebraic subgroup of G : $G/\text{rad}^u(G)$ is then reductive.

Proposition 6.2. *Let S be a prescheme, G and H two group S -preschemes of finite presentation over S , $i : H \rightarrow G$ a group S -monomorphism making H a subgroup of G , and let $P(s)$ be one of the following properties concerning the point s of S :*

- i) $(G_{\bar{s}})_{\text{red}}^0$ is an abelian variety (resp. is affine, is a torus, is unipotent).
- ii) $(H_{\bar{s}})_{\text{red}}^0$ is a maximal torus of $G_{\bar{s}}$.
- iii) $(H_{\bar{s}})_{\text{red}}^0$ is the centralizer in $(G_{\bar{s}})_{\text{red}}^0$ of a torus of $G_{\bar{s}}$ (resp. is a Cartan subgroup of $G_{\bar{s}}$).
- iv) $(H_{\bar{s}})_{\text{red}}^0$ is a Borel subgroup (resp. a parabolic subgroup) of $G_{\bar{s}}$.
- v) $(H_{\bar{s}})_{\text{red}}^0$ is the radical of $G_{\bar{s}}$ (resp. $(G_{\bar{s}})_{\text{red}}^0$ is semisimple).
- vi) $G_{\bar{s}}$ is affine, and $(H_{\bar{s}})_{\text{red}}^0$ is the unipotent radical of $G_{\bar{s}}$ (resp. $(G_{\bar{s}})_{\text{red}}^0$ is reductive).

Then the set E of points s of S such that $P(s)$ is true is locally constructible (EGA 0_{III} 9.1.11).

Remark 6.2.1. This proposition complements Exp. VIB § 10. Moreover, one can make the structure of E more precise using semicontinuity theorems (cf. Exp. X 8.7); we shall see an example a little further on.

Proof of 6.2. Note that if S is the spectrum of a field, E is invariant under extension of this field. A standard reduction (EGA IV 9) then allows one to reduce to the case where S is noetherian integral with generic point η . One must show that E or $\text{ens}(S) \setminus E$ contains a neighborhood of η (EGA IV 9.2.1). One may suppose S affine with ring A and fraction field K . If L is a finite extension of K , it is immediate that there exists a sub- A -algebra B of L , finite over A , with fraction field L . The canonical morphism $S' \rightarrow S$, where $S' = \text{Spec } B$, is dominant, of finite presentation, so the image of a nonempty open subset of S' contains a nonempty open subset of S (EGA IV 1.8.4). From the point of view of interest to us, we may therefore replace S by S' , hence replace K by a finite extension L . Thus we may choose L so that $(G_L)_{\text{red}}$ and $(H_L)_{\text{red}}$ are smooth over L (EGA IV 4.6.6). After restricting S' , we may suppose that G_{red} and H_{red} are group preschemes smooth over S (Exp. VIB § 10 and EGA IV 17). In view of the properties to be proved, we may replace G and H by their reduced neutral components (Exp. VIB 10.9), hence suppose G and H smooth over S , with connected fibers. Finally, we may suppose that H is a closed group subprescheme of G (Exp. VIB 10.4). *Proof of i).* After making a finite extension of K , we may suppose that G_{η} possesses a ‘‘Chevalley decomposition’’, i.e. is an extension of an abelian variety D_{η} by a smooth connected linear algebraic group F_{η} (Séminaire Bourbaki 1956/57 No. 145). By Exp. VIB

10.16, there exists an open neighborhood U of η such that this generic extension comes from an extension:

$$1 \longrightarrow F \longrightarrow G|_U \longrightarrow D \longrightarrow 1.$$

One may in addition suppose F and D smooth over U , with connected fibers, F affine over U and D proper over U (EGA IV 8, 9 and 17). For every point s of U , (D_s, F_s) is then the “Chevalley decomposition” of G_s . Consequently G_s is an abelian variety (resp. is affine) if and only if F_s (resp. D_s) is the unit group, which is a constructible property (EGA IV 9.2.6.1).

To establish the last two assertions of i), in view of what precedes, we may suppose G affine over S . Let q be a prime number invertible on S , and ${}_qG$ the “kernel” of raising to the q th power in G . It follows easily from the structure of affine algebraic groups that G_s is a torus (resp. is unipotent) if and only if a) ${}_qG_s$ is quasi-finite, which is a constructible property (EGA IV 9.3.2), and b) ${}_qG_s$ has r^q geometric points, where r denotes the relative dimension of G over S (resp. ${}_qG_s$ has a single point). Now the function $s \mapsto$ (number of geometric points of ${}_qG_s$) is constructible (EGA IV 9.7.9). This completes the proof of (i).

Proof of iii). a) Case of a centralizer of a torus. Suppose that $H_\eta = \text{Centr}_{G_\eta}(T_\eta)$, where T_η is a torus of G_η , and let us show that H_s is the centralizer of a subtorus of G_s for s in a neighborhood of η . By i), after restricting S , one may suppose that T_η comes from a subtorus T of G . But then $Z = \text{Centr}_G(T)$ is representable (Exp. XI 6.11) by a group subprescheme of G . Since H and Z agree generically, they agree over a neighborhood of η . This proves that the set E of points s of S such that H_s is the centralizer in G_s of a subtorus of G_s is ind-constructible (EGA IV 1), and this result will suffice to establish, in Lemma 6.6 below, that E is an open subset of S ; a fortiori, E will indeed be a locally constructible subset of S .
iii) b) Case of a Cartan subgroup. Suppose that H_η is a Cartan subgroup of G_η , and let us show that H_s is a Cartan subgroup of G_s at every point s of a neighborhood of η . The group H_η is the centralizer in G_η of a torus of G_η and is nilpotent (Exp. XII 6.6). By a) and Exp. VIB 8.4, H_s has the same properties at every point s of a neighborhood U of η . For every point s of U , the group H_s therefore has the same reductive rank as G_s , and its unique maximal torus is central (Exp. XII 6.7), so it is the centralizer of a maximal torus of G_s , i.e. a Cartan group of G_s .

Suppose now that H is not a Cartan subgroup of G , and let us show that H_s is not a Cartan subgroup of G_s for s in a neighborhood U of η . Taking into account the assertion provisionally granted in (a) above, we may restrict to the case where H is the centralizer in G of a subtorus T . But then H_η contains a Cartan subgroup C_η of H_η . We have just seen that, after restricting S , C_η extends to a Cartan subgroup C of G , which one may suppose contained in H . By hypothesis H_η strictly contains C_η , so H_s strictly contains C_s for s in a neighborhood U of η (EGA IV 9.5.2); a fortiori, H_s is not a Cartan subgroup of G_s for s in U .

Proof of ii). Suppose that H_η is a maximal torus of G_η , and let C_η be its centralizer in G_η . By i) and iii), H is a torus over a neighborhood U of η , and $C = \text{Centr}_{G|_U}(H|_U)$ is a Cartan subgroup of $G|_U$. To prove that H is a maximal torus of G over a neighborhood of η , one may then replace G by C , then by the linear component F of a Chevalley decomposition of C (cf. i)). Let q be an integer invertible on S , ${}_qF$ the kernel of raising to the q th power in F . Since F_s is affine, nilpotent, smooth and connected, F_s is the direct product of its maximal torus T_s by a unipotent group (BIBLE 6-04), so ${}_qF_s = {}_qT_s$. Since H is a maximal torus, ${}_qH_\eta = {}_qF_\eta$, and consequently ${}_qH = {}_qF$ over a neighborhood V of η . For every point s of V , ${}_qH_s = {}_qT_s$, so $H_s = T_s$ is a maximal torus.

Suppose now that H_η is not a maximal torus of G_η . By i), we may restrict to the case where H_η is a torus, then suppose that it is contained in a strictly larger torus T_η . The latter extends to a torus T strictly containing H over a neighborhood U of η . A fortiori, H_s is not a maximal torus for $s \in U$.

Proof of iv). After restricting S , one may suppose that the center Z of G is representable (Exp. VIB 10.11) and flat over S , as is the quotient G/Z (*loc. cit.*). The property “ H_s

contains Z_s is constructible (EGA IV 9.5.2), and every parabolic subgroup of G_s contains Z_s (Exp. XIV 4.9 a)); this allows one to replace G by G/Z , hence suppose G affine over S (Exp. XII 6.1 and i)). One may again suppose that G/H is representable, but then H_s is a parabolic subgroup of G_s if and only if $(G/H)_s$ is proper (BIBLE 6. Th. 4 b)), which is an ind-constructible property (EGA IV 9.3.5). Thus E is ind-constructible, and this will suffice to prove that E is open (Lemma 6.6), hence locally constructible.

Let us now examine the case of Borel subgroups. If H_η is a Borel subgroup of G_η , i.e. a solvable parabolic subgroup of G_η , what precedes and Exp. VIB 8.4 imply that these properties are still true at every point s of a neighborhood of η . If now H_η is not a Borel subgroup of G_η , to prove that the same is true at the points s of a neighborhood of η , we may restrict (in view of what precedes) to the case where H_η is a parabolic subgroup, then suppose that H_η contains a Borel subgroup B_η . We have just shown that the latter extends to a Borel subgroup B of G over a neighborhood U of η . Since H_η strictly contains B_η , H_s strictly contains B_s at every point of an open subset V , and H_s is not a Borel subgroup of G_s for $s \in V$.

Proof of v). Suppose that H_η is the radical of G_η . The group H_η is therefore invariant in G_η , solvable (smooth and connected); the same therefore holds for H_s for s belonging to a neighborhood U of η (Exp. VIB 8 and 10), so for $s \in U$, H_s is contained in the radical of G_s . Replacing G by G/H (Exp. VIB 10), one must prove that if G_η is semisimple, G_s is semisimple at every point of a neighborhood V of η . Thanks to i) and ii), one may suppose that G is affine over S and possesses a maximal torus T . Let W be the Weyl group of T (Exp. XII 2), which is quasi-finite and étale over S , hence finite and étale over an open subset V . It then follows from the elementary properties of roots (Exp. XIX 1.12) that G is semisimple over V .

Suppose now that H_η is not the radical of G_η . After replacing K by a finite extension L , one may suppose that G_η possesses a radical R_η . By what precedes, R_η extends to a group subscheme R of $G|_U$ such that for every $s \in U$, R_s is the radical of G_s . By hypothesis $R_\eta \neq H_\eta$. Thus $R_s \neq H_s$ for $s \in V$. It remains to prove that if G_η is not semisimple, the same holds for G_s at neighboring points, but this is a particular case of what precedes (take $H = \text{unit group}$).

Proof of vi). The proof is entirely analogous to that of v), taking i) into account, and is left to the reader.

Corollary 6.3. *Let S_0 be a quasi-compact prescheme, S_i ($i \in L$) a projective system of S_0 -preschemes affine over S_0 , $S = \varprojlim S_i$ (EGA IV 8.2), G_0 a group prescheme of finite presentation over S_0 , $G_i = G_0 \times_{S_0} S_i$, $G = G_0 \times_{S_0} S$, H a subgroup of G . Then if H is a subtorus of G (resp. a maximal subtorus, a Cartan subgroup, a Borel subgroup, a parabolic subgroup), there exist an index $i \in L$ and a subgroup H_i of G_i such that $H = H_i \times_{S_i} S$ and H_i is a subtorus of G_i (resp. ...).*

Indeed, H is smooth, with connected fibers, hence of finite presentation over S (Exp. VIB 5.3.3). By Exp. VIB § 10, there exist $i \in L$ and a subgroup H_i of G_i , smooth over S , such that $H = H_i \times_{S_i} S$. Corollary Exp. 6.3 then follows from Definition 6.1, 6.2 and EGA IV 9.3.3.

Corollary 6.3 bis. *Let S be a prescheme, G a group S -prescheme of finite presentation over S . Then the functions $\rho_n, \rho_r, \rho_u, \rho_{ab}$ (cf. 6.1 ter) are locally constructible functions on S .*

It suffices to show (EGA IV 9) that if S is an integral noetherian scheme with generic point η , the functions in question are constant on a neighborhood of η . After replacing S by a scheme S' , finite over S , dominating S , we may suppose that G_η possesses a Cartan subgroup C_η with a Chevalley decomposition $1 \rightarrow L_\eta \rightarrow C_\eta \rightarrow A_\eta \rightarrow 1$. The argument made in 6.2 i) proves that this decomposition extends to a Chevalley decomposition over a

neighborhood of η :

$$1 \longrightarrow L \longrightarrow C \longrightarrow A \longrightarrow 1.$$

Moreover, one may suppose that C is a Cartan subgroup of G (6.3), and that the maximal torus T_η of L_η extends to a maximal torus T of L (6.3). The corollary follows immediately from this and from the definitions.

6.4.0. — Let then S be a prescheme, G a group S -prescheme of finite presentation over S , $\mathcal{L}$ the functor of subgroups of G , smooth, with connected fibers, and equal to their connected normalizer (cf. § 5); $\mathcal{L}$ is representable (5.2 and 5.8). The rest of this section is devoted to the study of certain subfunctors of $\mathcal{L}$. More precisely, introduce the subfunctors of $\mathcal{L}$ denoted by $\mathcal{LC}$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$), defined as follows: for every S -prescheme S' , $\mathcal{LC}(S')$ (resp. $\mathcal{CT}(S')$, resp. $\mathcal{P}(S')$) is the set of subgroups H of $G_{S'}$, smooth over S' , with connected fibers, equal to their connected normalizer, and such that for every point s of S' , H_s contains a Cartan subgroup (6.1) of G_s (resp. is the centralizer in $(G_{\bar{s}})_{\text{red}}^0$ of a subtorus of $G_{\bar{s}}$, resp. is a parabolic subgroup (6.1) of G_s).

Theorem 6.4. *Let S be a prescheme, G a group S -prescheme of finite presentation over S . Then the S -functors $\mathcal{LC}, \mathcal{CT}, \mathcal{P}$ above (6.4.0) are representable by S -preschemes of finite presentation over S , quasi-projective over S .*

Remark 6.5. If G has smooth fibers, for example if G is smooth over S or the residual characteristics of S are zero (Exp. VIB 1.6.1), every subgroup H of G , smooth over S , with connected fibers, such that for every point s of S , H_s contains a Cartan subgroup of $(G_s)^0$, is necessarily equal to its connected normalizer (and consequently is an element of $\mathcal{LC}(S)$). Indeed, thanks to 5.1 iii), one may suppose that S is the spectrum of a field, in which case the property was pointed out at the end of the statement of Exp. XIII 2.1.

Note that one has natural monomorphisms:

$$\begin{array}{ccc} \mathcal{CT} & & \\ & \searrow & \\ & \mathcal{LC} & \longrightarrow \mathcal{L} \\ & \nearrow & \\ \mathcal{P} & & \end{array}$$

Let us show that these monomorphisms are open immersions of finite presentation (which will already prove, taking 5.2 into account, that $\mathcal{LC}, \mathcal{CT}, \mathcal{P}$ are representable by S -preschemes, unions of increasing sequences of open subpreschemes, quasi-projective and of finite presentation over S). Now this will follow, by the usual technique, from the following lemma:

Lemma 6.6. *Let S be a prescheme, G a group S -prescheme of finite presentation, H a subgroup of G , smooth with connected fibers. Then the set of points s of S such that $H_{\bar{s}}$ contains a Cartan subgroup of $G_{\bar{s}}$ (resp. is equal to the centralizer in $(G_{\bar{s}})_{\text{red}}^0$ of a torus of $G_{\bar{s}}$) is an open subset of S . If in addition H is equal to its connected normalizer^{*}, the set of points s of S such that H_s is a parabolic subgroup of G_s is also an open subset of S .*

The assertion to be proved is local on S , which allows one to suppose S affine, then, by EGA IV 8.1 and 6.3, S noetherian. Denote by E the set of points of S having the property in question. It then follows from the assertions actually proved in 6.2 that E is ind-constructible. But S is noetherian, so to prove that E is open, it then suffices to show that E is stable under generalization (EGA IV 1.10.1). Using EGA II 7.1.9, one is finally led to prove that if S is the spectrum of a discrete valuation ring and the closed point s belongs to E , then the same holds for the generic point t . We shall need the following lemma:

^{*}A hypothesis which is in fact superfluous, cf. XVII App. III 3.

Lemma 6.7. *Let A be a complete noetherian local ring, $S = \operatorname{Spec} A$, s the closed point of S , H a group S -prescheme, smooth with connected fibers, T_s a subtorus of H_s . Then:*

i) There exists a closed group subscheme C of H , smooth with connected fibers, such that $C_s = \operatorname{Centr}_{H_s}(T_s)$.

ii) For every point t of S , C_t is the centralizer in H_t of a subtorus T_t of H_t .

Proof of 6.7. Let T' be an S -torus such that there exists an isomorphism $u_0 : T'_s \xrightarrow{\sim} T_s$ (Exp. X 4.6). Let $\mathfrak{m}$ be the maximal ideal of A , $A_n = A/\mathfrak{m}^n$, $S_n = \operatorname{Spec} A_n$, $H_n = H \times_S S_n$, etc. Since H is smooth over S , for every integer $n > 0$, there exists a group S_n -morphism:

$$u_n : T'_n \longrightarrow H_n$$

lifting u_0 (Exp. IX 3.6), and by induction on n one may suppose that u_n lifts u_{n-1} . On the other hand, let q be a prime number invertible on S ; for every integer ℓ equal to a power of q , denote by ${}_\ell u_n$ the restriction of u_n to the subgroup ${}_\ell T'_n$. For fixed ℓ and variable n , the morphisms ${}_\ell u_n$ form a projective system, hence come from a unique group S -morphism ${}_\ell u : {}_\ell T' \longrightarrow H$ (1.6 a)). Since H is separated (Exp. VIB 5.2) and u_0 is a monomorphism, ${}_\ell u$ is a monomorphism (Exp. IX 6.8), and even a closed immersion, since it is finite, ${}_\ell T'$ being finite over S . Denote by $M(\ell)$ the image group. It is clear that the family of subgroups of multiplicative type $M(\ell)$ is coherent in the sense of 4.1. Let $C_\ell = \operatorname{Centr}_H(M(\ell))$, which is representable by a group subscheme (2.5), closed (H is separated), smooth over S (Exp. XI 2.4). The C_ℓ form a decreasing filtered family of closed subschemes, hence stationary, since H is noetherian. The stationary value is a subgroup C , smooth and closed, such that $C_s = \operatorname{Centr}_{H_s}(T_s)$ by the density theorem (Exp. IX 4.7). It remains to show that for every point t of S , C_t is the centralizer in H_t of a subtorus T_t (which will imply that C_t is connected). But this will follow from the following more precise lemma, applied to the family $M(\ell)_t$ of subgroups of H_t :

Lemma 6.8. *Let G be a connected algebraic group defined over a field k , r an integer > 0 , q an integer prime to the characteristic⁽⁷⁾ of k , $M(\ell)$ (ℓ ranging over the powers of q) a coherent family of subgroups of multiplicative type of G , of type $(\mathbf{Z}/\ell\mathbf{Z})^r$ (cf. 4.6), M the algebraic subgroup of G generated by the $M(\ell)$ (loc. cit.), T the unique maximal torus of M (cf. 3.4). Then one has*

$$\operatorname{Centr}_G(T) = \operatorname{Centr}_G(M) = \operatorname{Centr}_G(M(\ell)) \quad \text{for } \ell \text{ sufficiently large.}$$

The last equality is quite clear. To prove the first, introduce the center Z of G , $G' = G/Z$, M' (resp. T') the image of M (resp. T) in G' , K the inverse image of T' in G (i.e. the algebraic subgroup of G generated by T and Z). It plainly suffices to prove that $K \supset M$, hence that $T' = M'$. Now M is smooth and connected (4.6) and G' is affine (Exp. XII 6.1), so M' is the direct product of its maximal torus T' (Exp. XII 6.6 d)) and a unipotent group (BIBLÉ 4 Th. 4) (one may suppose k algebraically closed). The image of $M(\ell)$ in M' is therefore necessarily contained in T' . Thus the inverse image of T' in M contains $M(\ell)$ for every ℓ , hence is equal to M . Consequently $M' = T'$. This proves 6.8 and therefore 6.7.

This being so, let us prove 6.6. We have reduced to the case where S is the spectrum of a discrete valuation ring A , which one may moreover suppose complete with algebraically closed residue field. Replacing A by its normalization in a finite extension of its field of fractions, one may suppose that $(G_t)_{\text{red}}$ is smooth⁽⁸⁾. It is clear that, to prove 6.6, one may replace G by the identity component of the schematic closure in G of $(G_t)_{\text{red}}^0$, hence suppose that G is flat over S , with connected fibers, and that G_t is smooth.

a) Suppose that H_s is the centralizer in $(G_s)_{\text{red}}$ of a torus T_s and let us show that H_t is then the centralizer in G_t of a subtorus of G_t . By Lemma 6.7, there exists a group subscheme C of H , smooth over S , whose closed fiber is H_s and such that $C_t = \operatorname{Centr}_{H_t}(T_t)$, where T_t

⁽⁷⁾**N.D.E.** The word “residual” has been deleted.

⁽⁸⁾**N.D.E.** Details or references to be given here. . .

is a subtorus of H_t . Since H is smooth over S , with connected fibers, one concludes, for reasons of dimension, that $C = H$. Keeping the notation of 6.7, one has $H = C = C_\ell$ for large ℓ . Similarly consider $C'_\ell = \underline{\text{Centr}}_G(M(\ell))$ (2.5), and let C' be the stationary value of C'_ℓ for large ℓ (2.5 bis). The group scheme C' contains H and is such that $C'_t = \text{Centr}_{G_t}(T_t)$ (6.8) and $C'_s = \text{Centr}_{G_s}(T_s)$. The hypothesis on H_s implies $\dim H_s = \dim C'_s$. Moreover, $\dim H_s = \dim H_t$ (since H is smooth over S) and $\dim C'_t \leq \dim C'_s$ (Exp. VIB 4.1), so one has $\dim H_t = \dim C'_t$. But since G_t is smooth and connected, $C'_t = \text{Centr}_{G_t}(T_t)$ is smooth and connected, so finally one has $H_t = C'_t = \text{Centr}_{G_t}(T_t)$.

b) Suppose that H_s contains a Cartan subgroup C_s of G_s , i.e. the centralizer in $(G_s)_{\text{red}}$ of a maximal torus T_s of G_s . By 6.7, there exists a group subscheme C of H , smooth over S , with connected fibers, which lifts C_s . It plainly suffices to prove that C_t contains a Cartan subgroup of G_t . Now by a), applied with $H = C$, C_t is the centralizer in G_t of a subtorus of G_t , hence contains a Cartan subgroup of G_t .

c) Suppose that H_s is a parabolic subgroup of G_s . Let $N = \underline{\text{Norm}}_G(H)$. By hypothesis, H is equal to its connected normalizer, so N_s is smooth, and consequently is equal to $\text{Norm}_{(G_s)_{\text{red}}}(H_s) = H_s$ (Exp. XII 8 bis). But then N is flat over S . We shall see in Exp. XVI that under these conditions G/N is representable. Since $H_s = N_s$ is a parabolic subgroup of G , $(G/N)_s$ is proper. Since (G/N) has connected fibers and is flat over S , it follows from EGA III 5.5.1 that (G/N) is proper over S . Thus $(G/N)_t = G_t/N_t$ is proper over t , and so is G_t/H_t , since N_t/H_t is finite. It then suffices to apply the following lemma:

Lemma 6.9. *Let k be a field, G an algebraic k -group, H a smooth connected algebraic subgroup of G , $N = \text{Norm}_G H$; then if $\dim H = \dim N$ and G/H is proper, H is a parabolic subgroup of G .*

Indeed, one may suppose k algebraically closed and G smooth and connected. The center Z of G is contained in N , and the hypothesis $\dim H = \dim N$ implies that $Z' = (Z)_{\text{red}}^0$ is contained in H , so $G' = G/Z'$ is affine (Exp. XII 6.1). Replacing G by G' and H by its image in G' , one reduces to the case where G is affine (Exp. XIV 4.9), and Lemma 6.9 then follows from BIBLE 6 Th. 4. We have therefore proved Lemma 6.6.

To finish proving 6.4, we must prove that the S -preschemes representing $\mathcal{LC}$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$) are of finite presentation over S . This assertion is local on S , which allows one to suppose S affine, then, since G is of finite presentation, S noetherian (EGA IV 8.9). We have just seen that the natural inclusions $\mathcal{CT} \rightarrow \mathcal{L}$ (resp. $\mathcal{P} \rightarrow \mathcal{L}$) are immersions, so the same holds for the inclusions $\mathcal{CT} \rightarrow \mathcal{LC}$ (resp. $\mathcal{P} \rightarrow \mathcal{LC}$), and consequently it suffices to prove that $\mathcal{LC}$ is representable by an S -prescheme of finite presentation.

Let us return to the notation introduced in 5.2. For every integer $n \geq 0$, let $\mathcal{L}^n$ therefore be the subfunctor of $\mathcal{L}$ such that

$$\mathcal{L}^n(S') = \{H \in \mathcal{L}(S') \text{ such that } \underline{\text{Norm}}_{G_{S'}}(H) = \underline{\text{Norm}}_{G_{S'}} H^{(n)}\}.$$

The S -functor $\mathcal{L}^n$ is thus representable by an open subprescheme of $\mathcal{L}$, the sum of the $\mathcal{L}_r^n$. Each $\mathcal{L}_r^n$ is of finite presentation over S (5.3) and is empty for $r > \sup_{s \in S} \dim G_s$ (which is a finite number, S being quasi-compact), so $\mathcal{L}^n$ is of finite presentation over S . It suffices to prove that $\mathcal{LC}$ is contained in $\mathcal{L}^n$ for sufficiently large n .

For every point s of S , let $d(s)$ be the smallest integer n (finite or infinite) such that $\mathcal{LC}_{G_s} \subset \mathcal{L}_{G_s}^n$. It suffices to show that the function d is bounded on S , for if M is an upper bound, $\mathcal{LC}$ will be set-theoretically contained in $\mathcal{L}^M$, hence $\mathcal{LC}$ will be contained in $\mathcal{L}^M$, since the latter is an open subset of $\mathcal{L}$. An immediate constructibility argument reduces us to proving that if S is noetherian and integral with generic point t , then the function d is bounded on a neighborhood of t .

a) *Reduction to the case where G is smooth over S .* Proceeding as in 6.2, one sees that, after changing S , one may suppose that $(G)_{\text{red}}$ is a group prescheme smooth over S , which we shall denote by G' . Put $X = \mathcal{LC}_G$, $X' = \mathcal{LC}_{G'}$, and let H (resp. H') be the group subprescheme of G_X (resp. $G'_{X'}$) universal for the functor $\mathcal{LC}_G$ (resp. $\mathcal{LC}_{G'}$). Since H is smooth over

X , $H \times_X (X_{\text{red}})$ is reduced, hence contained in $G'_{X_{\text{red}}}$, and it is an element of $\mathcal{LC}_{G'}(X_{\text{red}})$, whence a canonical morphism $p : X_{\text{red}} \rightarrow X'$. It is clear that p is a monomorphism; let us show that p is even an immersion. Let N' be the normalizer of H' in $G_{X'}$. The set of points s of X' such that the immersion $H'_s \rightarrow N'_s$ is an open immersion is an open subset U , and $H'|_U \rightarrow N|_U$ is an open immersion (Exp. VIB 2.5 and EGA IV 17.9.5). It follows from 5.1 iii) that U is the largest open subset of X' over which H' is equal to its connected normalizer in $G_{X'}$, so $H'|_U \in \mathcal{LC}_G(U)$. One immediately deduces that p is an isomorphism of X_{red} onto U_{red} . If one can show that $\mathcal{LC}_G$ is of finite type when G is smooth, X' will be of finite type over S , hence X_{red} will be of finite type (S is noetherian), and consequently will be contained in $\mathcal{L}_G^M$ for sufficiently large M ; the same will hold for X .

b) *Case where S is the spectrum of an algebraically closed field k of characteristic $p > 0$ and G is a smooth algebraic group over k .* Instead of using the infinitesimal neighborhoods $H^{(n)}$ of a group subscheme H of G , we shall use the radicial subgroups $F^n(H)$, kernels of the iterates of the Frobenius morphism in H (Exp. VIIA 4), which is legitimate here, since $F^n(H)$ is contained in the infinitesimal neighborhood of order p^n of the unit section of H . If T is a subtorus of G , $F^n(T) = p^n(T)$, and it is immediate by duality that $\text{Centr}_G(p^n(T))$ is equal to $\text{Centr}_G(T)$ for sufficiently large n . One then has the following more precise proposition:

Proposition 6.10. *Let k be a field of characteristic $p > 0$, G a smooth algebraic k -group, T a maximal torus of G_k , m the smallest integer such that*

$$\text{Centr}_G(p^m(T))^0 = \text{Centr}_G(T)^0.$$

Then, for every k -prescheme S and every $H \in \mathcal{LC}(S)$, one has

$$\text{Norm}_{G_S}(H) = \text{Norm}_{G_S}(F^m(H)).$$

A fortiori, $\mathcal{LC}$ is contained in $\mathcal{L}^{p^m}$.

Since H is smooth over S , of finite presentation over S , and has constant nilpotent rank (namely that of G), we shall see in the next paragraph (7.3) that the functor $\mathcal{C}_H$ of Cartan subgroups of H is representable by an S -prescheme smooth over S (the reader will check that the proof given of this property does not use the fact that $\mathcal{LC}_H$ is of finite type over S). It then follows from Exp. XIII 3.1 that one may consider the open subset U of regular points of H .

Let S' be an S -prescheme, g an element of $G(S')$ normalizing $F^m(H)_{S'}$. To prove that g normalizes $H_{S'}$, it suffices to prove that $\text{int}(g)U_{S'}$ is contained in $H_{S'}$; indeed, $H_{S'} \cap \text{int}(g)H_{S'}$ will then contain an open subgroup of $\text{int}(g)H$, hence will be equal to $\text{int}(g)H$, since the latter has connected fibers. Replacing S' by a suitable S'' , then S'' by S , one reduces to proving that if $g \in G(S)$ normalizes $F^m(H)$ and $u \in U(S)$, then $\text{int}(g)u \in H(S)$.

Now let C be the unique Cartan subgroup of H_S which “contains” u (Exp. XIII 3.2). It suffices for us to show that $C' = \text{int}(g)C \subset H$. Since $H \in \mathcal{LC}_G(S)$, C is also a Cartan subgroup of G , but the latter has maximal tori, so (Exp. XII 7.1 (a)) C is the centralizer in G_S^0 of its unique maximal torus T . It follows from the definition of m (and the fact that two maximal tori of G are locally conjugate for fpqc (Exp. XII 7.1)) that one has

$$C = (\text{Centr}_G(T))^0 = \text{Centr}_G(p^m(T))^0,$$

whence, by conjugation by g ,

$$C' = \text{Centr}_G(\text{int}(g)(p^m(T)))^0.$$

But $\text{int}(g)(p^m(T))$ is a subgroup of multiplicative type of $\text{int}(g)(F^m(H))$, which is equal to $F^m(H)$ (g normalizes $F^m(H)$), hence is contained in H . It then follows from Exp. XIII 2.1 (which is proved when the base is a field, but extends immediately to the case of an arbitrary base) that, for C' to be contained in H , it suffices that one have

$$\text{Lie } C' \subset \text{Lie } H.$$

Now $\text{Lie } C' = \text{int}(g)(\text{Lie } C) \subset \text{int}(g)(\text{Lie } H)$.

On the other hand, if $m \geq 1$, which one is free to suppose, one has (using Exp. VIIA 4.1.2)

$$\mathrm{Lie} H = \mathrm{Lie}({}_{F^m}(H)) = \mathrm{Lie}(\mathrm{int}(g)({}_{F^m}(H))) = \mathrm{int}(g)(\mathrm{Lie} H).$$

Thus $\mathrm{Lie} C' \subset \mathrm{Lie} H$, which completes the proof of 6.10.

We shall need another definition of the integer m introduced in 6.10.

Lemma 6.11. *Let G be a smooth algebraic group defined over an algebraically closed field k of characteristic $p > 0$, T a maximal torus of G , $\mathfrak{g} = \mathrm{Lie} G$, and R the family of nonzero characters of T occurring in the representation of T in $\mathfrak{g}$ induced by the adjoint representation of G . For every element $r \in R$, denote by e_r the largest integer n such that p^n divides r in the group of characters of T . Then $m = \sup_{r \in R} (e_r + 1)$ if $R \neq \emptyset$, and $m = 0$ otherwise.*

Indeed, $\mathrm{Centr}_G(T)$ is smooth and contained in $\mathrm{Centr}_G({}_{p^m}(T))$, so

$$\begin{aligned} (\mathrm{Centr}_G T)^0 &= (\mathrm{Centr}_G({}_{p^m}(T)))^0 \iff \mathrm{Lie}(\mathrm{Centr}_G T) = \mathrm{Lie}(\mathrm{Centr}_G({}_{p^m}(T))) \\ &\iff \mathfrak{g}^T = \mathfrak{g}^{p^m(T)} \quad (\text{Exp. II 5.2.3}). \end{aligned}$$

Now with the usual notation, one has

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \coprod_{r \in R} \mathfrak{g}_r.$$

Thus $\mathfrak{g}^T = \mathfrak{g}_0$ and $\mathfrak{g}^{p^m(T)} = \mathfrak{g}_0 + \coprod_{r \in R'} \mathfrak{g}_r$, where R' is the part of R formed by the characters of T whose restriction to ${}_{p^m}(T)$ is zero. But a nonzero character r of T has zero restriction to ${}_{p^m}(T)$ if and only if $m \leq e_r$, whence the lemma.

c) Let us return to the proof of 6.4. We have reduced (according to point a) and the paragraph preceding it) to the case where S is a noetherian integral scheme and G is smooth over S . We must show that the function d is bounded on a neighborhood of the generic point t of S . After changing S , we may suppose that G has a split maximal torus (6.2) $T = D_S(M)$. Let $\mathfrak{g} = \coprod_{\lambda \in M} \mathfrak{g}^\lambda$ be the decomposition of the Lie algebra of G according to the characters of T , and let R be the finite set of nonzero characters of M such that $\mathfrak{g}^\lambda \neq 0$. Let us then distinguish two cases:

1st case: the point t , and consequently all points of S , have residue characteristic $p > 0$. It is clear, in view of the foregoing, that the function d is then bounded above by p^m , where m is defined as in 6.11.

2nd case: the point t has residue characteristic zero. For every $\lambda \in R$, let n_λ be the largest integer dividing λ in the group M , and put $n = \prod n_\lambda$, $\lambda \in R$. For every prime number q dividing n , denote by S_q the closed subset of S formed by the points of S whose residue characteristic is equal to q , and let U be the nonempty open subset (it contains t) complementary in S to the union of the S_q . If now s is a point of U , either s has residue characteristic zero, and then $d(s) = 1$ (5.6), or s has residue characteristic $p > 0$ which does not divide n , so the integer m for the group G_s , defined in 6.11, is less than or equal to one. Moreover, it follows from Exp. VII⁽⁹⁾ that one has

$$\underline{\mathrm{Norm}}_{G_{S'}}({}_{F^1}(H)) = \underline{\mathrm{Norm}}_{G_{S'}}(\mathrm{Lie} H) = \underline{\mathrm{Norm}}_{G_{S'}}(H^{(1)}).$$

Finally, it follows from 6.10 that if $H \in \mathcal{L}\mathcal{C}_{G_s}(S')$, one has $\underline{\mathrm{Norm}}_{G_{S'}}(H) = \underline{\mathrm{Norm}}_{G_{S'}}(H^{(1)})$, and consequently $d(s) \leq 1$, so d is bounded by 1 on U .

This completes the proof of 6.4.

Corollary 6.12. *Let S be a prescheme and G a group S -prescheme of finite presentation. Suppose that the nilpotent rank (resp. the dimension of the Borel subgroups) of the fibers of G is a locally constant function on S ; then the functor $\mathcal{C}$ of Cartan subgroups of G (resp. the functor $\mathcal{B}$ of Borel subgroups of G) is representable by an S -prescheme of finite presentation over S .*

⁽⁹⁾**N.D.E.** Argument to be made explicit...

Indeed, after restricting S , one may suppose that the nilpotent rank ν of the fibers is constant. But then it is clear that $\mathcal{C}$ is represented by the subprescheme, both open and closed, of the prescheme X representing $\mathcal{LC}$ (6.4), over which the universal subgroup of G_X for the functor $\mathcal{LC}$ has relative dimension ν . The proof is analogous for the functor $\mathcal{B}$, taking account of the representability of the functor $\mathcal{P}$.

7. Cartan subgroups of a smooth group

Proposition 7.1. *Let S be a prescheme, G a group S -prescheme of finite presentation, smooth over S , with connected fibers.*

i) Let $\mathcal{CT}$ be the S -functor $((\text{Sch})/S)^\circ \longrightarrow (\text{Ens})$ such that for every S -prescheme S' , one has

$\mathcal{CT}(S') = \text{set of group subpreschemes } H \text{ of } G_{S'}, \text{ smooth over } S', \text{ such that for every point } s' \text{ of } S', H_{s'} \text{ is the centralizer in } G_{s'} \text{ of a subtorus of } G_{s'}.$

Then $\mathcal{CT}$ is representable by an S -prescheme of finite presentation over S , smooth and quasi-projective over S .

ii) If S is artinian and $H \in \mathcal{CT}(S)$, H is the centralizer in G of a subtorus of G . If S is the spectrum of a henselian local ring and $H \in \mathcal{CT}(S)$, H is the centralizer in G of a subgroup of multiplicative type of G , étale over S .

iii) If L is a group S -prescheme, smooth and of finite presentation over S , $i : L \longrightarrow G$ a group S -monomorphism, and H an element of $\mathcal{CT}(S)$, then $\text{Transp}_G(H, L)$ and $\text{Transp}_G^{\text{str}}(H, L)$ (cf. Exp. VIII 6.5 e)) are representable by closed subpreschemes of G , smooth over S .

iv) If $H \in \mathcal{CT}(S)$, H is closed in G , $N = \text{Norm}_G(H)$ is representable by a group subprescheme of G , closed and smooth over S ; N/H is representable by a group S -prescheme, separated over S , étale and of finite type over S ; G/N is representable by an S -prescheme smooth and quasi-projective over S .

v) Let G' be a group S -prescheme of finite presentation over S , and $u : G \longrightarrow G'$ a faithfully flat group S -morphism, so that G' satisfies the same hypotheses as G (Exp. VIB 9). Then if $H \in \mathcal{CT}_G(S)$, the image of H under u is representable by a group subprescheme H' of G' which is an element of $\mathcal{CT}_{G'}(S)$. Moreover, $H \longrightarrow H'$ is faithfully flat, and if H is the centralizer in G of a torus T , H' is the centralizer in G' of the torus $T' = u(T)$.

vi) Under the conditions of v), consider the S -morphism

$$\tilde{u} : \mathcal{CT}_G \longrightarrow \mathcal{CT}_{G'}, \quad H \longmapsto H' = u(H).$$

Then $\tilde{u}$ is a faithfully flat quasi-compact morphism; if moreover $\text{Ker } u$ is central, $\tilde{u}$ is an isomorphism, the inverse isomorphism being $H' \mapsto u^{-1}(H')$.

Proof of ii). For the first assertion, one may suppose S local artinian with closed point s . Let $H \in \mathcal{CT}(S)$ and T_s the maximal central torus of H_s , which is already defined over $\kappa(s)$ (cf. 3.4). Since $H_s \in \mathcal{CT}(s)$, one has $H_s = \text{Centr}_{G_s} T_s$. The group H is smooth, so T_s lifts (uniquely) to a subtorus T of H , central in H (Exp. IX 3.6 bis and Exp. IX 5.6). But then $H' = \text{Centr}_G T$ contains H and has the same fiber as H ; since H is flat over S , one has $H = H'$ (Exp. VIB 2.5).

Suppose now that S is the spectrum of a henselian local ring, which one may suppose noetherian by the usual reductions. Denote by s the closed point of S , T_s the maximal central torus of H_s , q an integer invertible on S , ℓ a power of q , T an S -torus whose closed fiber is isomorphic to T_s (Exp. X 4.6). On the other hand, let ${}_\ell H$ be the “kernel” of raising to the ℓ th power in H , and let U_ℓ be the largest open subset of ${}_\ell H$ which is étale over S . It then follows from 1.3 and the fact that H is flat over S that U_ℓ contains ${}_\ell T_s$. Since S is henselian, there exists a unique S -morphism

$$u_\ell : {}_\ell T \longrightarrow U_\ell$$

which, on the closed fiber, induces the canonical immersion ${}_\ell T_s \longrightarrow (U_\ell)_s$. By uniqueness, one easily sees that u_ℓ is a central morphism of S -groups (Exp. IX 5.6 a)). Proceeding as in

6.6 and 6.7, one shows that u_ℓ is an immersion and that, if M_ℓ is the image group $u_\ell(\ell T)$, then $\underline{\text{Centr}}_G(M_\ell)$ is equal to H for sufficiently large ℓ (this is where the hypothesis that S is noetherian is used).

Proof of i). The group G is smooth over S , with connected fibers, so if $H \in \mathcal{CT}(S)$, H has connected fibers (Exp. XII 6.6 b)) and is equal to its connected normalizer (6.5), so that the functor $\mathcal{CT}$ defined in 7.1 i) coincides with the functor also denoted by $\mathcal{CT}$ which was defined in 6.4.0. Thus, by Theorem 6.4, $\mathcal{CT}$ is representable by an S -prescheme of finite presentation and quasi-projective over S . It remains to show that this prescheme is smooth over S . One first reduces, by EGA IV 8, to the case where S is affine noetherian. By Exp. XI 1.5, it then suffices to prove that if S is the spectrum of an artinian local ring, S_0 a subscheme defined by a nilpotent ideal, H_0 an element of $\mathcal{CT}(S_0)$, then H_0 lifts to a group subscheme H of G , smooth over S . Now by ii), $H_0 = \underline{\text{Centr}}_{G_0} T_0$, where T_0 is a subtorus of G_0 . Since G is smooth, T_0 lifts to a subtorus T of G (Exp. IX 3.6 bis), and it suffices to take $H = \underline{\text{Centr}}_G(T)$, which is indeed smooth over S (Exp. XI 2.4 and Lemma 2.5).

Proof of iii). Since H is smooth, with connected fibers, by Exp. XI 6.11, $\underline{\text{Transp}}_G(H, L)$ is representable by a closed subscheme of G , of finite presentation over S . To show that this transporter is smooth, one reduces, as above, to proving that if S is local artinian, S_0 a closed subscheme of S , $g_0 \in G(S_0)$ such that $\text{int}(g_0)H_0 \subset L_0$, then g_0 lifts to $g \in G(S)$ such that $\text{int}(g)H \subset L$. Since the group G is smooth over S , there exists a section g_1 of G lifting g_0 ; let $H' = \text{int}(g_1)H$. It is therefore an element of $\mathcal{CT}(S)$ such that $H'_0 \subset L_0$. By ii), H' is the centralizer in G of a torus T' of G . Since L is smooth, the torus T'_0 of L_0 lifts to a torus T'' of L (Exp. XI 3.6 bis). The group $\underline{\text{Centr}}_L T''$ is contained in $\underline{\text{Centr}}_G T''$, has the same fiber as the latter (namely H'_0), and is smooth, hence is equal to $\underline{\text{Centr}}_G T'' = H''$. The subtori T' and T'' of G are two liftings of T'_0 , hence are conjugate by an element h of $G(S)$ reducing to the unit section of G_0 (Exp. IX 3.3 bis); the same therefore holds for their centralizers H' and H'' in G . The section $g = hg_1$ lifts g_0 , and one indeed has $\text{int}(g)H \subset L$.

If now $g \in \underline{\text{Transp}}_G(H, L)(S)$, for $\text{int}(g)H = L$, it is necessary and sufficient that for every $s \in S$, $\dim H_s = \dim L_s$. It follows that if U denotes the subscheme of S , both open and closed, over which the fibers of H have the same dimension as those of L , the strict transporter of H into L , $\underline{\text{Transp}}_G^{\text{str}}(H, L)$, is representable by the S -prescheme

$$U \times_S \underline{\text{Transp}}_G(H, L).$$

Proof of iv). To see that if $H \in \mathcal{CT}(S)$, H is closed in G , one may suppose S affine noetherian, then S the spectrum of a complete local ring (EGA IV 8), but then H is the centralizer in G of a subgroup of multiplicative type (by ii)), hence is closed since G is separated over S (Exp. VIB 5.2).

By iii), $N = \underline{\text{Norm}}_G(H) = \underline{\text{Transp}}_G^{\text{str}}(H, H)$ is representable by a group subscheme smooth over S and closed in G . Consider the S -morphism

$$G \longrightarrow \mathcal{CT}, \quad g \longmapsto \text{int}(g)H.$$

It follows from iii) that this morphism is smooth, so its image is an open subset U of $\mathcal{CT}$. One then proves, as in Exp. XI 5.3, that G/N is representable by U , hence in particular is quasi-projective.

Let us now study the quotient N/H . By EGA IV 8, to prove that N/H is representable, one may suppose S affine noetherian, then S the spectrum of a local ring A , hence of finite dimension⁽¹⁰⁾. We shall proceed by increasing induction on the dimension of S . If $\dim S = 0$, the property follows from Exp. VIA § 4. Note now that if N/H is representable, it is separated over S (since H is closed in N by iv)), of finite type and étale over S (since N is smooth over S , of finite type, and H is open in N), so N/H is necessarily quasi-affine over S (SGA1 VIII.6.2). By effective descent of quasi-affine schemes (loc. cit. 7.9), one may replace A by its completion, hence suppose S the spectrum of a complete noetherian local ring. Let s

⁽¹⁰⁾ **N.D.E.** (Krull dimension)

be its closed point and $U = S \setminus s$. By the induction hypothesis, $(N|_U)/(H|_U)$ is representable by a U -group K . Let then T_s be the maximal central torus of H_s , q an integer invertible on S , ℓ a power of q , M_ℓ the unique central subgroup of multiplicative type of H lifting ${}_\ell T_s$ (cf. ii)). Choose ℓ sufficiently large that $\text{Centr}_G(M_\ell) = H$, and let $N' = \text{Norm}_G(M_\ell)$. Since $\text{Centr}_G(M_\ell) = H$, one has $N' \subset \text{Norm}_G(H)$. Moreover, one immediately checks that T_s (hence also $(M_\ell)_s$) is a characteristic subgroup of H_s (i.e. stable under $\text{Aut}_{\text{gr}}(H_s)$), so N_s normalizes $(M_\ell)_s$, and consequently $N_s = N'_s$. The proof of Exp. XI 5.9 then proves that the quotient $N'/H = \text{Norm}_G(M_\ell)/\text{Centr}_G(M_\ell)$ is representable by an S -group K' . Since N' is smooth over S and $N'_s = N_s$, N' is an open subgroup of N (Exp. VIB 2.5) containing H , so the image of $N'|_U$ in K is an open subgroup, isomorphic to $K'|_U$. Let L be the S -prescheme obtained by gluing K and K' by means of the preceding isomorphism, and let p be the S -morphism $N \rightarrow L$ obtained by gluing the canonical projections $N|_U \rightarrow K$ and $N' \rightarrow K'$. It is clear that (L, p) represents the quotient N/H .

Proof of v). Suppose first that S is the spectrum of a field k . The image H' of H is then a smooth subgroup of G' . We must show that $H' \in \mathcal{CT}_{G'}(S)$, which will follow from the following more precise lemma:

Lemma 7.2. *Let $u : G \rightarrow G'$ be an epimorphism of smooth connected algebraic k -groups, T a torus of G , T' its image in G' ; then*

$$u(\text{Centr}_G T) = \text{Centr}_{G'} T'.$$

Write $H = \text{Centr}_G T$, $H' = \text{Centr}_{G'} T'$, $H'' = u(H)$. One has $H'' \subset H'$. To prove that $H' = H''$, one may suppose the base field algebraically closed, and it suffices to prove that every Cartan subgroup of H' is contained in H'' . Indeed, H'' will then contain the open subset of regular points of H' , hence H'' will be an open subgroup of H' and consequently will be equal to H' , since the latter has connected fibers. Let therefore C' be a Cartan subgroup of H' ; C' is also a Cartan subgroup of G' , since H' is the centralizer of a torus T' , hence has the same reductive rank and the same nilpotent rank as G' . Put $K = (u^{-1}(C'))_{\text{red}}^0$. Since T' is in the center of H' , C' contains T' , so K contains T . Let then C be a Cartan subgroup of K containing T . The torus T is contained in the unique maximal torus of C , which is central in C (Exp. XII 6.6 c)), so C is contained in $H = \text{Centr}_G T$. Using now the fact that two Cartan subgroups of G are conjugate and that the image of a Cartan subgroup of G is a Cartan subgroup of G' (Exp. XII 6.6), one deduces that C is also a Cartan subgroup of G , so its image is a Cartan subgroup of G' ; since it is contained in C' , one has $u(C) = C'$, so C' is indeed contained in H'' .

We have therefore established v) when S is the spectrum of a field k . Let us now study the general case. Since G , H and G' are of finite presentation over S , to prove that $u(H)$ is representable and is an element of $\mathcal{CT}_{G'}(S)$, one reduces by the usual technique to the case where S is affine noetherian, then to the case where S is the spectrum of a local ring. By fpqc descent of subpreschemes of G' , one may even suppose that S is the spectrum of a complete noetherian local ring A .

Let us return to the notation of ii), i.e.: let T_s be the maximal central torus of H_s (s is the closed point of S), M_ℓ a subgroup of multiplicative type of H lifting ${}_\ell T_s$ and such that $H = \text{Centr}_G M_\ell$. Let T'_s be the image of T_s in G'_s . Since G' is separated over S (Exp. VIB 5.2), the image of M_ℓ under u is a subgroup of multiplicative type M'_ℓ of G' (Exp. IX 6.8). Put then $H' = \text{Centr}_{G'} M'_\ell$, which is a smooth group subprescheme of G' . For every integer ℓ' equal to a power of q , there exists ℓ such that $(M'_\ell)_s$ contains ${}_{\ell'} T'_s$, so one may suppose ℓ chosen sufficiently large that $H'_s = \text{Centr}_{G'_s} T'_s = u_s(H_s)$, where the last equality follows from 7.2. The restriction of u to H , say v , plainly factors through H' . Let us prove that $v : H \rightarrow H'$ is a flat morphism. Since H and H' are flat over S and $H_s \rightarrow H'_s$ is flat, v is flat on a neighborhood of H_s (EGA IV 11.3.10 and 11.3.1). The morphism v is therefore flat on an open subgroup of H (Exp. VIB 2.2), hence v is flat, H having connected fibers. The set-theoretic image of H is thus an open subset of H' (necessarily equal to $(H')^0$) which,

equipped with its induced structure, represents the image sheaf $u(H)$ (for the fpqc topology). The fact that $u(H)$ is an element of $\mathcal{CT}_{G'}(S)$ then follows from 7.2.

Suppose now that H is the centralizer in G of a torus T , and let T' be the image of T under u , which is a subtorus of G' (Exp. IX 6.8). The image of H under u is contained in $\text{Centr}_{G'}(T')$, coincides fiber by fiber with the latter (7.2), and is smooth over S , so $u(H) = \text{Centr}_{G'}(T')$.

Proof of vi). To show that $\tilde{u}$ is a faithfully flat S -morphism, knowing that $\mathcal{CT}_G$ and $\mathcal{CT}_{G'}$ are smooth over S (by i)), it suffices to check this on the geometric fibers. We are therefore reduced to the case where S is the spectrum of an algebraically closed field k . Let $H' \in \mathcal{CT}_{G'}(k)$, T' its maximal central torus, T a subtorus of G whose image is T' , $H = \text{Centr}_G(T)$, so that $H' = u(H)$ (7.2), $N = \underline{\text{Norm}}_G(H)$, $N' = \underline{\text{Norm}}_{G'}(H')$. We have shown in iv) that G/N (resp. G'/N') is canonically identified with an open neighborhood of H in $\mathcal{CT}_G$ (resp. of H' in $\mathcal{CT}_{G'}$). Under these identifications, the restriction of $\tilde{u}$ to G/N coincides with the natural morphism

$$w : G/N \longrightarrow G'/N'$$

deduced from u by passage to the quotient. Now w is an epimorphism of homogeneous spaces under G , hence is faithfully flat. This proves that $\tilde{u}$ is a flat morphism such that $u(\mathcal{CT}_G)$, which is therefore an open subset of $\mathcal{CT}_{G'}$, contains every point of $\mathcal{CT}_{G'}(k)$. Since $\mathcal{CT}_{G'}$ is of finite type over k , one deduces that $\tilde{u}$ is surjective, hence is faithfully flat.

The additional assertions in vi) in the case where $\text{Ker } u$ is central follow from Exp. XII 7.12.

The following theorem generalizes Theorem 7.1 of Exp. XII:

Theorem 7.3. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , smooth, with connected fibers, and consider the S -functor $\mathcal{C} : ((\text{Sch})/S)^\circ \longrightarrow (\text{Ens})$ such that*

$$\mathcal{C}(S') = \text{set of Cartan subgroups of } G_{S'}.$$

i) *The following conditions are equivalent:*

- a) *The functor $\mathcal{C}$ is representable.*
- b) *The functor $\mathcal{C}$ is representable by a smooth quasi-projective S -prescheme of finite presentation over S , with affine fibers.*
- c) *The group G has, locally for the étale topology, a Cartan subgroup.*
- d) *The group G has, locally for the faithfully flat topology, a Cartan subgroup.*
- e) *The nilpotent rank of the fibers of G is a locally constant function on S .*

ii) *If the preceding conditions hold, two Cartan subgroups of G are locally conjugate for the étale topology. The set of regular points of the fibers of G (Exp. XIII 2.7) is an open subset G_{reg} , of finite presentation over S , and every section of G_{reg} over S is contained in one and only one Cartan subgroup of G .*

iii) *Let G' be a group S -prescheme of finite presentation over S , and $u : G \longrightarrow G'$ a faithfully flat group S -morphism, so that G' is smooth over S , with connected fibers. Then if C is a Cartan subgroup of G , $u(C) = C'$ is representable by a Cartan subgroup of G' , and $C \longrightarrow C'$ is faithfully flat.*

iv) *Under the conditions of i) and iii), the morphism*

$$\tilde{u} : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}, \quad C \longmapsto u(C) = C'$$

is faithfully flat. If moreover $\text{Ker}(u)$ is central, $\tilde{u}$ is an isomorphism.

v) *For further information concerning transporters and relations with maximal tori, one may consult 7.1 and Exp. XII 7.1.*

PROOF. i) We shall show that b) $\Rightarrow$ c) $\Rightarrow$ d) $\Rightarrow$ e) $\Rightarrow$ b) $\Rightarrow$ a) $\Rightarrow$ d).

b) $\Rightarrow$ c). Let s be a point of S . Since $\mathcal{C}_s$ is smooth over $\kappa(s)$, there exist points of $\mathcal{C}_s$ whose residue field is a finite separable extension of $\kappa(s)$. Applying Exp. XI 1.10, one sees

that there exists an open neighborhood U of s and a surjective étale morphism $U' \rightarrow U$ such that $G_{U'}$ has a Cartan subgroup.

c) $\Rightarrow$ d) is quite clear.

d) $\Rightarrow$ e). Let $s \in S$. By hypothesis, there exists an S -prescheme S' , flat over S , whose image contains s , such that $G_{S'}$ has a Cartan subgroup. Let s' be a point of S' above s . The nilpotent rank of the fibers of $G_{S'}$ is therefore constant on $\text{Spec } \mathcal{O}_{S',s'}$, and consequently the nilpotent rank of the fibers of G is constant on $\text{Spec } \mathcal{O}_{S,s}$, which is the image of $\text{Spec } \mathcal{O}_{S',s'}$ (EGA IV 2.3.4 ii). Let r be its value. It follows from 6.3 bis that the set E_r formed by the points x of S such that the nilpotent rank of G_x is equal to r is an ind-constructible subset of S , hence contains a neighborhood of s (EGA IV 1.10.1).

e) $\Rightarrow$ b). The assertion is local on S , so one may suppose that the nilpotent rank of the fibers of G is constant and equal to r . For every S -prescheme S' , the Cartan subgroups of $G_{S'}$ are then precisely the group subschemes of $G_{S'}$, smooth over S' , of relative dimension r , whose geometric fibers are the centralizers of a torus. Since $\mathcal{CT}$ is representable by 7.1, $\mathcal{C}$ is representable by the subscheme, both open and closed, of $\mathcal{CT}$ representing the subfunctor of $\mathcal{CT}$ formed by the groups of relative dimension r . The other assertions in b) are contained in 7.1 i), except the fact that the fibers of $\mathcal{C}$ are affine, which follows from Exp. XII 7.1 d).

It is clear that b) $\Rightarrow$ a). Let us show that a) $\Rightarrow$ d). It follows from 6.2 iii) that the functor $\mathcal{C}$ commutes with filtered inductive limits of rings, so if $\mathcal{C}$ is representable, it is necessarily representable by an S -prescheme locally of finite presentation (EGA IV 8.14.2). To prove that $\mathcal{C}$ is smooth over S , one reduces to showing that if S is affine and S_0 is the closed subscheme defined by a nilpotent ideal J , then every Cartan subgroup C_0 of $G_0 = G \times_S S_0$ lifts to a Cartan subgroup C of G . But the existence of C_0 implies that condition e) is satisfied for S_0 , hence for S , which has the same underlying space, and one concludes from the fact that d) $\Rightarrow$ b). Since $\mathcal{C}$ is smooth over S and $\mathcal{C} \rightarrow S$ is surjective, one sees that a) $\Rightarrow$ d).

ii) Let C and C' be two Cartan subgroups of G . Then $\text{Transp}_G(C, C')$ is representable by a prescheme smooth over S (7.1 iii)), with nonempty fibers (cf. Exp. XII 6.6 a) and c)). The fact that C and C' are locally conjugate for the étale topology is then a consequence of Hensel's lemma (Exp. XI 1.10).

The other assertions of ii) are consequences of XIII 3.1 and XIII 3.2, taking account of i).

iii) Let C be a Cartan subgroup of G . One knows (7.1 v)) that $u(C)$ is representable by a smooth subgroup C' of G' . Since the fibers of C' are Cartan subgroups of the fibers of G' (Exp. XII 6.6 d)), C' is a Cartan subgroup of G' .

iv) To prove that the morphism $\tilde{u}$ is faithfully flat, one proceeds as in 7.1 vi).

If now $\text{Ker } u$ is central and C' is a Cartan subgroup of G' , $u^{-1}(C') = C$ is smooth, with connected fibers (7.1 vi)), and its fibers are Cartan subgroups (Exp. XII 6.6 f)), so C is a Cartan subgroup of G , which completes the proof of iv), taking account of 7.1 vi). $\square$

8. Representability criterion for the subtori functor of a smooth group

8.0. — In this paragraph, if S is a prescheme and G a group S -prescheme, $\mathcal{T}_G$ (or simply $\mathcal{T}$ if there is no ambiguity) denotes the S -functor $((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ such that, for every S -prescheme S' , one has

$$\mathcal{T}(S') = \text{set of subtori of } G_{S'}.$$

One similarly defines $\mathcal{TC}$ as the functor of central subtori of G .

Proposition 8.1. *Let S be a locally noetherian prescheme and G a group S -prescheme of finite type. Then the following conditions are equivalent:*

i) *The functor $\mathcal{T}_G$ “commutes with adic limits of artinian local rings”, i.e. for every S -prescheme of the form $\text{Spec}(A) = S'$, where A is a noetherian local ring complete for the topology defined by its maximal ideal $\mathfrak{m}$, the canonical map*

$$\mathcal{T}(S') \rightarrow \varprojlim_n \mathcal{T}(S'_n) \quad (\text{where } S'_n = \text{Spec}(A/\mathfrak{m}^n))$$

is bijective.

ii) As in i), but one restricts to the case where A is a complete discrete valuation ring with algebraically closed residue field.

iii) As in ii), but one restricts to the subfunctor $\mathcal{T}^{(1)}$ of $\mathcal{T}$ corresponding to the subtori of G of relative dimension 1.

i bis) For every S -prescheme S' as in i) and every S' -torus T , the canonical map

$$\mathrm{Hom}_{S'-\mathrm{gr}}(T, G_{S'}) \longrightarrow \varprojlim_n \mathrm{Hom}_{S'_n-\mathrm{gr}}(T_{S'_n}, G_{S'_n})$$

is bijective.

ii bis) As in i bis), but one restricts to the case where A is a complete discrete valuation ring with algebraically closed residue field.

iii bis) As in ii bis), but one restricts to the case where T is the multiplicative group $\mathbf{G}_m$.

Remark 8.2. One has an analogous proposition on restricting to the central subtori of G and to central homomorphisms from a torus to G .

Proof. We shall use the following lemma:

Lemma 8.3. *Let S be a prescheme, G a group S -prescheme, T an S -torus, $u : T \rightarrow G$ a group S -morphism. Suppose moreover that G is of finite presentation over S , or that S is locally noetherian and G locally of finite type. Then:*

a) $\mathrm{Ker} u$ is a subgroup of multiplicative type of T .

b) The quotient $T' = T / \mathrm{Ker} u$ is a torus.

c) The canonical monomorphism $T' \rightarrow G$, deduced from u by passage to the quotient, is an immersion.

This lemma is a consequence of Exp. IX 6.8 when G is separated over S . In the general case, one reduces as usual to the case where S is noetherian. Let us first show that $K = \mathrm{Ker} u$ is flat. We may suppose that S is the spectrum of an artinian local ring (EGA 0_{III} 10.2.6), in which case G is separated (Exp. VIB 5.2), so K is of multiplicative type (Exp. IX 6.8), and a fortiori flat over S . Let us now prove that $\mathrm{Ker} u$ is closed in T , which reduces us to the case where S is the spectrum of a discrete valuation ring (EGA II 7.2.1). Since T is flat, with connected fibers, u factors through the identity component of the schematic closure in G of the generic fiber of G . We may therefore suppose G flat with connected fibers, but then G is separated (Exp. VIB 5.2), so K is closed. This being so, it follows from Exp. X 4.8 b) that K is a subgroup of multiplicative type of T . The quotient $T' = T/K$ is then representable, and T' is a group of multiplicative type (Exp. IX 2.7) whose fibers are tori, hence is a torus. The fact that the monomorphism $T' \rightarrow G$ is an immersion then follows from Exp. VIII 7.9.

Proof of 8.1. i) $\Rightarrow$ i bis). Put $T_n = T_{S'_n}$, $G_n = G_{S'_n}$, and let $(u_n)_{n \in \mathbf{N}'}$ be an element of $\varprojlim_n \mathrm{Hom}_{S'_n-\mathrm{gr}}(T_n, G_n)$. For every integer n , $u_n(T_n)$ is therefore a subtorus T'_n of G_n (Lemma 8.3). By hypothesis, there exists a unique subtorus T' of G lifting T'_n for every n . Since T' has affine fibers, one concludes by 4.4.

i bis) $\Rightarrow$ i). Let $(T_n)_{n \in \mathbf{N}}$ be an element of $\varprojlim_n \mathcal{T}(S_n)$. By Exp. X 4.6, there exists an S' -torus T' and an S'_0 -isomorphism

$$u_0 : T'_0 \xrightarrow{\sim} T_0.$$

Since T_n is smooth over S'_n , u_0 lifts to an S'_n -morphism

$$u_n : T'_n \longrightarrow T_n \quad (\text{Exp. IX 3.6}),$$

and one may suppose the family $(u_n)_{n \in \mathbf{N}}$ coherent, hence coming from a morphism $u : T' \rightarrow G$. The image of T' under u is then a subtorus of G (Lemma 8.3) lifting T_n for every n .

The implications i) $\Rightarrow$ ii) $\Rightarrow$ iii), on the one hand, and i bis) $\Rightarrow$ ii bis) $\Rightarrow$ iii bis), on the other, are evident. The implication iii) $\Rightarrow$ iii bis) is proved as i) $\Rightarrow$ i bis). It therefore suffices for us to prove iii bis) $\Rightarrow$ ii bis) $\Rightarrow$ i bis).

ii bis) $\Rightarrow$ i bis). With the terminology introduced in 4.3, assertion i bis) is true if and only if every element $(u_n)_{n \in \mathbb{N}}$ of $\varprojlim_n \text{Hom}_{S_n\text{-gr}}(T_n, G_n)$ is “admissible”. For every point t of S' distinct from the closed point s of S' , there exists an S -scheme S'' , the spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point maps to t and whose closed point maps to s (EGA II 7.1.9). One immediately deduces a valuative criterion for a family of morphisms to be admissible. This says that ii bis) $\Rightarrow$ i bis).

iii bis) $\Rightarrow$ ii bis). Let T be an S -torus, where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. The torus T is then split (Exp. X 4.6), i.e. isomorphic to $(\mathbf{G}_{mS})^r$ for a suitable integer r . Let $(u_n)_{n \in \mathbb{N}}$ be an element of $\varprojlim_n \text{Hom}_{S_n\text{-gr}}((\mathbf{G}_m)_{S_n}^r, G_n)$. By hypothesis, the restrictions of the u_n to each factor of $(\mathbf{G}_m)_S^r$ come from a group morphism

$$\mathbf{G}_{mS} \longrightarrow G.$$

Whence a product morphism $(\mathbf{G}_m)_S^r \longrightarrow G^r$ which, composed with the morphism $G^r \longrightarrow G$ defined by the composition law in G , provides a morphism

$$v : (\mathbf{G}_m)_S^r \longrightarrow G.$$

In view of the existence of the group morphism u_n , it is clear that $u_n = v_n$. It remains to see that v is a group morphism, which means that two evident morphisms $f, g : X = (\mathbf{G}_m)_S^r \times_S (\mathbf{G}_m)_S^r \longrightarrow G$ coincide. Let Z be the coincidence subscheme of f and g . Since $(\mathbf{G}_m)_S^r$ has connected fibers and $v(\mathbf{G}_m)_S^r$ contains the unit section, one sees as in 8.3 that v factors through the identity component of G , which allows us to suppose G separated (Exp. VIB 5.2), hence Z closed. Moreover, since v_n is a group morphism, one has $f_n = g_n$ for every n , so Z contains a neighborhood of the closed fiber of X (EGA I 10.9.4), hence is schematically dense in X , X having smooth irreducible fibers (Exp. IX 4.6), and consequently $Z = X$, so $f = g$.

Definition 8.4. Let S be a locally noetherian prescheme, G a group S -prescheme of finite type, and $\mathcal{S}$ the set of S -schemes S' , spectra of complete discrete valuation rings with algebraically closed residue field, with closed point s , generic point t , such that $(G_t)_{\text{red}}^0$ is smooth and has a Chevalley decomposition, i.e. is an extension of an abelian variety by a smooth connected linear algebraic subgroup L_t (this decomposition is then unique). If $S' \in \mathcal{S}$, denote by G' (resp. L') the schematic closure in $G_{S'}$ of G_t (resp. L_t).

Under these conditions, we shall say that

“the abelian part of G does not degenerate into a toric part”, or more briefly that G satisfies property AT,

if for every $S' \in \mathcal{S}$, L_s has the same reductive rank as G'_s . (Intuitively, suppose that the quotient $A = G'/L$ is representable, in which case A is a flat group prescheme such that $(A_t)_{\text{red}}^0$ is an abelian variety. Condition “AT” then means that A_s has reductive rank zero, hence that $(A_s)_{\text{red}}^0$ is an extension of an abelian variety by a unipotent group.)

Similarly, supposing moreover that G has connected fibers, we shall say that

G satisfies property ATC

if for every $S' \in \mathcal{S}$, the schematic closure Z of the center Z_t of G_t satisfies AT.

These technical definitions are justified by the following proposition:

Proposition 8.5. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type. Then:*

- a) *For the functor $\mathcal{T}$ of subtori of G to “commute with adic limits of artinian local rings” (cf. 8.1), it is necessary and sufficient that G satisfy property AT (8.4).*
- b) *If G has connected fibers, for the functor $\mathcal{TC}$ of central subtori of G to commute with adic limits of artinian local rings, it is necessary and sufficient that G satisfy property ATC (8.4).*

We shall need the following technical lemma:

Lemma 8.6. *Let S be the spectrum of a discrete valuation ring, s the closed point of S , G a group S -prescheme flat and of finite type, T_s a subtorus of G_s . Then there exists an S -scheme S' , the spectrum of a discrete valuation ring, faithfully flat over S , with closed point s' , and a group subscheme C of $G_{S'}$, flat over S' , commutative, with connected fibers, such that $C_{s'}$ contains $T_s \times_s s'$.*

Replacing S by S' , the spectrum of a discrete valuation ring faithfully flat over S , one may suppose that T_s is equal to $(\mathbf{G}_m)_s^r$ and that the transcendence degree of $\kappa(s)$ over the prime field is $\geq r$ (EGA 0_{III} 10.3.1 and EGA II 7.1.9). There then exists an element x of $T_s(s)$ such that every algebraic subgroup of G_s which “contains” x contains T_s (cf. Exp. XIII proof of 2.1 (ii) $\Rightarrow$ (vii)). Since G is flat over S , a quasi-section passes through x (EGA IV 14.5.8), and consequently, replacing S by the spectrum of a discrete valuation ring faithfully flat over S , one may suppose that there exists a section x of G over S lifting x . Let C_t be the commutative algebraic subgroup of G_t generated by x_t (Exp. VIB 7), and let C be the schematic closure of C_t in G . It is clear that C_s contains x , hence contains T_s , and consequently the “identity component” of C will be a flat commutative group scheme answering the question.

Proof of 8.5 a). Suppose that the functor $\mathcal{T}$ commutes with adic limits of artinian rings, and let us show that G satisfies property AT. Let therefore $S' \in \mathcal{S}$, and let T_s be a maximal torus of G'_s . We must prove that T_s is contained in L_s . The formation of L and of G' plainly commutes with faithfully flat extensions $S'' \rightarrow S'$ of discrete valuation rings. After changing S' , we may therefore, by Lemma 8.6, suppose that there exists a flat commutative group subscheme C of $G'_{S'}$ such that C_s contains T_s . But then T_s is a central subtorus of C_s , and consequently lifts infinitesimally to a central subtorus (Cor. 2.3). In view of the hypothesis on G , T_s lifts to a subtorus T of G . Plainly T_t is contained in the linear component L_t of G_t , so T is contained in L .

Suppose now that G satisfies property AT, and let us show that condition 8.1 iii) bis is verified. Let therefore S be the spectrum of a discrete valuation ring A , complete, with algebraically closed residue field, $\mathfrak{m}$ the maximal ideal of A , $S_n = \text{Spec}(A/\mathfrak{m}^n)$, u_n , $n \in \mathbf{N}$, a coherent system of group morphisms $u_n : (\mathbf{G}_{mS})_n \rightarrow G_n = G \times_S S_n$. Let q be a prime number invertible on S . With the integer ℓ ranging over the powers of q , there exists a unique group S -morphism

$$\ell u : \ell \mathbf{G}_{mS} \rightarrow G$$

lifting $u_n|_{\ell(\mathbf{G}_{mS})_n}$ for every n (Prop. 1.6 a)). Consequently, if there exists a group S -morphism $u : \mathbf{G}_{mS} \rightarrow T$ lifting u_n for every n , its restriction to $\ell \mathbf{G}_{mS}$ is uniquely determined. Taking account of the density theorem (Exp. IX 4.8 (a)), this proves uniqueness of u and the fact that, to prove existence of u , we may allow ourselves a faithfully flat extension of the base.

Now $(G_t)_{\text{red}}^0$ has a Chevalley decomposition, and this is already defined over a finite extension L of the field of fractions K of A . Replacing S by the normalization of S in L , we may therefore suppose that $S \in \mathcal{S}$. The morphism ℓu_t factors through $(G_t)_{\text{red}}$, so ℓu factors through the schematic closure G'' of $(G_t)_{\text{red}}$ in G' . Again by the density theorem, one deduces that u_n factors through G''_n for every n . Since G has property AT, every subtorus of G'_s , and a fortiori of G''_s , is contained in L_s , so u_s factors through L_s . I say that for every n , u_n factors through L_n . Indeed, since L_t is invariant in $(G'')_t$, L is invariant in G'' ; moreover, L is flat over S , so for every integer n , the quotient $H_n = G''_n/L_n$ is representable (Exp. VIA § 4). The image of $(\mathbf{G}_{mS})_n$ in H_n is a subtorus of H_n (Exp. IX 6.8), whose closed fiber is zero, so this image is zero, and consequently u_n factors through L_n . But since L_t is affine, one deduces that the family u_n is admissible (Prop. 4.3), which completes the proof.

The proof of 8.5 b) is entirely analogous to that of 8.5 a), taking account of 8.2 and Exp. IX 5.6 a).

Proposition 8.7. *Let S be a locally noetherian prescheme, G a group S -prescheme of finite type. Then:*

- i) If G has connected fibers and satisfies AT, it satisfies ATC.
- ii) If G satisfies AT, every group subprescheme of G satisfies AT.
- iii) If G is flat over S and the abelian rank of the fibers of G is a locally constant function on S , then G satisfies AT.

i) follows immediately from 8.5 and Exp. IX 5.6 a).

ii) Let H be a group subprescheme of G . To prove that H satisfies property AT, we may suppose that S is the spectrum of a complete discrete valuation ring, and we must show that every coherent family of group S_n -morphisms

$$u_n : (\mathbf{G}_{mS})_n \longrightarrow H_n$$

is admissible (8.5 and 8.1 iii bis)). Now by hypothesis, G satisfies property AT, so there exists a group S -morphism

$$u : (\mathbf{G}_m)_S \longrightarrow G$$

lifting u_n for every n . Proceeding as in the proof of 8.5, one sees that $u|_{\ell \mathbf{G}_{mS}}$ (where ℓ ranges over the powers of a prime number q invertible on S) factors through H . By density, one deduces that on the generic fiber, u_t factors through H_t . Since $(\mathbf{G}_m)_S$ is reduced, it indeed follows that u factors through H .

iii) Let $S' \in \mathcal{S}$ (notation of 8.4). The group L is flat over S , and its generic fiber L_t is affine; under these conditions, one can show that L_s is necessarily affine (XVII App. III, 2). Since G and L are flat, one has $G = G'$, and the dimension of the fibers of G and L is constant on S (Exp. VIB 4), so that one has the inequalities

$$\text{abelian rank } G_s \leq \dim G_s - \dim L_s = \dim G_t - \dim L_t = \text{abelian rank } G_t.$$

The hypothesis implies that these inequalities are equalities. It follows that $(G_s/L_s)_{\text{red}}^0$ is an abelian variety, hence has reductive rank zero, and consequently L_s has the same reductive rank as G_s .

We are now in a position to prove the main theorems of this paragraph:

Theorem 8.8. *Let G be a group prescheme of finite type over a locally noetherian prescheme S . Suppose G flat over S and with connected fibers. Then:*

- a) *For the functor $\mathcal{T}\mathcal{C}$ of central subtori of G to be representable, it is necessary and sufficient that G have property ATC (8.4). Moreover, in this case, $\mathcal{T}\mathcal{C}$ is representable by an S -prescheme étale and separated over S .*
- b) *Under the conditions of a), for every S -torus, the functor $\underline{\text{Hom}}_{S\text{-gr}}^{\text{cent}}(T, G)$ of central homomorphisms from T to G is representable by an S -prescheme étale and separated over S .*

Theorem 8.9. *Let S be a locally noetherian prescheme and G a group S -prescheme of finite type, smooth over S .*

- a) *For the functor $\mathcal{T}$ of subtori of G to be representable, it is necessary and sufficient that G have property AT (8.4). Moreover, in this case, $\mathcal{T}$ is representable by an S -prescheme smooth and separated over S ; more precisely, the structural morphism $\mathcal{T} \longrightarrow S$ admits a canonical factorization*

$$\mathcal{T} \xrightarrow{u} Y \xrightarrow{v} S$$

where v is a smooth quasi-projective morphism (hence of finite type) and u is a separated étale morphism.

- b) *Under the conditions of a), for every S -torus T , the functor $\underline{\text{Hom}}_{S\text{-gr}}(T, G)$ of homomorphisms from T to G is representable by an S -prescheme smooth and separated over S .*

Proof of 8.8 a). If the functor $\mathcal{T}\mathcal{C}$ is representable, it commutes with adic limits of artinian rings, and consequently (8.2 and 8.5 b)) G has property ATC. To establish the converse, we shall use the following result, which will be proved in EGA VI, and is also found in Murre's exposé: Sémin. Bourbaki, May 1965, no. 294, Theorem 1, Corollary 2.

Lemma 8.10. *Let S be a locally noetherian prescheme, and $\mathcal{F}$ a contravariant functor defined on $(\text{Sch})/S$, with values in the category of sets. For $\mathcal{F}$ to be representable by an étale separated S -prescheme, (it is necessary and) it suffices that $\mathcal{F}$ satisfy the following five properties:*

- i) $\mathcal{F}$ is a sheaf for the fpqc topology (Exp. IV).
- ii) $\mathcal{F}$ commutes with inductive limits of rings (Exp. XI 3.2).
- iii) $\mathcal{F}$ commutes with adic limits of artinian local rings (8.1 i)).
- iv) $\mathcal{F}$ satisfies the “valuative criterion of separation”, i.e. for every S -scheme S' which is the spectrum of a discrete valuation ring, with generic point t , the canonical map

$$\mathcal{F}(S') \longrightarrow \mathcal{F}(t)$$

is injective.

- v) $\mathcal{F}$ is infinitesimally étale (Exp. XI 1.8).

Let us show that the functor $\mathcal{TC}$ of 8.8 satisfies the five conditions of 8.10.

i) The functor $\mathcal{T}$ (resp. $\mathcal{TC}$) is a sheaf for the fpqc topology as soon as G is of finite presentation over S . Indeed, every monomorphism from a torus to G is then an immersion (Exp. VIII 7.9), and the property follows from fpqc descent of subpreschemes.

ii) Corollary 6.3 proves that the functor $\mathcal{T}$ commutes with inductive limits of rings if G is of finite presentation over S ; one immediately deduces that the same holds for $\mathcal{TC}$.

iii) By 8.5, condition iii) is precisely equivalent to property ATC.

iv) follows simply from the fact that if S is the spectrum of a discrete valuation ring, two subtori of G having the same generic fiber coincide, and more precisely coincide with the identity component of the schematic closure in G of their generic fiber.

v) follows from 2.3, since G is flat over S .

Proof of 8.8 b). Proceeding as in Exp. XI 4.2, one sees that it suffices to prove that the product group $T \times_S G$ still satisfies property ATC, which is immediate from the definition.

Proof of 8.9 a). Replacing G by its identity component (Exp. VIB 3.10), one may suppose G with connected fibers. If T is a subtorus of G , its centralizer $\text{Centr}_G(T)$ is then representable (Exp. XI 6.11) by a group subprescheme of G , smooth over S (Exp. XI 2.4), with connected fibers (Exp. XII 6.6 b)), hence is an element of $\mathcal{CT}(S)$, where $\mathcal{CT}$ is the functor defined in 7.1 i). It is clear that the map

$$T \longmapsto \text{Centr}_G(T)$$

defines an S -morphism

$$u : \mathcal{T} \longrightarrow \mathcal{CT}.$$

Since $\mathcal{CT}$ is representable by an S -prescheme smooth and quasi-projective over S (7.1 i)), it suffices for us to prove that the morphism u is representable by a separated étale morphism.

After a suitable base change $S' \longrightarrow S$ (with $S' = \mathcal{CT}$, hence S' locally noetherian), we are reduced to the following problem: let S be a locally noetherian prescheme, G a group S -prescheme, smooth and of finite type over S , with connected fibers, H a subgroup of G , smooth over S and with connected fibers. Consider the subfunctor X of $\mathcal{TC}_H$ such that, for every S -prescheme S' , $X(S')$ is the set of central subtori T of $H_{S'}$ such that $\text{Centr}_{G_{S'}}(T) = H_{S'}$. We shall show that X is representable by an S -prescheme étale and separated over S .

Indeed, by hypothesis, G satisfies property AT, so the same holds for H (8.7 ii)), and since H has connected fibers, H also satisfies property ATC (8.7 i)). On the other hand, H is smooth over S , hence flat. By 8.8 a), $\mathcal{TC}_H$ is representable by an S -prescheme étale and separated over S . It then suffices for us to show that the canonical monomorphism $X \longrightarrow \mathcal{TC}_H$ is representable by an open immersion.

Put $S' = \mathcal{TC}_H$, and let K be the centralizer in $G_{S'}$ of the “universal” central torus of $H_{S'}$. The group K is a group scheme smooth over S' , with connected fibers, containing $H_{S'}$. By definition, one has $X = \prod_{K/S'} H_{S'}/K$, which is indeed representable by the subprescheme, both open and closed, of S' over which $H_{S'}$ and K have the same relative dimension.

One proves 8.9 b) analogously to 8.8 b).

Corollary 8.11. *Let S be a prescheme, G a group S -prescheme smooth over S and of finite presentation. Then, if the abelian rank of the fibers of G is a locally constant function on S (in particular if the fibers of G are affine), the functor of subtori of G is representable by an S -prescheme smooth and separated over S , and the same holds for the functor $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, G)$, for every S -torus T .*

Indeed, the assertion is local on S , so we may suppose S affine and the abelian rank of the fibers of G constant. One can then find (Exp. VIB § 10) an affine noetherian scheme S_0 and a group S_0 -prescheme G_0 , smooth and of finite type over S_0 , such that G is S -isomorphic to $G_0 \times_{S_0} S$. Moreover, the abelian rank of the fibers of a group S -prescheme of finite presentation over S is an ind-constructible function (6.3 bis). By a standard argument (EGA IV 8), one concludes that in the present case, one may suppose the abelian rank of the fibers of G_0 constant on S_0 . But then G_0 has property AT (8.7 iii)), and consequently (8.10) the functor of subtori of G_0 is representable by an S_0 -prescheme smooth and separated over S_0 , whence the asserted property for G . As for the functor $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, G)$, one proceeds analogously.

Generalization of 8.9. Since the functor $\mathcal{T}$ of subtori of a smooth group G is not necessarily representable, we shall state sufficient conditions for a subfunctor of $\mathcal{T}$ to be representable.

Proposition 8.12. *Let S be a locally noetherian prescheme, G a group S -prescheme, smooth over S , with connected fibers, and $\mathcal{F}$ a sub- S -functor of the functor $\mathcal{T}$ of subtori of G , satisfying the following properties:*

- i) $\mathcal{F}$ is a sheaf for the fpqc topology (Exp. IV).
- ii) $\mathcal{F}$ commutes with inductive limits of rings (Exp. XI 3.2).
- iii) $\mathcal{F}$ commutes with adic limits of artinian local rings (Exp. XI 3.3).
- iv) $\mathcal{F}$ is infinitesimally smooth over S (Exp. XI 1.8).
- v) $\mathcal{F}$ is stable under inner automorphisms of G , i.e. for every S -prescheme S' , one has

$$T \in \mathcal{F}(S') \text{ and } g \in G(S') \Rightarrow \mathrm{int}(g)T \in \mathcal{F}(S').$$

Then $\mathcal{F}$ is representable by an S -prescheme smooth and separated over S .

Proposition 8.12 bis. *Let S and G be as above, T an S -torus and $\mathcal{F}$ a subfunctor of $\underline{\mathrm{Hom}}_{S\text{-gr}}(T, G)$, satisfying the following properties:*

- i), ii), iii), iv) as above.
- v) $\mathcal{F}$ is stable under inner automorphisms of G , i.e. for every S -prescheme S' , one has

$$u \in \mathcal{F}(S') \text{ and } g \in G(S') \Rightarrow \mathrm{int}(g)u \in \mathcal{F}(S').$$

Then $\mathcal{F}$ is representable by an S -prescheme smooth and separated over S .

Proof of 8.12. (The proof of 8.12 bis is entirely analogous and is left to the reader.)

Let $u : \mathcal{F} \rightarrow \mathcal{CT}$ (7.1 i)) be the S -morphism associating to every element T of $\mathcal{F}(S')$ the element $\underline{\mathrm{Centr}}_{G_{S'}}(T)$ of $\mathcal{CT}(S')$. Since $\mathcal{CT}$ is representable by a smooth quasi-projective S -prescheme (7.1 i)), we are reduced to proving representability of u . After the base change $\mathcal{CT} \rightarrow S$, we are reduced to the following problem: given S and G as above, H a smooth subgroup of G with connected fibers, we must represent the functor $\mathcal{F}_H$ such that $\mathcal{F}_H(S') =$ set of elements $T \in \mathcal{F}(S')$ such that

$$H_{S'} = \underline{\mathrm{Centr}}_{G_{S'}}(T).$$

We shall show that $\mathcal{F}_H$ is étale and separated over S . To do so, it suffices for us to verify the five conditions of 8.10.

Conditions i), ii) and iii) of 8.10 follow easily from 8.12 i), ii) and iii). It has already been remarked that $\mathcal{TC}_H$ is a separated infinitesimally étale functor, so $\mathcal{F}_H$, which is a subfunctor of $\mathcal{TC}_H$, is separated and infinitesimally unramified (Exp. XI 1.8). It therefore

suffices for us to show that $\mathcal{F}_H$ is infinitesimally smooth (loc. cit.). Let S be the spectrum of an artinian local ring, S_0 a subscheme of S defined by a nilpotent ideal, T_0 an element of $\mathcal{F}_H(S_0)$; let us prove that T_0 lifts to an element T of $\mathcal{F}_H(S)$. By hypothesis (8.12 iv)), T_0 lifts to an element T' of $\mathcal{F}(S)$. On the other hand, H being smooth, T_0 lifts to a subtorus T'' of H_S (Exp. IX 3.6 bis) conjugate to T' by an element $g \in G(S)$ (loc. cit.), so $T'' \in \mathcal{F}(S)$ (8.12 v)). Since $\text{Centr}_{G_S}(T'')$ is smooth over S and coincides with H_{S_0} over S_0 , T'' is in the center of H_S (Exp. IX 5.6 a)), and its centralizer in G is equal to H_S ; in brief, $T'' \in \mathcal{F}_H(S)$, and one takes $T = T''$.

Corollary 8.13. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, and let $\mathcal{T}\mathcal{D}$ be the subfunctor of $\mathcal{T}$ whose set of points with values in an S -prescheme S' is the set of subtori T of $G_{S'}$ such that, for every point s' of S' , $T_{s'}$ is contained in the derived subgroup (Exp. VIB 7) of $G_{s'}$. Then $\mathcal{T}\mathcal{D}$ is representable by an S -prescheme smooth and separated over S .*

Corollary 8.13 bis. *Let S and G be as above, T an S -torus, and let*

$$\underline{\text{Hom}}_{S\text{-gr}}^{\text{der}}(T, G)$$

be the subfunctor of $\underline{\text{Hom}}_{S\text{-gr}}(T, G)$ whose set of points with values in S' is the set of S' -morphisms $u : T_{S'} \rightarrow G_{S'}$ such that, for every point s' of S' , $u_{s'}$ factors through the derived group of $G_{s'}$. Then this functor is representable by an S -prescheme smooth and separated over S .

Corollary 8.13 and Corollary 8.13 bis are proved analogously; let us prove, for example, 8.13 bis. By the usual procedure, we reduce to the case where S is noetherian. Let us verify that the five conditions of 8.12 bis are satisfied:

Conditions i) and iv) follow immediately from the corresponding properties of the functor $\underline{\text{Hom}}_{S\text{-gr}}(T, G)$. Condition v) is satisfied since the derived group of an algebraic group is invariant (Exp. VIB § 7). To establish ii), we are reduced by a standard reduction (EGA IV 8) to proving that if S is a noetherian integral scheme with generic point η and $u : T \rightarrow G$ is a group S -morphism which on the generic fiber factors through the derived group of G_η , then there exists a neighborhood U of η such that, for every point s of U , u_s factors through the derived group of G_s . But this follows immediately from Exp. VIB 10.12. To establish iii), let us return to the notation of 4.3. To show that an element $(u_m)_{m \in \mathbf{N}}$ of $\varprojlim_m \underline{\text{Hom}}_{S_m\text{-gr}}^{\text{der}}(T_m, G_m)$ is “admissible” in the sense of 4.3 and comes from an element of $\underline{\text{Hom}}_{S\text{-gr}}^{\text{der}}(T, G)$, one immediately reduces to the case where S is the spectrum of a complete discrete valuation ring (cf. 8.1). Let t be the generic point and s the closed point of S , D the schematic closure in G of the derived group D_t of G_t .

a) D_s contains the derived group of G_s . Indeed, since D_t is invariant in G_t and G is flat over S , D is invariant in G . Moreover, the morphism

$$G \times_S G \rightarrow G, \quad (x, y) \mapsto xyx^{-1}y^{-1}$$

factors through D_t on the generic fiber, so it factors through D . Consequently the algebraic group G_s/D_s is commutative, whence assertion a).

b) If $u_m \in \underline{\text{Hom}}_{S_m\text{-gr}}^{\text{der}}(T_m, G_m)$, u_m factors through D_m . Indeed, by hypothesis, u_s factors through the derived group of G_s , hence a fortiori factors through D_s , by a). Since D_m is flat over S_m and invariant in G_m , the quotient group $H_m = G_m/D_m$ is representable (Exp. VIA § 4). Since the image of T_m in H_m is a torus (Exp. IX 6.8) whose closed fiber is zero, the image of T_m is zero; this says that u_m factors through D_m .

c) The family $(u_m)_{m \in \mathbf{N}}$ is “admissible” and lifts to a morphism $T \rightarrow G$ belonging to $\underline{\text{Hom}}_{S\text{-gr}}^{\text{der}}(T, G)$. In view of the foregoing and 4.1 bis, it suffices to prove that D_t is an affine algebraic group, which follows from the following lemma:

Lemma 8.14. *Let G be a smooth connected algebraic group defined over a field k . Then the derived group D of G is affine.*

Since formation of D commutes with extension of the base field (Exp. VIB 7), one may suppose k algebraically closed. But then G is an extension of an abelian variety A by a linear group L . Since A is commutative, D is necessarily contained in L , hence is affine.

Maximal tori.

Theorem 8.15. *Let S be a locally noetherian prescheme and G a group S -prescheme smooth and of finite type over S . Then the following conditions are equivalent:*

- i) *The S -functor $\mathcal{T}\mathcal{M}$ whose set of points with values in an S -prescheme S' is the set of maximal tori of $G_{S'}$ (Exp. XII 1.3) is representable.*
- ii) *The preceding functor $\mathcal{T}\mathcal{M}$ is representable by an S -prescheme smooth and quasi-projective over S , with affine fibers.*
- iii) *The group G has, locally for the étale topology, a maximal torus.*
- iv) *The group G has, locally for the faithfully flat topology, a maximal torus.*
- v) *The group G has property AT (8.4), and the reductive rank of its fibers is a locally constant function on S .*

Proof: ii) $\Rightarrow$ i) is clear.

i) $\Rightarrow$ iii). Indeed, since G is of finite presentation over S , it follows from 6.3 that $\mathcal{T}\mathcal{M}$ commutes with inductive limits of rings, hence is locally of finite presentation if it is representable (EGA IV 8.14). Moreover, $\mathcal{T}\mathcal{M}$ is formally smooth (Exp. XI 2.1 bis). Thus if it is representable, it is representable by a prescheme smooth over S , and iii) then follows from Hensel's lemma (Exp. XI 1.10).

iii) $\Rightarrow$ iv) is clear.

iv) $\Rightarrow$ v). Let s be a point of S . By hypothesis, there exists an S -prescheme S' , flat over S , whose image contains s , such that $G_{S'}$ has a maximal torus T' . Let s' be a point of S' above s . The reductive rank of the fibers of $G_{S'}$ is therefore constant on $\text{Spec } \mathcal{O}_{S',s'}$, and consequently the reductive rank of the fibers of G is constant on the image of $\text{Spec } \mathcal{O}_{S',s'}$, which is $\text{Spec } \mathcal{O}_{S,s}$ (EGA IV 2.3.4 ii)). Now the reductive rank of the fibers of G is a locally constructible function on S (6.3 bis), hence this rank is constant on a neighborhood of s (EGA IV 1.10.1).

To see that G has property AT, consider an S -scheme S_1 , the spectrum of a discrete valuation ring, the closed point s_1 of S_1 mapping to the preceding point s . The prescheme $S'_1 = S_1 \times_S S'$ is faithfully flat over S_1 , and $G_{S'_1}$ has a maximal torus. Let A (resp. A') be the ring of S_1 (resp. S'_1). Considering A' as the inductive limit of its A -subalgebras of finite type, it follows from 6.3 that there exists an S_1 -scheme S''_1 such that $G_{S''_1}$ has a maximal torus and such that the structural morphism $S''_1 \rightarrow S_1$ is of finite type and surjective. Using now EGA II 7.1.9, we may suppose that S'_1 is the spectrum of a discrete valuation ring. But then it is clear that $G_{S'_1}$, hence also G_{S_1} , has property AT. Since this is true for every S -prescheme S_1 , the spectrum of a discrete valuation ring, G has property AT.

v) $\Rightarrow$ i). Indeed, by 8.9 the functor $\mathcal{T}$ of subtori of G is representable, and it is clear that $\mathcal{T}\mathcal{M}$ is representable by the subprescheme, both open and closed, of $\mathcal{T}$ representing the subfunctor of tori of rank r , where r denotes the reductive rank of G (which one may suppose constant). iii) $\Rightarrow$ ii). Indeed, if condition iii) holds, we may use the results of Exp. XII 7.1. The functor $\mathcal{T}\mathcal{M}$ is therefore canonically isomorphic to the functor of Cartan subgroups of G , and it suffices to apply 7.3 i).

Remark 8.16. One can show that the prescheme $\mathcal{T}\mathcal{M}$ of maximal tori of G is affine over S^* , which generalizes Exp. XII 5.4.

* Cf. M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes* (thesis, to appear)(N.D.E.: see Lecture Notes Math. 119 (1970), Springer), especially IX 2.9.

Corollary 8.17. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S . Suppose that the abelian rank and the reductive rank of the fibers of G are locally constant functions on S ; then G satisfies the (equivalent) properties i) to iv) of 8.15.*

We may suppose that the abelian rank and the reductive rank of the fibers of G are constant. Proceeding as in 8.11 and taking account of 6.3 bis, one reduces to the case where S is noetherian. But then G has property AT (8.7), and one concludes by 8.15 v).

Corollary 8.18. *Let S be a prescheme, G a group S -prescheme smooth over S . Then the following conditions are equivalent:*

- a) *The unipotent rank ρ_u and the abelian rank ρ_{ab} (6.1 ter) of the fibers of G are locally constant functions on S .*
- b) *The unipotent rank ρ_u is locally constant, and G has, locally for the fpqc topology, maximal tori.*
- c) *The reductive rank ρ_r (6.1 ter) and the abelian rank ρ_{ab} of the fibers of G are locally constant functions on S .*

Remarks 8.19. Under the hypotheses of 8.18, a more precise argument, using the lower semicontinuity of the abelian rank (announced in Exp. X 8.7), shows that if two of the three ranks ρ_u , ρ_r , ρ_{ab} are locally constant, then the same holds for the third.

Proof of 8.18. Replacing G by its identity component, we may suppose G of finite presentation over S (Exp. VIB 5.3.3).

a) $\Rightarrow$ c). Let s be a point of S . Since ρ_{ab} is locally constant, it follows from 8.11 that, after making an étale extension covering S , one may suppose that there exists a subtorus T of G whose fiber T_s is a maximal torus of G_s . Let $C = \underline{\text{Centr}}_G(T)$, which is a group subscheme of G , smooth over S , with connected fibers. For every point t of S , C_t plainly contains a Cartan subgroup C'_t of G_t . After restricting S , one may suppose ρ_u , ρ_{ab} and the relative dimension of C over S constant.

One then has the inequalities

$$\begin{aligned} \dim C_s &= \dim C_t \geq \dim C'_t \geq \rho_u(t) + \rho_{ab}(t) + \dim T_t \\ &= \rho_u(s) + \rho_{ab}(s) + \dim T_s = \rho_\nu(s) = \dim C_s. \end{aligned}$$

One deduces that $C_t = C'_t$, hence that T is a maximal torus of G , and a fortiori $\rho_r(t) = \rho_r(s)$.

c) $\Rightarrow$ b) by 8.17.

b) $\Rightarrow$ a). Indeed, since G has, locally for the fpqc topology, maximal tori, it has, locally for the fpqc topology, Cartan subgroups, and consequently (Exp. XII 7.3) the nilpotent rank $\rho_\nu = \rho_u + \rho_r + \rho_{ab}$ is locally constant. Since ρ_r and ρ_{ab} are locally constant, ρ_u is locally constant.